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# Hybrid Neural–MGARCH Models for Volatility and Financial Risk Forecasting

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## Abstract

In this thesis, three classes of volatility models are examined. First, a classical econometric framework based on multivariate Generalized Autoregressive Conditional Heteroskedasticity (MGARCH) models is considered. Second, purely data-driven recurrent neural network-based volatility models are analyzed. Third, hybrid Neural-MGARCH models are investigated through two strategies: integrating an MGARCH covariance kernel into a recurrent architecture and using a recurrent controller to produce time-varying MGARCH parameter matrices.

Specifically, a fully asymmetric Neural-BEKK model is implemented, allowing for leverage effects and (a)symmetric volatility spillovers across assets through both conditional variances and covariances. To the author’s knowledge, neither a Neural-BEKK with full time-varying asymmetric parameter matrices nor the integration of a fully asymmetric BEKK kernel into a custom LSTM-style recurrent cell has previously been studied.

In the main design, the leading hybrid specifications attain the lowest mean out-of-sample joint VaR–ES losses, improve dynamic VaR calibration relative to the classical benchmarks, and remain competitive with the pure recurrent models on calibration, although the model confidence set retains almost all specifications. Within the Neural-BEKK family, the diagonal restriction records lower mean losses than the full variant, so the additional off-diagonal spillover flexibility shows no observed out-of-sample benefit on these data.

Furthermore, this thesis provides a bidirectional empirical comparison of hybrid neural–MGARCH architectures against both classical econometric (BEKK) and purely data-driven (LSTM/GRU) baselines, addressing a gap in the existing literature on neural-augmented multivariate volatility models, where comparisons against purely neural benchmarks are typically absent.

**Keywords:** Risk, Volatility Modeling, MGARCH, Neural Networks

## 1 Introduction

Modeling time-varying second moments has become a cornerstone of statistical time series analysis. Although conditional heteroskedasticity was first formalised in macroeconometrics to describe inflation uncertainty (Engle, 1982), it has since proven to be a generic feature of stochastic systems rather than a peculiarity of economic or financial data: volatility models are nowadays routinely employed in fields as diverse as energy systems engineering, where they capture the mean and volatility dynamics of wind speed (H. Liu et al., 2011), and hydrology, where they describe the variance dynamics of daily streamflow (W. Wang et al., 2005). In finance, however, the problem takes on a particular structural form: not only the variance of a single series but the entire conditional covariance matrix of an asset portfolio is of immediate economic relevance for portfolio allocation, derivative pricing, and, most explicitly, for regulatory capital under the Fundamental Review of the Trading Book, whose internal-models approach uses Expected Shortfall at the 97.5% confidence level (Basel Committee on Banking Supervision, 2019, January).

The dominant econometric framework for this task is the multivariate GARCH (MGARCH) class, of which the BEKK parametrization (Engle and Kroner, 1995) remains a central representative: it guarantees positive definiteness of the conditional covariance matrix by construction and admits interpretable parameter restrictions. Its resilience as a benchmark is well documented, but not universal. In the univariate case, Hansen and Lunde (2005) find that GARCH(1,1) is not outperformed for exchange-rate volatility, whereas leverage specifications perform better for the equity series in their comparison. Likewise, the large-scale multivariate comparison of Laurent et al. (2012) finds no uniformly dominant MGARCH specification: the gains from richer parameterizations depend on the market regime, loss function, and competing benchmark. Yet the very features that make MGARCH models tractable also constrain them. Their recursion is linear in past outer products of returns and lagged covariances, their parameter matrices are time-invariant, and the conditional covariance dynamics depend exclusively on the portfolio’s own return history. This closed-universe assumption abstracts from shocks originating outside the modelled asset universe and from slowly moving macroeconomic conditions. Volatility transmission has been documented across US equity, bond, foreign-exchange and commodity markets (Diebold and Yilmaz, 2012). Moreover, inflation and industrial-production growth help explain long-run equity volatility (Engle, Ghysels, and Sohn, 2013), while business-cycle conditions, inflation, monetary policy and macroeconomic uncertainty have been linked to time variation in stock–bond dependence (Asgharian et al., 2015; Yang et al., 2009). The strength of this macroeconomic channel is not uniform across studies: Baele et al. (2010) find that liquidity proxies explain stock–bond correlations better than conventional macroeconomic fundamentals.

This thesis addresses the restrictions imposed by linear recursions and time-invariant coefficient matrices by integrating recurrent neural networks into the BEKK framework. It does not relax the closed-universe information set: all implemented models are estimated from the same portfolio-return features without exogenous macroeconomic inputs, which is treated explicitly as a limitation and direction for future work. Two hybridisation strategies are pursued. The first retains the classical BEKK recursion as a covariance kernel inside a recurrent cell and del-

egates temporal adaptation to a learnt modulation of the kernel output. The second — the Neural-BEKK proposed in this thesis — goes one step further and makes the BEKK parameter matrices themselves time-varying functions of the unfolding information set, generated at each step by a recurrent controller. The specification is fully asymmetric, accommodating leverage effects and asymmetric volatility spillovers through both conditional variances and covariances. A targeted literature search conducted through June 2026 identified a diagonal symmetric Neural-BEKK and an additive LSTM-BEKK hybrid as the closest precedents (H. Wang et al., 2025; Yin and Barucca, 2022), but did not identify a fully asymmetric Neural-BEKK with full time-varying parameter matrices. Both a full off-diagonal parameterization and a diagonal restriction are evaluated; the diagonal variant records lower mean out-of-sample losses, so the additional spillover flexibility is not responsible for the leading result. The empirical evaluation is deliberately bidirectional: the hybrid architectures are benchmarked against classical econometric specifications (symmetric and asymmetric BEKK, DCC) *and* against purely data-driven LSTM and GRU covariance models — a comparison that the literature on neural-augmented multivariate volatility models typically omits. Forecast quality is assessed at the portfolio level through Value-at-Risk (VaR) and Expected Shortfall (ES) backtests, strictly consistent joint VaR-ES scoring functions, and model confidence sets, with neural and hybrid models re-trained across eleven random seeds to quantify sensitivity to training noise. The leading hybrid specifications attain the lowest mean out-of-sample joint VaR-ES losses and generally reduce the dynamic predictability of VaR exceedances. This ranking is not decisive, however: the model confidence set retains almost all specifications, and some leading hybrids are rejected by two-sided tests of exact ES calibration. Among the classical benchmarks, the symmetric BEKK, DCC, and ADCC fail both conditional-coverage and dynamic-quantile tests, whereas the asymmetric BEKK fails the dynamic-quantile test only.

The remainder of this thesis is organised as follows. Section 2 introduces the econometric, neural, and hybrid model classes considered in this work, together with the risk measures and backtesting procedures used for evaluation. Section 3 describes the multi-asset portfolio that serves as the empirical testbed and discusses its statistical properties. Section 4 details the model specifications, estimation strategies, and the unified train-validation-test protocol under which all model classes are estimated and evaluated on a common information footing. Section 5 reports the comparative risk-forecasting results, robustness checks, and formal inference on forecast superiority. Section 6 discusses the limitations of the empirical design and outlines directions for future research, and Section 7 concludes.

## 2 Theoretical Background

### 2.1 Volatility Models

#### 2.1.1 GARCH

The seminal contribution of Bollerslev (1986) extends the ARCH framework of Engle (1982) by allowing the conditional variance to depend not only on past squared innovations but also on its own past realizations. The conditional variance equation of the workhorse GARCH(1,1)

specification is

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (1)$$

where  $\omega > 0$ ,  $\alpha \geq 0$ ,  $\beta \geq 0$ , and  $\alpha + \beta < 1$  jointly ensure positivity of the conditional variance and covariance stationarity of the process. Despite its parsimonious structure, the GARCH(1, 1) model captures key stylized facts of financial return series, most notably volatility clustering and persistence, and remains a demanding forecasting benchmark. In a comparison of 330 volatility models, Hansen and Lunde (2005) find that it is not outperformed for the exchange-rate series, while leverage specifications are superior for the equity series. The evidence therefore supports its use as a benchmark rather than universal forecast dominance.

A substantial univariate literature has developed extensions that relax specific assumptions of the baseline specification. Most prominently, the EGARCH model of Nelson (1991) and the GJR-GARCH model of Glosten et al. (1993) introduce asymmetric responses to negative shocks—the leverage effect—through, respectively, a logarithmic variance recursion and an indicator-augmented quadratic term. Further extensions include power-GARCH formulations, long-memory variants such as FIGARCH, and specifications allowing for non-Gaussian innovations; a comprehensive survey is provided by Bollerslev (2010). The leverage effect is taken up explicitly in the multivariate setting below, where it forms one of the structural building blocks of the asymmetric BEKK specification.

However, from both an economic and a practical perspective, modeling conditional variances alone is often insufficient. In portfolio allocation and risk management, the full conditional covariance matrix of asset returns is a central input (Bauwens et al., 2006). In the conditional CAPM framework, the time-varying beta coefficient is directly determined by conditional covariances between asset and market returns (Bollerslev, Engle, and Wooldridge, 1988). Consequently, extending univariate GARCH models to a multivariate framework becomes indispensable when analyzing interconnected financial assets or systemic risk.

### 2.1.2 MGARCH

Extending the univariate GARCH framework to a multivariate setting is motivated by economic applications in which the joint dynamics of asset returns matter. Portfolio selection, dynamic hedging, asset pricing, and risk management require the full conditional covariance matrix of returns rather than the marginal conditional variances alone (Bauwens et al., 2006). The conditional mean is assumed to be zero throughout, so that the return vector  $\mathbf{r}_t \in \mathbb{R}^N$  coincides with the innovation vector,

$$\mathbf{r}_t = \boldsymbol{\varepsilon}_t = \boldsymbol{\Sigma}_t^{1/2} \mathbf{z}_t, \quad \mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N),$$

where  $\boldsymbol{\Sigma}_t$  denotes the conditional covariance matrix given the information set  $\Omega_{t-1}$ . The assumption of conditional Gaussianity is made for the construction of the (quasi-)likelihood and does not need to hold in finite samples; standard QMLE arguments deliver consistent estimates of the covariance parameters under mild regularity conditions even when the Gaussian assumption is violated (Bollerslev and Wooldridge, 1992).

A vector of  $N$  asset returns implies  $N(N+1)/2$  distinct elements in the conditional covariance matrix, namely  $N$  conditional variances and  $\binom{N}{2} = N(N-1)/2$  conditional covariances. Since  $\boldsymbol{\Sigma}_t$  is symmetric, only its lower triangular part needs to be parameterized. This is captured by the half-vectorization operator  $\text{vech}(\boldsymbol{\Sigma}_t) \in \mathbb{R}^{N(N+1)/2}$ , which stacks the columns of the lower-triangular part of  $\boldsymbol{\Sigma}_t$  into a single vector.

The general multivariate ARCH framework of Engle and Kroner (1995) specifies  $\text{vec}(\boldsymbol{\Sigma}_t)$  as a linear function of past outer products of innovations and lagged covariances,

$$\text{vec}(\boldsymbol{\Sigma}_t) = \mathbf{c} + \sum_{i=1}^q \mathbf{A}_i \text{vec}(\boldsymbol{\varepsilon}_{t-i} \boldsymbol{\varepsilon}_{t-i}^\top) + \sum_{j=1}^p \mathbf{G}_j \text{vec}(\boldsymbol{\Sigma}_{t-j}), \quad (2)$$

where  $\mathbf{c} \in \mathbb{R}^{N^2}$  and  $\mathbf{A}_i, \mathbf{G}_j \in \mathbb{R}^{N^2 \times N^2}$ . Although the  $\text{vec}$  representation provides analytical convenience for characterizing the general MGARCH class, it does not remove the redundancy implied by symmetry. Imposing symmetry restrictions and eliminating duplicate elements yields the  $\text{vech}$ -form used in early empirical MGARCH applications such as Bollerslev, Engle, and Wooldridge (1988). Subject to these restrictions,  $\text{vec}$  and  $\text{vech}$  can represent the same conditional second-moment dynamics;  $\text{vech}$  makes the symmetry explicit and avoids duplicate equations.

While the  $\text{vec}$  and  $\text{vech}$  representations are sufficiently flexible to nest all linear MGARCH models, they suffer from two practical limitations. First, the number of parameters grows rapidly with  $N$ , leading to a severe curse of dimensionality. Second, neither representation guarantees that the resulting  $\boldsymbol{\Sigma}_t$  is positive definite without imposing complicated nonlinear restrictions on  $\mathbf{A}_i$  and  $\mathbf{G}_j$ . Structured parameterizations have therefore been developed to address one or both of these limitations: the BEKK parameterization of Engle and Kroner (1995) ensures positive definiteness by construction through a quadratic-form representation; the constant conditional correlation (CCC) model of Bollerslev (1990) and the dynamic conditional correlation (DCC) model of Engle (2002) decompose the covariance dynamics into separate variance and correlation components, which substantially reduces the parameter count. The DCC model is widely used as a benchmark in empirical work. Detailed surveys of the MGARCH literature, including a unified taxonomy and discussion of identification and estimation, can be found in Bauwens et al. (2006) and Silvennoinen and Teräsvirta (2009).

### 2.1.3 CCC & DCC

An alternative class of MGARCH models avoids parameterizing the conditional covariance matrix directly and instead decomposes it into a conditional standard-deviation component and a conditional correlation component. The constant conditional correlation (CCC) model of Bollerslev (1990) writes

$$\boldsymbol{\Sigma}_t = \mathbf{D}_t \mathbf{R} \mathbf{D}_t, \quad \mathbf{D}_t = \text{diag}(\sigma_{1,t}, \dots, \sigma_{N,t}),$$

where each conditional standard deviation  $\sigma_{i,t}$  follows a univariate GARCH process and  $\mathbf{R}$  is a time-invariant correlation matrix. A common estimator proceeds in two steps—univariate GARCH fits per asset, followed by a sample correlation matrix of the standardized residuals—

which sidesteps the high-dimensional joint optimization of full MGARCH specifications. The empirical plausibility of constant correlations is, however, frequently rejected, particularly during periods of financial stress when cross-asset correlations tend to rise.

The dynamic conditional correlation (DCC) model of Engle (2002) relaxes this restriction by letting the correlation matrix vary over time. The decomposition becomes

$$\boldsymbol{\Sigma}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t,$$

with  $\mathbf{R}_t$  obtained from a GARCH-type recursion in the standardized residuals  $\mathbf{u}_t = \mathbf{D}_t^{-1} \boldsymbol{\varepsilon}_t$ ,

$$\mathbf{Q}_t = (1 - \alpha - \beta) \bar{\mathbf{Q}} + \alpha \mathbf{u}_{t-1} \mathbf{u}_{t-1}^\top + \beta \mathbf{Q}_{t-1}, \quad \mathbf{R}_t = \text{diag}(\mathbf{Q}_t)^{-1/2} \mathbf{Q}_t \text{diag}(\mathbf{Q}_t)^{-1/2},$$

where  $\bar{\mathbf{Q}}$  denotes the positive-definite unconditional correlation matrix of  $\mathbf{u}_t$ . With a positive-definite initialization,  $\alpha, \beta \geq 0$  preserve positive definiteness, while  $\alpha + \beta < 1$  is the usual mean-reversion restriction; formal stationarity additionally requires the regularity conditions of the model. Estimation typically proceeds in two steps as well, with univariate GARCH fits in the first stage and joint estimation of the scalar DCC parameters  $(\alpha, \beta)$  in the second (Engle, 2002).

The DCC specification is widely used in empirical finance because of its parsimonious parameterization—only two scalar parameters govern the correlation dynamics regardless of  $N$ —and its computational tractability. The model is, however, structurally restrictive: by construction, all pairwise correlations follow the same dynamic process, ruling out asset-specific persistence in correlations. The cross-products  $\mathbf{u}_{t-1} \mathbf{u}_{t-1}^\top$  allow shocks to affect conditional correlations, but the separate univariate variance recursions contain no direct cross-asset shock or variance terms. Full BEKK instead permits such variance spillovers through the off-diagonal elements of  $\mathbf{A}$  and  $\mathbf{G}$ , as introduced below. The scalar DCC model serves as the parsimonious classical benchmark.

#### 2.1.4 BEKK

**Full BEKK.** The BEKK( $p, q$ ) specification of Engle and Kroner (1995) parameterizes the conditional covariance recursion through quadratic forms of past return outer products and lagged conditional covariances. The full BEKK(1, 1) model takes the form

$$\boldsymbol{\Sigma}_t = \mathbf{C} \mathbf{C}^\top + \mathbf{A}^\top \boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}^\top \mathbf{A} + \mathbf{G}^\top \boldsymbol{\Sigma}_{t-1} \mathbf{G}, \quad (3)$$

where  $\mathbf{C} \in \mathbb{R}^{N \times N}$  is lower triangular and  $\mathbf{A}, \mathbf{G} \in \mathbb{R}^{N \times N}$  are unrestricted. The quadratic-form construction guarantees that  $\boldsymbol{\Sigma}_t$  is symmetric and positive definite for any admissible parameter values, provided  $\mathbf{C}$  has full rank. The off-diagonal elements of  $\mathbf{A}$  and  $\mathbf{G}$  govern volatility spillovers across assets. Under the convention in (3), a non-zero entry  $A_{ji}$  allows the past shock in asset  $j$  to enter the conditional variance of asset  $i$ ; analogously, column  $i$  of  $\mathbf{G}$  determines how the lagged covariance matrix contributes to that variance. Covariance stationarity holds whenever all eigenvalues of  $\mathbf{A} \otimes \mathbf{A} + \mathbf{G} \otimes \mathbf{G}$  lie inside the unit circle (Engle and Kroner, 1995).

Estimation of (3) is performed by quasi-maximum likelihood under the conditionally Gaussian density implied by  $\mathbf{r}_t \mid \Omega_{t-1} \sim \mathcal{N}(\mathbf{0}, \Sigma_t)$ . The number of free parameters in the unrestricted full BEKK(1, 1) is  $N(N+1)/2 + 2N^2$ , which still grows quadratically in the cross-sectional dimension and motivates the more parsimonious specifications discussed below.

**Asymmetric BEKK.** Empirically, negative shocks tend to have a larger impact on conditional variances than positive shocks of equal magnitude—the so-called leverage effect documented in the univariate setting by Glosten et al. (1993). The asymmetric BEKK specification of Kroner and Ng (1998) extends (3) by adding a quadratic term in the negative parts of the lagged shocks,

$$\Sigma_t = \mathbf{C}\mathbf{C}^\top + \mathbf{A}^\top \boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}^\top \mathbf{A} + \mathbf{G}^\top \Sigma_{t-1} \mathbf{G} + \mathbf{B}^\top \boldsymbol{\eta}_{t-1} \boldsymbol{\eta}_{t-1}^\top \mathbf{B}, \quad (4)$$

with  $\boldsymbol{\eta}_{t-1} = \mathbf{1}\{\boldsymbol{\varepsilon}_{t-1} < \mathbf{0}\} \odot \boldsymbol{\varepsilon}_{t-1}$  collecting the negative components of the lagged innovations, and  $\mathbf{B} \in \mathbb{R}^{N \times N}$  governing the propagation of negative shocks within and across assets. The quadratic-form structure preserves positive definiteness by construction; covariance stationarity requires an analogous eigenvalue condition that now also involves  $\mathbf{B}$ . This component-wise construction is the asymmetric building block of the Neural-BEKK and related hybrid architectures developed below. The asymmetric econometric benchmark instead uses the joint-sign formulation described in section 4.1.

**Diagonal BEKK.** The diagonal BEKK model restricts  $\mathbf{A}$ ,  $\mathbf{G}$ , and, in the asymmetric case,  $\mathbf{B}$  to be diagonal. The conditional variance of asset  $i$  then depends only on its own past squared innovation and its own past variance, and the conditional covariance between assets  $i$  and  $j$  depends only on the lagged cross-product  $\varepsilon_{i,t-1}\varepsilon_{j,t-1}$  and the lagged covariance. The number of parameters reduces from order  $N^2$  to order  $N$ , at the cost of ruling out cross-equation transmission through off-diagonal coefficient entries. Covariances can still evolve through pair-specific lagged cross-products and lagged covariances. For the symmetric diagonal BEKK with  $\mathbf{A} = \text{diag}(a_1, \dots, a_N)$  and  $\mathbf{G} = \text{diag}(g_1, \dots, g_N)$ , the general BEKK stationarity condition reduces to  $\max_{i,j} |a_i a_j + g_i g_j| < 1$ , equivalently  $\max_i (a_i^2 + g_i^2) < 1$  (Engle and Kroner, 1995).

**Scalar BEKK.** The scalar BEKK model imposes the even stronger restriction  $\mathbf{A} = a \mathbf{I}_N$ ,  $\mathbf{G} = g \mathbf{I}_N$  (and  $\mathbf{B} = b \mathbf{I}_N$  in the asymmetric case) for scalar parameters  $a, g, b \in \mathbb{R}$ . This specification is highly parsimonious but rules out any asset-specific or cross-asset heterogeneity in the volatility dynamics. For the symmetric scalar BEKK, covariance stationarity requires  $a^2 + g^2 < 1$ , which is the scalar specialization of the BEKK eigenvalue condition (Engle and Kroner, 1995).

The (a)symmetric BEKK(1, 1) likelihood is maximized using the BEKKs R package of Fülle et al. (2024), which provides an efficient implementation of the quasi-maximum likelihood estimator with analytical derivatives and eigenvalue-based stationarity checks.

### 2.1.5 Artificial Neural Networks

In the volatility forecasting literature, two classes of neural network architectures have been used most prominently: feed-forward multilayer perceptrons (MLPs) for non-sequential mappings, and recurrent neural networks (RNNs) for modelling temporal dependence in financial time series. This thesis focuses on recurrent architectures—specifically the long short-term memory (LSTM) cell of Hochreiter and Schmidhuber (1997) and the gated recurrent unit (GRU) of Cho et al. (2014)—because they are designed to capture long-range temporal dependencies of the kind that characterize conditional second-moment dynamics.

**Multilayer Perceptron.** A multilayer perceptron defines a deterministic mapping  $\mathbf{y} = f_\theta(\mathbf{x})$  from an input vector  $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$  to an output  $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$  through a sequence of affine transformations interleaved with elementwise nonlinearities,

$$\mathbf{h}^{(\ell)} = \phi^{(\ell)}\left(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\right), \quad \ell = 1, \dots, L,$$

with  $\mathbf{h}^{(0)} = \mathbf{x}$  and  $\mathbf{y} = \mathbf{h}^{(L)}$ . Common activation functions are the rectified linear unit  $\text{ReLU}(x) = \max(0, x)$  and the hyperbolic tangent  $\tanh(x)$ . They correspond to the PyTorch modules `torch.nn.ReLU` and `torch.nn.Tanh` used in the implementation. These elementwise transformations introduce the nonlinearity that distinguishes the MLP from a linear model. In the models evaluated here, feed-forward output heads map recurrent hidden states to parameter or covariance outputs; their exact depth and activations vary by architecture and are specified in section 4.1.

**Recurrent Neural Network.** Recurrent neural networks process sequential inputs  $\mathbf{x}_{1:T} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$  by maintaining a hidden state  $\mathbf{h}_t \in \mathbb{R}^h$  that summarizes the relevant history up to time  $t$ . A vanilla RNN applies the same state-transition map at every time step,

$$\mathbf{h}_t = \phi(\mathbf{W}_x \mathbf{x}_t + \mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{b}),$$

where  $\mathbf{W}_x$ ,  $\mathbf{W}_h$ , and  $\mathbf{b}$  are shared over all  $t$ . This temporal weight sharing allows the same transition rule to process sequences of arbitrary length. While this construction is in principle universal in the sense that it can approximate arbitrary sequence-to-sequence mappings, the vanilla RNN suffers from well-documented optimization problems: gradients backpropagated through long sequences tend to either vanish or explode, which severely limits the network’s ability to learn long-range temporal dependencies in practice (Bengio et al., 1994).

**Long Short-Term Memory.** The long short-term memory (LSTM) architecture introduced by Hochreiter and Schmidhuber (1997) addresses the vanishing gradient problem through an additive cell state  $\mathbf{c}_t$ . The modern three-gate formulation used here additionally incorporates the forget gate of Gers et al. (2000). The forget gate  $\mathbf{f}_t$  determines which parts of the previous cell state are retained, the input gate  $\mathbf{i}_t$  controls how much of the candidate update enters the cell state, and the output gate  $\mathbf{o}_t$  controls which parts of the cell state become visible in the

hidden state. Formally,

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f), \quad (5)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i), \quad (6)$$

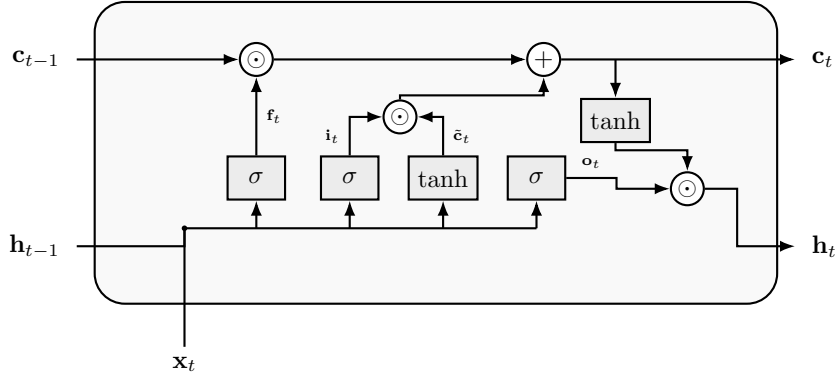
$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o), \quad (7)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c), \quad (8)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (9)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \quad (10)$$

where  $\sigma(\cdot)$  denotes the elementwise logistic sigmoid and  $\odot$  the Hadamard product. All gate and candidate-state weights and biases are shared across time steps; only the states  $\mathbf{c}_t$  and  $\mathbf{h}_t$  evolve with  $t$ . The additive structure of the cell state recursion  $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \dots$  preserves gradients across long temporal distances when the forget gate is close to one, which is the central mechanism that distinguishes LSTMs from vanilla RNNs and explains their success on long sequences. The LSTM serves as the building block for the kernel-replacement architectures of section 2.2.1 below; the reinterpretation of the output gate  $\mathbf{o}_t$  as a BEKK kernel is discussed there in detail.



**Figure 1:** Architecture of the long short-term memory (LSTM) cell.

**Gated Recurrent Unit.** The gated recurrent unit (GRU) of Cho et al. (2014) simplifies the LSTM architecture by merging the forget and input gates into a single update gate  $\mathbf{g}_t^{(z)}$  and by eliminating the separate cell state, retaining only the hidden state  $\mathbf{h}_t$ . The resulting recursion uses two gates,

$$\mathbf{g}_t^{(z)} = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z), \quad (11)$$

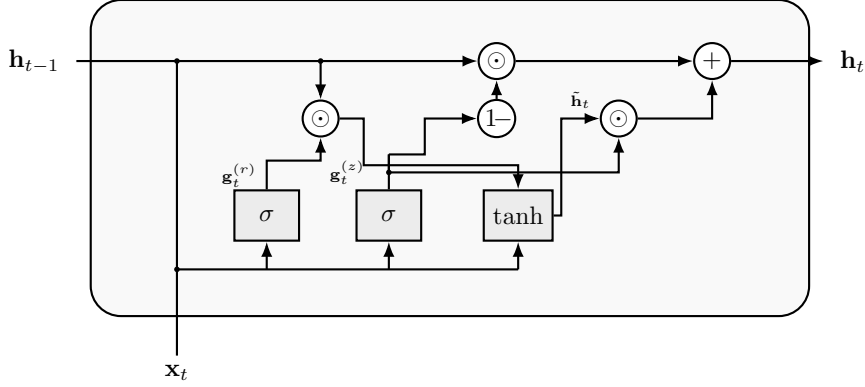
$$\mathbf{g}_t^{(r)} = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r), \quad (12)$$

together with a candidate hidden state and the actual update,

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h (\mathbf{g}_t^{(r)} \odot \mathbf{h}_{t-1}) + \mathbf{b}_h), \quad (13)$$

$$\mathbf{h}_t = \mathbf{g}_t^{(z)} \odot \mathbf{h}_{t-1} + (\mathbf{1} - \mathbf{g}_t^{(z)}) \odot \tilde{\mathbf{h}}_t. \quad (14)$$

The update gate  $\mathbf{g}_t^{(z)}$  interpolates between the previous hidden state and the candidate, providing the same additive gradient pathway that allows LSTMs to learn long-range dependencies. Chung et al. (2014) provide an empirical comparison of LSTM and GRU architectures across sequence modelling tasks and report competitive predictive performance. Because the GRU also uses fewer gate parameters than the LSTM, it is particularly relevant for financial return data, where effective sample sizes are limited. The GRU is therefore included alongside the LSTM in the evaluation.



**Figure 2:** Architecture of the gated recurrent unit (GRU) cell. The update gate  $\mathbf{g}_t^{(z)}$  interpolates between  $\mathbf{h}_{t-1}$  and  $\tilde{\mathbf{h}}_t$ , while  $\mathbf{g}_t^{(r)}$  controls candidate formation.

**From hidden states to covariance forecasts.** A central question for the application of recurrent networks to multivariate volatility forecasting is how the hidden state  $\mathbf{h}_T \in \mathbb{R}^h$  at the forecast origin is mapped to a valid conditional covariance matrix  $\Sigma_{T+1} \in \mathbb{R}^{N \times N}$ . Two standard parameterizations achieve positive definiteness by construction.

The first approach is direct Cholesky parameterization. A linear output layer maps the hidden state to a vector of length  $N(N+1)/2$ ,

$$\mathbf{v}_T = \mathbf{W}_{\text{out}} \mathbf{h}_T + \mathbf{b}_{\text{out}} \in \mathbb{R}^{N(N+1)/2},$$

which is reshaped into a lower-triangular Cholesky factor  $\mathbf{L}_T \in \mathbb{R}^{N \times N}$  via  $\mathbf{L}_T = \text{unvech}_{\text{lt}}(\mathbf{v}_T)$ . Here  $\text{unvech}_{\text{lt}}$  fills the lower-triangular entries column by column and sets the upper triangle to zero; unlike the usual inverse half-vectorization of a symmetric matrix, it does not mirror the off-diagonal entries. A softplus transformation is applied to the diagonal elements to enforce strict positivity. The conditional covariance matrix is then obtained as

$$\Sigma_{T+1} = \mathbf{L}_T \mathbf{L}_T^\top,$$

which is symmetric and positive definite by construction for any  $\mathbf{v}_T$ . In contrast to the classical Cholesky decomposition of a given matrix, the network here *directly parameterizes* the Cholesky factor, which is differentiable with respect to the network parameters and therefore amenable to gradient-based training. This parameterization is the standard choice for the data-driven LSTM and GRU benchmarks considered in the empirical section.

Two multivariate precedents in the literature adopt this direct, decomposition-based philosophy and are therefore the closest relatives of the neural benchmarks used here. Bucci (2020) lets a neural network forecast the Cholesky elements of the realized covariance matrix,  $\mathbf{RC}_t = \mathbf{C}_t \mathbf{C}_t^\top$ , and trains the network by mean squared error between predicted and observed Cholesky-factor vectors; matrix distances are used for forecast evaluation. This is the direct analogue of the plain-Cholesky parameterization above. W. K. Liu et al. (2022) replace the plain Cholesky factor by the *modified* Cholesky decomposition  $\mathbf{T}_t \boldsymbol{\Sigma}_t \mathbf{T}_t^\top = \mathbf{D}_t$ , in which the unit-lower-triangular matrix  $\mathbf{T}_t$  collects generalized autoregressive coefficients  $\phi_{t,j,i}$  and the diagonal matrix  $\mathbf{D}_t$  collects innovation variances  $\sigma_{t,j}^2$ , whose dynamics are modeled by an LSTM. The modified decomposition trades the lack of interpretability of the raw Cholesky factor for an explicit regression reading of  $\phi_{t,j,i}$ , at the cost of an ordering dependence among the assets. For the taxonomy used here, the model of W. K. Liu et al. (2022) is closest to the direct neural covariance benchmarks because the LSTM outputs the parameters of the modified Cholesky decomposition. This differs from the hybrid architectures of section 2.2, which retain an explicit BEKK recursion and embed neural components within or around that recursion.

The second approach is appropriate when only conditional variances are of interest. Restricting the output to  $\boldsymbol{\Sigma}_{T+1} = \text{diag}(\sigma_{1,T+1}^2, \dots, \sigma_{N,T+1}^2)$ , the linear output layer produces an  $N$ -dimensional vector that is mapped to strictly positive variances via a softplus transformation. No Cholesky decomposition is required in this case, but cross-asset covariances are excluded by construction. This option is mentioned for completeness; all models evaluated here produce full conditional covariance matrices.

The kernel-replacement and Neural-BEKK architectures introduced in section 2.2 below replace one or both of these output mappings with an econometrically structured layer, while retaining the recurrent backbone introduced in this section.

## 2.2 Hybrid Volatility Models

Two hybrid designs are considered. The kernel-replacement models retain time-invariant BEKK matrices  $\mathbf{C}$ ,  $\mathbf{A}$ ,  $\mathbf{G}$ , and, where applicable,  $\mathbf{B}$ , while a recurrent state modulates the resulting covariance kernel. The Neural-BEKK models instead let a recurrent controller produce time-varying matrices  $\mathbf{C}_t$ ,  $\mathbf{A}_t$ ,  $\mathbf{G}_t$ , and  $\mathbf{B}_t$ . Both designs differ from the direct Cholesky benchmarks of the previous section, which map a hidden state directly to a covariance matrix without a BEKK recursion.

### 2.2.1 MGARCH Kernel Function

A first strand of hybrid neural-MGARCH models replaces an internal component of a recurrent cell with an econometrically motivated volatility recursion, while leaving the surrounding controller architecture intact. A recent contribution in this direction is Zhao et al. (2024), who reinterpret the LSTM output gate  $o_t$  of a univariate volatility model as a GARCH(1,1) kernel,

$$o_t := \mathcal{K}_{\text{GARCH}}(\varepsilon_{t-1}, \sigma_{t-1}^2; \theta) = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

and let the LSTM cell state act as a controller that modulates the kernel output. The forget and input gates as well as the cell state recursion remain those of a standard LSTM, so that the model retains the temporal expressiveness of recurrent networks while imposing the parsimonious GARCH structure on the volatility output itself.

**Related univariate developments.** Beyond this output-gate reinterpretation, two univariate refinements inform the design choices made below. Wei et al. (2025) embed the GARCH(1,1) recursion into the multiplicative gating structure of both LSTM and GRU cells and find that the GARCH-GRU variant attains an accuracy-efficiency trade-off at least competitive with, and on their data superior to, its LSTM counterpart. Since the relative ranking of the two gating architectures is therefore not settled a priori, both are retained as controllers rather than committing to the LSTM by default. Buczyński and Chlebus (2024) combine recurrent networks with GARCH-inspired conditional-variance dynamics and estimate the resulting models under Gaussian, Student- $t$ , and skewed Student- $t$  likelihoods. Their out-of-sample VaR evaluation shows that neural-GARCH hybrids can be assessed in a tail-risk setting, although VaR-sensitive losses are evaluation criteria rather than training objectives in their study.

**Multivariate extension.** While these contributions remain univariate, the kernel-replacement principle extends naturally to the multivariate setting: the univariate kernel function can be replaced by a multivariate kernel function  $\mathcal{K}_{\text{MGARCH}}$ , e.g., a BEKK kernel, while retaining the recurrent controller. In its symmetric form, this kernel reads

$$\mathcal{K}_{\text{BEKK}}(\boldsymbol{\varepsilon}_{t-1}, \boldsymbol{\Sigma}_{t-1}) = \mathbf{C}\mathbf{C}^\top + \mathbf{A}^\top \boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}^\top \mathbf{A} + \mathbf{G}^\top \boldsymbol{\Sigma}_{t-1} \mathbf{G}. \quad (15)$$

A central design choice in this construction concerns the input set of the recurrent controller. The minimal specification follows the econometric kernel exactly and lets the gates depend only on past shocks  $\boldsymbol{\varepsilon}_{t-1}$  and the lagged conditional covariance matrix  $\boldsymbol{\Sigma}_{t-1}$ . More generally, however, the controller can be defined on an extended information set  $\tilde{\Omega}_{t-1} \supseteq \Omega_{t-1}^{\text{portfolio}}$  that includes any predetermined variable available at time  $t-1$ , such as cross-product features  $\text{vech}(\boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}^\top)$  that summarize realized covariation, broader market or factor returns, macroeconomic indicators, or sentiment proxies. This flexibility is one of the conceptual advantages of hybrid neural-MGARCH architectures over classical BEKK models, whose closed form restricts the conditioning set of the portfolio covariance dynamics to the portfolio’s own return history. This flexibility is not exercised in the empirical design, which restricts every model to portfolio-return features without exogenous inputs. Throughout this work, the controller input at time  $t-1$  is denoted generically by  $\mathbf{x}_{t-1}$ , and the BEKK kernel itself continues to depend exclusively on  $\boldsymbol{\varepsilon}_{t-1}$  and  $\boldsymbol{\Sigma}_{t-1}$ , so that the structural interpretation of the econometric component is preserved.

Because the conditional covariance matrix is symmetric, the natural covariance feature for the

gates is its half-vectorization. The gates and cell state are then defined as

$$\mathbf{f}_t = \sigma_g(\mathbf{W}_f \mathbf{x}_{t-1} + \mathbf{U}_f \text{vech}(\boldsymbol{\Sigma}_{t-1}) + \mathbf{b}_f), \quad (16)$$

$$\mathbf{i}_t = \sigma_g(\mathbf{W}_i \mathbf{x}_{t-1} + \mathbf{U}_i \text{vech}(\boldsymbol{\Sigma}_{t-1}) + \mathbf{b}_i), \quad (17)$$

$$\tilde{\mathbf{c}}_t = \sigma_c(\mathbf{W}_c \mathbf{x}_{t-1} + \mathbf{U}_c \text{vech}(\boldsymbol{\Sigma}_{t-1}) + \mathbf{b}_c), \quad (18)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (19)$$

with  $\mathbf{x}_{t-1} \in \mathbb{R}^p$  as defined above. The cell state  $\mathbf{c}_t$  subsequently enters the modulation function  $\text{Mod}$  that maps the BEKK kernel into the final conditional covariance prediction,

$$\mathbf{K}_t = \mathcal{K}_{\text{BEKK}}(\boldsymbol{\varepsilon}_{t-1}, \boldsymbol{\Sigma}_{t-1}), \quad \boldsymbol{\Sigma}_t = \text{Mod}(\mathbf{K}_t, \mathbf{c}_t).$$

This construction inherits the structural advantages of the BEKK specification—positive definiteness by construction and an explicit econometric interpretation of  $\mathbf{C}$ ,  $\mathbf{A}$ , and  $\mathbf{G}$ —while delegating the temporal modulation of the covariance level to a recurrent controller. A GRU variant uses the same inputs  $\mathbf{x}_{t-1}$  and  $\text{vech}(\boldsymbol{\Sigma}_{t-1})$  and the same covariance modulation head, but replaces the LSTM state equations above with the GRU update and reset gates of figure 2. The LSTM notation below is therefore a compact presentation of the shared kernel-modulation design, not a restriction to one controller type.

**Related multivariate hybrid.** H. Wang et al. (2025) provide a related but structurally distinct multivariate hybrid. Their model augments a symmetric scalar-BEKK recursion with an additive positive-semidefinite term  $\mathbf{C}_t \mathbf{C}_t^\top$  generated by an LSTM,

$$\boldsymbol{\Sigma}_t = \mathbf{C} \mathbf{C}^\top + \mathbf{C}_t \mathbf{C}_t^\top + a \boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}^\top + b \boldsymbol{\Sigma}_{t-1}.$$

They report gains over classical multivariate GARCH benchmarks in density forecasting and global minimum-variance portfolio allocation. Unlike the construction developed here, however, their recurrent term is added to the BEKK recursion rather than used to modulate its output. The asymmetric kernels and scalar, vector, and convex-mixture modulation schemes below are therefore distinct extensions rather than generalizations of a scalar modulation proposed by Wang et al.

**Asymmetric BEKK kernel.** To address the first limitation, the symmetric kernel of (15) is augmented, following the asymmetric BEKK specification of Kroner and Ng (1998), with a leverage component,

$$\mathcal{K}_{\text{asBEKK}}(\boldsymbol{\varepsilon}_{t-1}, \boldsymbol{\Sigma}_{t-1}) = \mathbf{C} \mathbf{C}^\top + \mathbf{A}^\top \boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}^\top \mathbf{A} + \mathbf{G}^\top \boldsymbol{\Sigma}_{t-1} \mathbf{G} + \mathbf{B}^\top \boldsymbol{\eta}_{t-1} \boldsymbol{\eta}_{t-1}^\top \mathbf{B}, \quad (20)$$

where  $\boldsymbol{\eta}_{t-1} = \mathbf{1}\{\boldsymbol{\varepsilon}_{t-1} < \mathbf{0}\} \odot \boldsymbol{\varepsilon}_{t-1}$  collects the negative components of the lagged shocks; componentwise,  $\eta_{i,t-1} = \varepsilon_{i,t-1}$  if  $\varepsilon_{i,t-1} < 0$  and zero otherwise. The matrix  $\mathbf{B} \in \mathbb{R}^{d \times d}$  governs how negative shocks propagate into the conditional covariance matrix, both within and across assets. The recursion remains positive definite by construction.

**Modulation schemes.** To provide different levels of recurrent adaptation, the mapping  $\text{Mod}$  from the kernel output and the cell state to the final conditional covariance matrix ranges from scalar rescaling through asset-specific diagonal rescaling to a convex combination of the BEKK kernel and a Cholesky-parameterized neural covariance proposal. The exact modulation heads are specified in section 4.1; the shared mini-batch training step is summarized in the algorithm box below.

---

**Algorithm 1** BEKK-LSTM Mini-Batch Training Step

---

One AdamW update for a kernel-replacement model with an asymmetric BEKK recursion and LSTM-based modulation.

---

**Require:** input sequences  $\{\mathbf{X}_{0:T-1}^{(b)}\}_{b=1}^{B_{\text{mb}}}$ , one-step-ahead target returns  $\{\mathbf{r}_T^{(b)}\}_{b=1}^{B_{\text{mb}}}$ , initial covariances  $\{\mathbf{\Sigma}_0^{(b)}\}_{b=1}^{B_{\text{mb}}}$ , parameters  $\boldsymbol{\theta} = (\boldsymbol{\theta}_B, \boldsymbol{\theta}_{\text{LSTM}}, \boldsymbol{\theta}_{\text{mod}})$ , and AdamW optimizer state.

- 1: Reset gradients and set the mini-batch objective  $\mathcal{J} \leftarrow 0$ .
  - 2: **for** each sequence  $b = 1, \dots, B_{\text{mb}}$  **do**
  - 3:   Initialize  $\mathbf{c}_0^{(b)} \leftarrow \mathbf{0}$  and  $\mathbf{\Sigma}_0^{(b)}$ .
  - 4:   **for**  $t = 1, \dots, T$  **do**
  - 5:     Set  $\mathbf{x}_{t-1}^{(b)} \leftarrow \mathbf{X}_{t-1}^{(b)}$ , and extract  $\boldsymbol{\varepsilon}_{t-1}^{(b)}$ .
  - 6:     Compute  $\tilde{\mathbf{s}}_{t-1}^{(b)} \leftarrow \text{LN}(\text{vech}(\mathbf{\Sigma}_{t-1}^{(b)}))$ .
  - 7:     Update the LSTM state  $\mathbf{c}_t^{(b)} \leftarrow \text{LSTMCell}_{\boldsymbol{\theta}_{\text{LSTM}}}(\mathbf{x}_{t-1}^{(b)}, \tilde{\mathbf{s}}_{t-1}^{(b)}, \mathbf{c}_{t-1}^{(b)})$ .
  - 8:     Compute the BEKK kernel  $\mathbf{K}_t^{(b)} \leftarrow \mathcal{K}_{\text{asBEKK}}(\boldsymbol{\varepsilon}_{t-1}^{(b)}, \mathbf{\Sigma}_{t-1}^{(b)})$ .
  - 9:     Map kernel and state to covariance  $\mathbf{\Sigma}_t^{(b)} \leftarrow \text{Mod}(\mathbf{K}_t^{(b)}, \mathbf{c}_t^{(b)}; \boldsymbol{\theta}_{\text{mod}})$ .
  - 10:    Symmetrize and stabilize  $\mathbf{\Sigma}_t^{(b)} \leftarrow \frac{1}{2}(\mathbf{\Sigma}_t^{(b)} + (\mathbf{\Sigma}_t^{(b)})^\top) + \eta \mathbf{I}$ .
  - 11:   **end for**
  - 12:   Accumulate  $\mathcal{J} \leftarrow \mathcal{J} + B_{\text{mb}}^{-1} \ell(\mathbf{r}_T^{(b)}, \mathbf{\Sigma}_T^{(b)})$ .
  - 13: **end for**
  - 14: Compute  $\nabla_{\boldsymbol{\theta}} \mathcal{J}$  by backpropagation through time and clip gradients if required.
  - 15: Update  $\boldsymbol{\theta} \leftarrow \text{AdamWStep}(\boldsymbol{\theta}, \nabla_{\boldsymbol{\theta}} \mathcal{J})$ .
  - 16: **return** updated parameters and mini-batch objective  $\mathcal{J}$ .
- 

## 2.2.2 Neural-BEKK

A second strand of hybrid neural-MGARCH models retains the standard MGARCH recursion in full and lets a neural network generate its parameter matrices as time-varying functions of past information. This contrasts with the kernel-replacement architectures of the previous section, in which the controller modulates the output of an econometric kernel: in a Neural-BEKK model, the econometric recursion itself is preserved, while its parameters become dynamic. The resulting specification can be viewed as a nonlinear, data-driven extension of classical MGARCH models, with positive definiteness of the conditional covariance matrix guaranteed by the BEKK parameterization of Engle and Kroner (1995).

Formally, a recurrent neural network maps past information to a time-varying BEKK parameter vector

$$\boldsymbol{\gamma}_t = (\text{vech}(\mathbf{C}_t)^\top, \text{vec}(\mathbf{A}_t)^\top, \text{vec}(\mathbf{G}_t)^\top, \text{vec}(\mathbf{B}_t)^\top)^\top, \quad (21)$$

where  $\text{vec}(\cdot)$  stacks matrix columns and  $\text{vech}(\mathbf{C}_t)$  collects the lower-triangular elements of  $\mathbf{C}_t$ .

The parameter vector is reshaped into the BEKK matrices  $\mathbf{C}_t$ ,  $\mathbf{A}_t$ ,  $\mathbf{B}_t$ , and  $\mathbf{G}_t$ , which enter a time-varying-coefficient extension of the asymmetric BEKK recursion of Kroner and Ng (1998),

$$\boldsymbol{\Sigma}_t = \mathbf{C}_t \mathbf{C}_t^\top + \mathbf{A}_t^\top \boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}^\top \mathbf{A}_t + \mathbf{G}_t^\top \boldsymbol{\Sigma}_{t-1} \mathbf{G}_t + \mathbf{B}_t^\top \boldsymbol{\eta}_{t-1} \boldsymbol{\eta}_{t-1}^\top \mathbf{B}_t, \quad (22)$$

with  $\boldsymbol{\eta}_{t-1} = \mathbf{1}\{\boldsymbol{\varepsilon}_{t-1} < \mathbf{0}\} \odot \boldsymbol{\varepsilon}_{t-1}$  collecting the negative components of the lagged shocks, in line with the multivariate GJR-style leverage formulation of Kroner and Ng (1998) and Glosten et al. (1993).

As in the kernel-replacement architectures, the controller that produces  $\boldsymbol{\gamma}_t$  may use the extended information set described in section 2.2.1. The BEKK recursion in (22) itself, however, continues to be evaluated on  $\boldsymbol{\varepsilon}_{t-1}$ ,  $\boldsymbol{\eta}_{t-1}$ , and  $\boldsymbol{\Sigma}_{t-1}$ . The quadratic-form structure therefore preserves positive definiteness. The classical covariance-stationarity condition does not carry over automatically to time-varying matrices and must be enforced separately, as discussed below.

**Existing literature.** Time-varying BEKK coefficients predate neural implementations. Wu et al. (2013) propose a Bayesian multivariate dynamic covariance model in which BEKK parameter matrices evolve through stochastic state transitions and are inferred with a regularized auxiliary particle filter. The BEKK quadratic form preserves positive definiteness, but the parameter dynamics and inference mechanism are Bayesian rather than neural.

A targeted literature search conducted through June 2026 identified the Neural GARCH framework of Yin and Barucca (2022) as the closest neural precedent. They consider a univariate GARCH(1,1) specification and a *diagonal* BEKK(1,1) specification in the multivariate case. Their time-varying parameters are treated as latent random variables and inferred via stochastic gradient variational Bayes, which delivers a probabilistic Neural-BEKK model. Two structural restrictions characterize their multivariate specification: the dynamic shock and persistence matrices are diagonal, while the intercept factor remains triangular, eliminating cross-equation transmission through off-diagonal dynamic coefficients; and the recursion is symmetric, ruling out explicit leverage effects.

**Contribution of this thesis.** Building on this literature, this thesis proposes a time-varying asymmetric Neural-BEKK specification that (i) admits both the *full*, non-diagonal BEKK parameterization of Engle and Kroner (1995)—allowing for symmetric and asymmetric volatility spillovers across assets in both conditional variances and covariances—and a parsimonious diagonal restriction thereof; (ii) incorporates the asymmetric leverage term of Kroner and Ng (1998) through the time-varying matrix  $\mathbf{B}_t$  in (22); and (iii) adopts a deterministic estimation framework in which the parameter vector  $\boldsymbol{\gamma}_t$  is produced as a deterministic function of past information by a recurrent neural network, with all parameters estimated jointly by quasi-maximum likelihood. To the author’s knowledge, this is the first hybrid neural–MGARCH model that combines a (fully) asymmetric BEKK structure with time-varying parameter matrices generated by a neural controller. Empirically (section 5), the diagonal restriction emerges as the preferred specification: the additional off-diagonal flexibility of the full parameterization does not improve—and

marginally degrades—out-of-sample forecast accuracy across both the joint VaR–ES (FZ0) loss and the QLIKE portfolio-variance loss, consistent with the econometric structure acting as a regularizer for the neural controller.

**Deterministic Neural-BEKK.** In the deterministic formulation adopted here, the neural network maps past information to time-varying BEKK parameters,

$$\boldsymbol{\gamma}_t = f_\theta(\Omega_{t-1}),$$

which are inserted into the recursion in (22). Estimation is carried out by quasi-maximum likelihood, where the gradient of the Gaussian or Student- $t$  log-likelihood is propagated jointly through the BEKK recursion and the recurrent network using automatic differentiation.

**Probabilistic Neural-BEKK.** Alternatively, a probabilistic formulation treats the time-varying parameters as latent random variables,

$$\boldsymbol{\gamma}_t \sim q_\phi(\boldsymbol{\gamma}_t \mid \Omega_{t-1}),$$

so that parameter uncertainty is captured by a variational distribution and inference proceeds by maximizing an evidence lower bound (ELBO). This approach is not pursued here; its estimation principle is summarized in section 2.3.1.

## 2.3 Parameter Estimation

The estimation of volatility models requires specifying both a statistical objective function and a numerical procedure to solve the resulting optimization problem. While econometric and neural approaches differ in their structural assumptions, they can be unified within this perspective: model parameters are obtained by optimizing an objective function that reflects the conditional distribution of returns.

The following discussion distinguishes estimation principles, model-specific procedures, and numerical optimization methods.

### 2.3.1 Estimation Principles

**Likelihood-Based Estimation** In classical econometric volatility modeling, parameter estimation is typically based on the principle of maximum likelihood. Let  $\mathbf{r}_t$  denote a vector of asset returns with conditional covariance matrix  $\boldsymbol{\Sigma}_t$ . Under the assumption of conditionally Gaussian innovations,

$$\mathbf{r}_t \mid \Omega_{t-1} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t),$$

the log-likelihood function is given by

$$\mathcal{L}(\theta) = -\frac{1}{2} \sum_{t=1}^T \left( \log |\boldsymbol{\Sigma}_t(\theta)| + \mathbf{r}_t^\top \boldsymbol{\Sigma}_t(\theta)^{-1} \mathbf{r}_t \right),$$

where  $\theta$  collects all model parameters.

In practice, financial return series often exhibit heavy tails and excess kurtosis, as documented for the series in this study in table 1, violating the Gaussian assumption. Nevertheless, maximizing the Gaussian likelihood yields consistent parameter estimates under fairly general conditions, a procedure commonly referred to as quasi maximum likelihood (QML) estimation. This robustness property explains the widespread use of QML in the GARCH and MGARCH literature.

**Variational Inference** In probabilistic neural-MGARCH models, the time-varying parameter vectors  $\gamma_t$  are treated as latent random variables rather than deterministic functions of past information. As a result, parameter estimation no longer reduces to maximizing a likelihood function, but instead requires inference over the latent parameter process.

Let  $\gamma_{1:T}$  denote the sequence of latent parameter vectors and  $\mathbf{r}_{1:T}$  the observed return series. The probabilistic model defines a joint distribution

$$p_\theta(\mathbf{r}_{1:T}, \gamma_{1:T}) = \prod_{t=1}^T p_\theta(\mathbf{r}_t \mid \gamma_t) p_\theta(\gamma_t \mid \Omega_{t-1}),$$

where  $p_\theta(\mathbf{r}_t \mid \gamma_t)$  corresponds to the Gaussian observation density implied by the BEKK recursion.

Inference requires evaluating the posterior distribution

$$p_\theta(\gamma_{1:T} \mid \mathbf{r}_{1:T}),$$

which is generally intractable due to the nonlinear and recursive dependence of  $\Sigma_t$  on  $\gamma_t$ .

Variational inference addresses this problem by introducing an approximate posterior distribution

$$q_\phi(\gamma_{1:T} \mid \mathbf{r}_{1:T}),$$

parameterized by a neural network. This object is a smoothing posterior for offline latent-state inference, not a one-step-ahead forecasting rule; an online forecasting variant would require a filtered approximation conditional on  $\Omega_{t-1}$ . The model parameters  $\theta$  and variational parameters  $\phi$  are jointly estimated by maximizing the evidence lower bound (ELBO),

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi} \left[ \sum_{t=1}^T \log p_\theta(\mathbf{r}_t \mid \gamma_t) \right] - \text{KL}(q_\phi(\gamma_{1:T}) \parallel p_\theta(\gamma_{1:T})).$$

The first term corresponds to the expected log-likelihood under the approximate posterior, while the second term regularizes the solution by penalizing deviations from the prior distribution over the latent parameters. The deterministic formulation can be interpreted as a limiting case in which the posterior collapses to a point mass, reducing the ELBO to the standard likelihood function.

### 2.3.2 Model-Specific Estimation

**Estimation of BEKK Models** The estimation of BEKK models is typically performed via quasi maximum likelihood using gradient-based optimization methods. The nonlinear and recursive structure of the covariance equation leads to a high-dimensional and potentially complex optimization problem, particularly as the number of assets increases.

A key advantage of the BEKK parametrization is that it guarantees positive definiteness of the conditional covariance matrix by construction. However, this structural property comes at the cost of increased computational complexity. Efficient estimation therefore relies on the availability of analytical gradients, appropriate initialization strategies, and parameter restrictions to ensure numerical stability and convergence.

**Training of Neural Volatility Models** Neural network-based volatility models are estimated using gradient-based learning algorithms. Rather than imposing a fixed parametric structure, neural networks learn flexible nonlinear mappings from past information to model outputs.

Depending on the specification, neural models can be trained using distribution-free loss functions, such as mean squared error on volatility proxies, or likelihood-based objectives derived from distributional assumptions on returns. In the latter case, neural networks can be interpreted as directly parameterizing the conditional covariance matrix within a probabilistic framework.

Parameter optimization is typically carried out using stochastic gradient descent (SGD) or an adaptive variant, combined with backpropagation through time for recurrent architectures. The empirical models in this thesis use AdamW, defined below. While neural optimization offers substantial flexibility, the direct neural benchmarks lack the structural constraints and interpretability of econometric models.

**Estimation of Hybrid Neural-MGARCH Models** In the deterministic hybrid models used here, estimation minimizes the negative log-likelihood by propagating gradients through both the covariance recursion and the neural network. This is computationally more demanding than estimating the classical MGARCH benchmarks. The probabilistic alternative and its variational objective have already been described in section 2.3.1.

### 2.3.3 Numerical Optimization

The optimization problems arising in volatility modeling are typically nonlinear and do not admit closed-form solutions. Consequently, parameter estimation relies on numerical optimization techniques.

**BHHH algorithm** In econometric models, algorithms such as the BHHH algorithm or quasi-Newton methods (e.g., BFGS) are commonly used. The BHHH algorithm approximates the Hessian using the outer product of gradients, while quasi-Newton methods iteratively construct Hessian approximations based on gradient information.

**Gradient-Based Learning in Neural Networks** In neural network-based models, parameter estimation is performed using gradient-based learning algorithms that operate on a computational graph. Given a loss function  $\mathcal{L}(\theta)$ , gradients with respect to all model parameters are computed using automatic differentiation, which systematically applies the chain rule to propagate derivatives through the network.

This approach enables efficient optimization even in highly complex models, such as hybrid neural-MGARCH architectures, where gradients must be propagated through both the neural network and the recursive covariance structure.

**Training Procedure** Neural networks are trained iteratively using a procedure commonly referred to as the training loop. At each iteration, the following steps are performed:

1. *Forward pass*: The model processes a batch of input data and produces predictions, e.g., the conditional covariance matrix  $\Sigma_t$ .
2. *Loss computation*: The discrepancy between predictions and observed data is quantified using a loss function, such as the negative log-likelihood.
3. *Backward pass*: Gradients of the loss with respect to the model parameters are computed via automatic differentiation.
4. *Parameter update*: The model parameters are updated using an optimization algorithm.

This procedure is repeated over multiple passes through the dataset, referred to as epochs, until convergence is achieved.

**Mini-Batch Optimization with AdamW** For update  $k$ , let  $\mathcal{B}_k$  denote a mini-batch of training sequences and let

$$\mathbf{g}_k = \frac{1}{|\mathcal{B}_k|} \sum_{s \in \mathcal{B}_k} \nabla_{\theta} \ell_s(\theta_k)$$

be the average gradient of the conditional negative log-likelihood. Adam forms exponentially weighted first- and second-moment estimates of these stochastic gradients (Kingma and Ba, 2015),

$$\mathbf{m}_k = \beta_1 \mathbf{m}_{k-1} + (1 - \beta_1) \mathbf{g}_k, \quad \mathbf{v}_k = \beta_2 \mathbf{v}_{k-1} + (1 - \beta_2) \mathbf{g}_k^{\odot 2},$$

and uses their bias-corrected versions  $\hat{\mathbf{m}}_k$  and  $\hat{\mathbf{v}}_k$ . AdamW applies weight decay separately from the adaptive gradient step (Loshchilov and Hutter, 2019),

$$\theta_{k+1} = (1 - \eta_k \lambda) \theta_k - \eta_k \frac{\hat{\mathbf{m}}_k}{\sqrt{\hat{\mathbf{v}}_k} + \epsilon},$$

where  $\eta_k$  is the learning rate and  $\lambda$  the decoupled weight-decay coefficient. This is the optimizer used for the neural and hybrid models.

Mini-batching partitions only the training sequences used for gradient updates; temporal order is retained within every sequence. After an epoch, the validation block is evaluated without

parameter updates and is used for early stopping and hyperparameter selection. The test block is not consulted until the final out-of-sample evaluation. The concrete chronological 70%/15%/15% split and the distinct treatment of the econometric benchmarks are specified in section 4.2.

**Learning Rate and Scheduling** The learning rate  $\eta_k$  in the AdamW update determines the step size. Values that are too large can destabilize the recursive covariance path, whereas very small values slow optimization. The empirical models use a cosine schedule, which decreases the initial learning rate smoothly along a cosine curve as training progresses. The initial rate and early-stopping point are selected from validation performance and never from the test sample; the operational details are given in sections 4.2 and 4.3.

## 2.4 Risk Measures

Regulatory capital requirements for market risk have evolved substantially over the past decades. While Value at Risk (VaR) was the dominant metric under Basel II and the Basel 2.5 market-risk revisions, the Fundamental Review of the Trading Book (FRTB) replaced VaR with Expected Shortfall (ES) as the primary risk measure within the internal-models approach, using a 97.5% confidence level and liquidity-horizon adjustments (Basel Committee on Banking Supervision, 2019, January). The regulatory evolution across Basel II, Basel 2.5, FRTB, and the EU implementation through CRR II/III is summarized in table 14.

### 2.4.1 Value at Risk

Value at Risk is the most widely used quantile-based measure of downside risk in financial institutions. Let  $r_t$  denote the scalar portfolio return. For a lower-tail probability  $\alpha \in (0, 1)$ , define the signed return quantile

$$q_\alpha = F_r^{-1}(\alpha) = \inf\{x \in \mathbb{R} : \Pr(r_t \leq x) \geq \alpha\}.$$

The reported Value at Risk is its positive loss magnitude,

$$\text{VaR}_\alpha = -q_\alpha, \quad \Pr(r_t \leq -\text{VaR}_\alpha) = \alpha,$$

where the probability equality holds for a continuous return distribution. Thus  $\alpha \in \{0.01, 0.05\}$  corresponds to the 99% and 95% confidence levels. Conditional on the information set  $\Omega_{t-1}$ , the corresponding quantities are

$$q_{\alpha,t} = F_{r_t|\Omega_{t-1}}^{-1}(\alpha), \quad \text{VaR}_{\alpha,t} = -q_{\alpha,t}.$$

Throughout this thesis,  $q_{\alpha,t}$  denotes the signed return threshold used in backtesting, whereas  $\text{VaR}_{\alpha,t}$  denotes the same risk forecast reported as a positive magnitude. Jorion (1996) emphasizes that the resulting VaR estimate is itself subject to material sampling and estimation uncertainty.

Three classes of methods are commonly used to compute the conditional VaR: parametric, historical simulation, and filtered historical simulation. The parametric method imposes a distributional

assumption on the standardized innovations and combines it with a model for the conditional volatility, while filtered historical simulation relaxes this assumption by drawing from the empirical distribution of standardized residuals. The analysis relies on the latter; both methods are described below.

**Parametric Method.** The parametric approach to VaR relies on a distributional assumption for the latent standardized innovation  $\mathbf{z}_t = \boldsymbol{\Sigma}_t^{-1/2} \boldsymbol{\varepsilon}_t$ . After estimating the covariance model, its observable counterpart  $\hat{\mathbf{z}}_t = \hat{\boldsymbol{\Sigma}}_{t|t-1}^{-1/2} \mathbf{r}_t$  is a *standardized residual*. Parametric VaR specifies a law for  $\mathbf{z}_t$ , whereas filtered historical simulation uses the empirical distribution of  $\hat{\mathbf{z}}_t$ . Under conditional Gaussianity, the one-step-ahead VaR for an asset with conditional standard deviation  $\sigma_{t+1}$  takes the closed form

$$\text{VaR}_{\alpha,t+1} = -\sigma_{t+1} \Phi^{-1}(\alpha),$$

where  $\Phi^{-1}$  denotes the quantile function of the standard normal distribution. For a portfolio with weights  $\mathbf{w}$ , the analogous expression uses the conditional portfolio variance  $\sigma_{p,t+1}^2 = \mathbf{w}^\top \boldsymbol{\Sigma}_{t+1} \mathbf{w}$ . Under a conditional Student- $t$  distribution with  $\nu$  degrees of freedom—which better accommodates the heavy tails documented in financial return data—the VaR expression becomes

$$\text{VaR}_{\alpha,t+1} = -\sigma_{t+1} \sqrt{(\nu-2)/\nu} t_\nu^{-1}(\alpha),$$

where  $t_\nu^{-1}$  denotes the quantile function of the conventional Student- $t$  distribution with  $\nu$  degrees of freedom; the preceding factor rescales it to unit variance. The parametric approach is computationally cheap and analytically tractable, but its accuracy depends critically on the validity of the distributional assumption. Empirically, financial return series exhibit heavy tails, skewness, and excess kurtosis, as documented for the assets in this study in table 1. Neither the Gaussian nor the symmetric Student- $t$  specification captures all such distributional features. This motivates the semi-parametric filtered historical simulation approach described below.

**Filtered Historical Simulation.** To relax the parametric assumption on the innovation distribution while retaining the time-varying second moments implied by the volatility model, filtered historical simulation (FHS) is employed, following the GARCH-filtered empirical-distribution approach evaluated by Adcock et al. (2012). All risk forecasts are constructed at the portfolio level. Given the conditional covariance forecast  $\boldsymbol{\Sigma}_t$  and fixed weights  $\mathbf{w}$ , the portfolio return  $r_{p,t} = \mathbf{w}^\top \mathbf{r}_t$  has conditional variance  $\sigma_{p,t}^2 = \mathbf{w}^\top \boldsymbol{\Sigma}_t \mathbf{w}$ . Filtering the portfolio return by its fitted conditional volatility yields the standardized portfolio residuals

$$\hat{z}_{p,t} = \frac{r_{p,t}}{\sigma_{p,t}},$$

which, under correct model specification, are approximately i.i.d. and whose empirical distribution provides a nonparametric estimate of the portfolio innovation law. As an empirical design choice, the distribution is estimated from a rolling window rather than from the full history. This avoids look-ahead bias and allows for slow structural change in the innovation distribution. The empirical  $\alpha$ -quantile at time  $t$  is therefore computed from the  $W = 1000$  most recent

standardized residuals,

$$\hat{q}_{\alpha,t}^z = \text{Quantile}_{\alpha}(\hat{z}_{p,t-W+1}, \dots, \hat{z}_{p,t}).$$

Because the evaluation uses a one-step-ahead horizon, the next-period portfolio return is represented as  $r_{p,t+1} = \sigma_{p,t+1} z_{p,t+1}$ , with the innovation law approximated by the empirical residual distribution. The conditional quantile is therefore obtained directly from  $\hat{q}_{\alpha,t}^z$  without a bootstrap resampling step. The superscript  $z$  distinguishes this standardized-residual quantile from the conditional return quantile  $q_{\alpha,t}$ . The one-step-ahead FHS Value-at-Risk is therefore

$$\text{VaR}_{\alpha,t+1}^{\text{FHS}} = -\sigma_{p,t+1} \hat{q}_{\alpha,t}^z.$$

FHS preserves the model-implied volatility dynamics while admitting heavy tails and asymmetry through a nonparametric estimate of the conditional quantile, and the same construction is used for every model. Forecast differences arise from each model's conditional covariance path, both directly through  $\sigma_{p,t+1}$  and indirectly through the standardized-residual history.

#### 2.4.2 Expected Shortfall

While VaR provides a widely used quantile-based measure of downside risk, it does not convey information about the magnitude of losses beyond the VaR threshold. In particular, VaR only specifies a cutoff point such that losses exceed this level with probability  $\alpha$ , but it remains silent about the severity of these tail events. Two VaR-equivalent models may therefore imply very different exposures to extreme losses.

To address this limitation, Expected Shortfall (ES), also referred to as Conditional Value-at-Risk (CVaR), summarizes the mean return in the lower tail. Under a continuous conditional return distribution, define the signed tail mean

$$e_{\alpha,t} = \mathbb{E}[r_t \mid r_t \leq q_{\alpha,t}, \Omega_{t-1}], \quad \text{ES}_{\alpha,t} = -e_{\alpha,t}. \quad (23)$$

Thus  $e_{\alpha,t} \leq q_{\alpha,t}$  is the signed return-space quantity used by the backtests, while  $\text{ES}_{\alpha,t}$  is reported as a positive loss magnitude. Equivalently,

$$e_{\alpha,t} = \frac{1}{\alpha} \int_0^{\alpha} q_{u,t} du, \quad \text{ES}_{\alpha,t} = -\frac{1}{\alpha} \int_0^{\alpha} q_{u,t} du.$$

In contrast to VaR, Expected Shortfall is a coherent risk measure in the sense of Artzner et al. (1999), satisfying in particular the subadditivity axiom  $\rho(X_1 + X_2) \leq \rho(X_1) + \rho(X_2)$ . Subadditivity is the property that captures, at the level of a risk measure, the risk reduction implied by diversification: combining two positions can never produce more risk than the sum of the individual position risks. VaR satisfies subadditivity only under restrictive distributional assumptions, such as joint ellipticity of returns, and may fail it for portfolios with heavy-tailed or asymmetric dependence structures.

The one-step-ahead FHS Expected Shortfall reuses the standardized portfolio residuals  $\hat{z}_{p,s}$  and

their empirical  $\alpha$ -quantile  $\hat{q}_{\alpha,t}^z$ . It is the conditional mean of the residuals in the  $\alpha$ -tail, rescaled by the conditional portfolio volatility,

$$\text{ES}_{\alpha,t+1}^{\text{FHS}} = -\sigma_{p,t+1} \frac{1}{|\mathcal{T}_{\alpha,t}|} \sum_{s \in \mathcal{T}_{\alpha,t}} \hat{z}_{p,s}, \quad \mathcal{T}_{\alpha,t} = \{s \in \{t-W+1, \dots, t\} : \hat{z}_{p,s} \leq \hat{q}_{\alpha,t}^z\}.$$

## 2.5 Back Testing

Once a model has produced a sequence of conditional VaR and ES forecasts, statistical procedures are required to assess whether those forecasts are consistent with the realized returns. Backtesting procedures can be broadly divided into two groups: *absolute* tests that evaluate the statistical adequacy of a single model's forecasts, and *relative* tests that compare predictive performance across competing models. The first group addresses the question whether a given model is correctly specified; the second addresses whether one model significantly outperforms another. Both questions are addressed in the empirical evaluation.

The risk measures of sections 2.4.1 and 2.4.2 are reported as positive loss magnitudes. The backtests and scoring functions use the corresponding signed return-space quantities introduced above,

$$q_{\alpha,t} := -\text{VaR}_{\alpha,t} = F_{r_t|\Omega_{t-1}}^{-1}(\alpha), \quad e_{\alpha,t} := -\text{ES}_{\alpha,t},$$

i.e. the conditional  $\alpha$ -quantile and lower tail mean of the return distribution (both non-positive for small  $\alpha$ ); a return  $r_t \leq q_{\alpha,t}$  constitutes a VaR exceedance. This is the lower-tail return convention of Patton et al. (2019), who use exactly these signed quantities (their  $\text{VaR}_t$  and  $\text{ES}_t$  coincide with  $q_{\alpha,t}$  and  $e_{\alpha,t}$ ) and which the **esback** implementation of the Nolde and Ziegel (2017) tests likewise adopts. Note that Nolde and Ziegel (2017) themselves present the equivalent tests in the positive-loss convention (loss  $-r_t$ , level  $1 - \alpha$ ); writing the identification and scoring functions in terms of  $(q_{\alpha,t}, e_{\alpha,t})$  keeps them in the form actually implemented here.

### 2.5.1 VaR Backtesting

**Unconditional Coverage (Kupiec).** The simplest and most widely used VaR backtest, the unconditional coverage (proportion-of-failures) test of Kupiec (1995), which was later embedded by Christoffersen (1998) in the conditional-coverage framework below, evaluates whether the empirical exceedance frequency matches the nominal level  $\alpha$ . Define the exceedance indicator

$$I_t = \mathbf{1}\{r_t \leq q_{\alpha,t}\},$$

and let  $x = \sum_{t=1}^N I_t$  denote the total number of exceedances over an evaluation period of length  $N$ . Under correct unconditional coverage and independent Bernoulli exceedances,  $x \sim \text{Binomial}(N, \alpha)$ . The test compares the null hypothesis  $H_0 : \pi = \alpha$  against the alternative  $H_1 : \pi \neq \alpha$ , using the maximum likelihood estimator  $\hat{\pi} = x/N$ . The likelihood-ratio statistic is

$$LR_{UC} = -2 \ln \left( \frac{\alpha^x (1-\alpha)^{N-x}}{\hat{\pi}^x (1-\hat{\pi})^{N-x}} \right) \stackrel{H_0}{\sim} \chi^2(1),$$

which has a chi-squared limit distribution with one degree of freedom under correct specification. Rejection indicates that the model produces either too many or too few exceedances relative to the nominal level.

**Conditional Coverage (Christoffersen).** The unconditional coverage test ignores the temporal structure of exceedances. A model that produces the correct *average* number of exceedances may still cluster them in time, for example during stress episodes whose volatility dynamics are not captured by the fitted model. Such clustering means that violations arrive in bursts precisely when reliable risk forecasts are most important. Christoffersen (1998) addresses this by jointly testing correct coverage and serial independence of the exceedance sequence. Modelling  $(I_t)$  as a two-state Markov chain with transition probabilities  $p_{ij} = \Pr(I_t = j \mid I_{t-1} = i)$ , the independence hypothesis is

$$H_0^{\text{IND}} : p_{01} = p_{11} = \pi,$$

under which the next exceedance indicator does not depend on the previous one. The corresponding likelihood-ratio statistic  $LR_{\text{IND}}$  has a  $\chi^2(1)$  limit distribution. The joint conditional coverage statistic combines unconditional coverage and independence,

$$LR_{\text{CC}} = LR_{\text{UC}} + LR_{\text{IND}} \stackrel{H_0}{\sim} \chi^2(2).$$

The decomposition  $LR_{\text{CC}} = LR_{\text{UC}} + LR_{\text{IND}}$  allows the unconditional coverage and independence components to be reported separately, which is diagnostically useful: rejection driven by  $LR_{\text{UC}}$  alone points to a level misspecification, rejection driven by  $LR_{\text{IND}}$  alone points to time-dependence in the violation process, and joint rejection indicates both. Accordingly, all three statistics  $(LR_{\text{UC}}, LR_{\text{IND}}, LR_{\text{CC}})$  are reported with their associated  $p$ -values.

**Dynamic Quantile Test.** The Kupiec and Christoffersen tests evaluate the exceedance sequence in isolation and do not exploit potentially informative conditioning variables, such as the VaR forecast itself or past returns. The dynamic quantile (DQ) test of Engle and Manganelli (2004) addresses this limitation by regressing the centered exceedance indicator  $\tilde{I}_t = I_t - \alpha$  on a vector of predetermined regressors  $\mathbf{x}_{t-1}$ ,

$$\tilde{I}_t = \mathbf{x}_{t-1}^\top \boldsymbol{\delta} + u_t,$$

which may include an intercept, lagged exceedance indicators, the current VaR forecast, and functions of past returns. Under correct VaR specification, all coefficients in  $\boldsymbol{\delta}$  are zero and the residuals are uncorrelated, which is tested jointly by a Wald-type statistic with a chi-squared limit distribution. With suitable regressor choices, the DQ framework can impose moment restrictions corresponding to unconditional coverage and exceedance dynamics, while remaining a distinct Wald-type test rather than the likelihood-ratio tests of Kupiec and Christoffersen. It additionally detects misspecification that manifests as systematic dependence of exceedances on the magnitude of the forecast or on lagged volatility shocks. The test is implemented via the `segMGarch` R package, using the chosen regressor set of intercept, four lagged exceedance

indicators, the current VaR forecast, and one lagged squared return.

### 2.5.2 Expected Shortfall and Joint VaR–ES Backtesting

Backtesting Expected Shortfall is conceptually more challenging than backtesting VaR because ES is not an elicitable functional in isolation (Fissler and Ziegel, 2016). There is no scoring function that uniquely minimizes ES in expectation, so classical exceedance-frequency arguments do not extend directly. Three strategies have been developed to address this: testing conditional moment conditions implied by joint VaR–ES elicibility, testing the magnitude of exceedance residuals, and testing the joint VaR–ES pair through an auxiliary regression. All three are implemented through the R package `esback` of Bayer and Dimitriadis (2022).

**Conditional Calibration (CC).** Nolde and Ziegel (2017) formulate ES backtests in terms of *conditional calibration*: under correct specification of the joint VaR–ES forecast pair, the conditional expectation of suitably defined identification functions vanishes at every point in time. Concretely, define

$$V_t^V = \alpha - \mathbb{1}\{r_t \leq q_{\alpha,t}\}, \quad V_t^E = e_{\alpha,t} - q_{\alpha,t} + \frac{1}{\alpha}(q_{\alpha,t} - r_t) \mathbb{1}\{r_t \leq q_{\alpha,t}\},$$

which capture, respectively, exceedance frequency and conditional tail mean. Under correct VaR–ES forecasts, both have conditional mean zero. Two-sided CC variants test joint zero-moment restrictions and are therefore tests of exact calibration. The empirical summary in this thesis instead reports the `esback` one-sided general CC test: it forms four standardized, instrumented identification moments and combines the upper-tail  $p$ -values for positive moment means with the Hommel adjustment (Hommel, 1988). Its null is that all four selected moment means are non-positive; rejection indicates that at least one is positive. Because the directions have to be interpreted component by component, this result is kept distinct from both a two-sided test of exact calibration and the ES-specific underestimation tests below.

**Exceedance Residual (ER).** An alternative ES backtest, originally due to McNeil and Frey (2000), focuses directly on the tail residuals of the model. Define the *exceedance residuals*,

$$\xi_t = \frac{r_t - e_{\alpha,t}}{\sigma_t}, \quad t \text{ such that } r_t \leq q_{\alpha,t},$$

which under correct ES specification have mean zero conditional on a VaR exceedance occurring. The ER test evaluates this prediction by means of a one-sided test on the sample mean of the exceedance residuals, with null hypothesis  $\mathbb{E}[\xi_t \mid r_t \leq q_{\alpha,t}] = 0$ . Because the number of exceedances is small in typical samples and the tail distribution of  $\xi_t$  is non-Gaussian, the `esback` implementation obtains the  $p$ -value from an ordinary nonparametric bootstrap that resamples the exceedance residuals independently with replacement. The test is intuitive and computationally inexpensive, but its interpretation is conditional on the accompanying VaR forecast and is therefore not a standalone diagnosis when VaR itself is misspecified. The implementation uses  $B = 1000$  bootstrap replications.

**Expected Shortfall Regression (ESR).** The expected shortfall regression backtest of Bayer and Dimitriadis (2022) combines the joint VaR–ES elicibility result of Fissler and Ziegel (2016) with a regression-based testing strategy. The realized return is regressed on the forecasted Expected Shortfall through an auxiliary regression of the form

$$r_t = \beta_0 + \beta_1 e_{\alpha,t} + u_t,$$

where the coefficients are estimated by minimizing a strictly consistent joint VaR–ES scoring function from the family of Fissler and Ziegel (2016). Under correct ES specification, the population intercept and slope satisfy  $\beta_0 = 0$  and  $\beta_1 = 1$ , which is tested with a Wald-type statistic that converges in distribution to a chi-squared random variable. Three variants of the test are reported, following the version numbering of the **esback** implementation:

- **ESR-v1 (strict).** The two-sided joint test of  $H_0 : (\beta_0, \beta_1) = (0, 1)$  in the ES-only regression  $r_t = \beta_0 + \beta_1 e_{\alpha,t} + u_t$ . It assesses *exact* ES calibration and rejects for both under- and over-estimation.
- **ESR-v2 (auxiliary).** The two-sided auxiliary variant estimates the linked quantile and ES regressions  $r_t = \beta_0^q + \beta_1^q q_{\alpha,t} + u_t^q$  and  $r_t = \beta_0^e + \beta_1^e e_{\alpha,t} + u_t^e$  jointly. It tests the ES-equation restriction  $(\beta_0^e, \beta_1^e) = (0, 1)$  while accounting for the supplied quantile forecast;  $q_{\alpha,t}$  is not an additional regressor in the ES equation. Like v1 it is a two-sided test of exact calibration.
- **ESR-v3 (strict intercept).** A *one-sided*, intercept-only test that fixes the slope at one and examines the intercept restriction alone. It is directed against systematic ES *underestimation* (risk that is too small), the economically non-conservative direction.

Standard errors are constructed using a sparsity-based estimator under the assumption of independent and identically distributed residuals, with a misspecification-robust correction enabled throughout. The one-sided ER and ESR-v3 tests thus diagnose whether a model *systematically underestimates* tail risk, whereas the two-sided v1 and v2 tests reject any departure from exact ES calibration in either direction.

### 2.5.3 Forecast Comparison

The backtests of section 2.5.1 and section 2.5.2 evaluate the statistical adequacy of a single model in isolation. To assess whether one model produces *significantly* better forecasts than another, a relative test is required.

The empirical comparison uses the zero-homogeneous FZ0 score of Fissler and Ziegel (2016), operationalized by Patton et al. (2019),

$$S_{FZ}(r_{p,t}, q_{\alpha,t}, e_{\alpha,t}) = \mathbf{1}\{r_{p,t} \leq q_{\alpha,t}\} \frac{r_{p,t} - q_{\alpha,t}}{\alpha e_{\alpha,t}} + \frac{q_{\alpha,t}}{e_{\alpha,t}} + \log(-e_{\alpha,t}) - 1, \quad (24)$$

where  $r_{p,t} = \mathbf{w}^\top \mathbf{r}_t$  is the portfolio return from section 3.3, and  $q_{\alpha,t} = -\text{VaR}_{\alpha,t}$  and  $e_{\alpha,t} = -\text{ES}_{\alpha,t}$

are the signed return-space tail forecasts. The logarithm is well defined because  $e_{\alpha,t} < 0$ . The score is strictly consistent for the joint elicitation of VaR and ES and is used only for out-of-sample comparison, never as a training objective.

**Diebold–Mariano Test.** The Diebold–Mariano test (Diebold and Mariano, 1995) compares the out-of-sample predictive accuracy of two competing forecasting models on the basis of a chosen loss function. Let  $\mathcal{L}_t^{(1)}$  and  $\mathcal{L}_t^{(2)}$  denote the realized forecast-loss values of models 1 and 2 at time  $t$ —for example, the negative log-likelihood, the joint VaR–ES scoring function of Fissler and Ziegel (2016), or a squared-error loss on a volatility proxy. Define the loss differential  $d_t = \mathcal{L}_t^{(1)} - \mathcal{L}_t^{(2)}$  and let  $\bar{d} = N^{-1} \sum_t d_t$  denote its sample mean over the out-of-sample window of length  $N$ . Under the null hypothesis of equal predictive accuracy,  $\mathbb{E}[d_t] = 0$ . The Diebold–Mariano statistic is

$$\text{DM} = \frac{\bar{d}}{\sqrt{\hat{\sigma}_d^2/N}} \xrightarrow{H_0} \mathcal{N}(0, 1),$$

where  $\hat{\sigma}_d^2$  is a heteroskedasticity- and autocorrelation-consistent estimator of the long-run variance of  $d_t$ , typically obtained via the Newey–West estimator with a truncation lag growing in  $N$ . A significantly negative DM statistic indicates that model 1 outperforms model 2 under the chosen loss; a significantly positive statistic indicates the reverse.

The choice of loss function determines which dimension of predictive accuracy is being compared. The comparison uses the strictly consistent joint VaR–ES scoring function of Fissler and Ziegel (2016) (equation (24)), which evaluates both tail risk measures jointly under a single decision-theoretic criterion. This score is used purely for out-of-sample forecast comparison; it does not enter the estimation of any model. The DM test is therefore the natural complement to the absolute backtests in sections 2.5.1 and 2.5.2: while those diagnose whether a given model is internally well-calibrated, the DM test ranks competing models against each other on the same tail-risk-aware loss. In finite samples the normal approximation can be liberal; the small-sample modification of Harvey et al. (1997), which rescales the statistic and compares it against Student- $t$  critical values, is therefore employed.

**Model Confidence Set.** The DM test compares exactly two models. When a larger collection of  $m$  models is evaluated simultaneously, pairwise testing induces a multiple-comparison problem: the probability of spuriously rejecting at least one true null grows with the number of comparisons. The model confidence set (MCS) procedure of Hansen, Lunde, and Nason (2011) addresses this by constructing, for a prespecified confidence level  $1 - \alpha_{\text{MCS}}$ , a set of models that contains the best-performing model with at least this probability asymptotically. Starting from the full collection  $\mathcal{M}_0$ , the procedure iterates two steps: an equivalence test of the null hypothesis

$$H_{0,\mathcal{M}} : \mathbb{E}[d_{ij,t}] = 0 \quad \text{for all } i, j \in \mathcal{M},$$

where  $d_{ij,t}$  denotes the loss differential between models  $i$  and  $j$ , and—upon rejection—an elimination rule that removes the model contributing most to the rejection. The algorithm terminates at the first non-rejection, and the surviving set  $\widehat{\mathcal{M}}_{1-\alpha_{\text{MCS}}}^*$  satisfies  $\lim_{N \rightarrow \infty} \mathbb{P}(\mathcal{M}^* \subseteq \widehat{\mathcal{M}}_{1-\alpha_{\text{MCS}}}^*) \geq$

$1 - \alpha_{\text{MCS}}$ , where  $\mathcal{M}^*$  is the set of models with minimal expected loss. The distribution of the test statistic is approximated by a stationary bootstrap, which preserves the serial dependence of the loss differentials. The MCS is computed on the joint VaR–ES loss of Fissler and Ziegel (2016) at the 90% confidence level. Since the loss differentials and all other forecast diagnostics are produced in `Python`, the MCS is evaluated directly in the same environment using the `arch` package (Sheppard, 2025, October), with 1,000 stationary-bootstrap resamples, the package-default block length  $\lfloor \sqrt{N} \rfloor = 30$ , bootstrap seed 42, and the range statistic  $T_R$ , which keeps the entire evaluation pipeline in a single toolchain.

### 3 Data

Financial return series exhibit recurring empirical features that motivate the volatility models considered in this thesis. Cont (2001) summarizes the univariate regularities, while Bauwens et al. (2006) survey multivariate GARCH models for dynamic covariances; multivariate spillovers and asymmetric comovement are studied more specifically by Kroner and Ng (1998) and Cappiello et al. (2006). The box below is therefore a schematic summary of commonly observed tendencies, not a set of universal restrictions holding for every asset pair, lag, and sample period. The modeling framework is designed to accommodate these tendencies.

#### Stylized Facts of Multivariate Asset Returns

Let  $\mathbf{r}_t \in \mathbb{R}^N$  denote a vector of asset returns with

$$\mathbb{E}[\mathbf{r}_t] = \mathbf{0}, \quad \text{Cov}(\mathbf{r}_t \mid \Omega_{t-1}) = \boldsymbol{\Sigma}_t.$$

- |                                              |                                                                                                             |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| (1) <b>Time-varying covariance matrix</b>    | $\boldsymbol{\Sigma}_t \neq \boldsymbol{\Sigma}$                                                            |
| (2) <b>Volatility clustering</b>             | $\text{Corr}(r_{i,t}^2, r_{i,t-k}^2) > 0, \quad \text{at relevant lags } k > 0$                             |
| (3) <b>Dynamic conditional correlations</b>  | $\rho_{ij,t} = \frac{\Sigma_{ij,t}}{\sqrt{\Sigma_{ii,t} \Sigma_{jj,t}}}$ varies over $t$                    |
| (4) <b>Volatility spillovers</b>             | $\frac{\partial \Sigma_{ii,t}}{\partial r_{j,t-1}^2} \neq 0, \quad i \neq j$                                |
| (5) <b>Asymmetric spillovers</b>             | $\frac{\partial \Sigma_{ii,t}}{\partial r_{j,t-1}^-} > \frac{\partial \Sigma_{ii,t}}{\partial r_{j,t-1}^+}$ |
| (6) <b>Weak linear cross-autocorrelation</b> | $\text{Corr}(r_{i,t}, r_{j,t-k}) \approx 0, \quad k > 0$                                                    |

#### 3.1 Data Description

The portfolio consists of four daily financial time series covering the period from August 2000 to May 2026, resulting in 6,427 observations per series. The dataset includes representative assets from the major macroeconomic asset classes: the S&P 500 equity index ( $\sim \text{GSPC}$ ), gold futures ( $\text{GC}=\text{F}$ ), West Texas Intermediate crude oil futures ( $\text{CL}=\text{F}$ ), and the 10-year U.S. Treasury yield ( $\sim \text{TNX}$ ). The selection covers equities, precious metals, energy commodities, and fixed income, allowing the analysis of cross-asset volatility dynamics and spillover effects across structurally heterogeneous economic sectors.

The unadjusted daily closing series were downloaded from Yahoo Finance through the `yfinance`

interface using the data snapshot of 3 May 2026; the common complete-case sample ends on 1 May 2026.

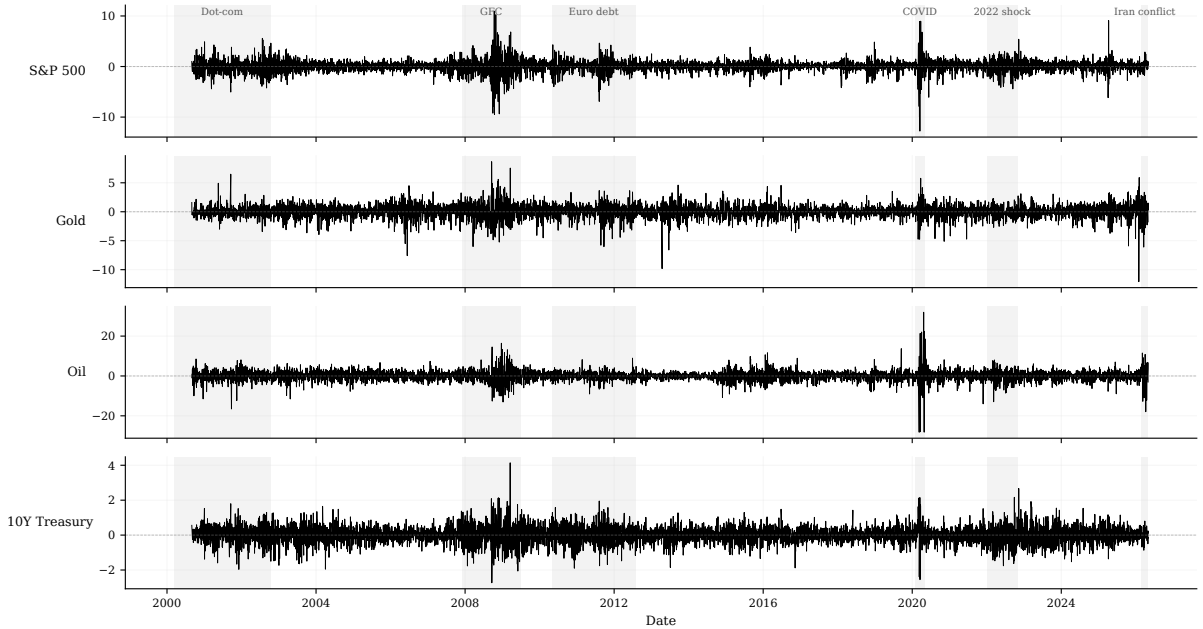
For the three price series (S&P 500, gold, crude oil), log returns are computed as

$$r_{i,t} = \log(P_{i,t}/P_{i,t-1}) \times 100,$$

expressed in percent. The 10-year U.S. Treasury series is quoted as a yield-to-maturity ( $\hat{\text{TNX}}$ ) rather than a tradable bond price. Consequently, a proxy holding return for a synthetic 10-year par bond is constructed as an empirical design choice. It adds approximate one-day carry to the standard second-order duration–convexity approximation for the price change (CFA Institute, 2026). The observed  $\hat{\text{TNX}}$  value  $y_t$  is quoted in percent (for example, 4.25 for a yield of 4.25%), whereas the approximation operates on decimal yields. Hence,  $\tilde{y}_t = y_t/100$  and the corresponding decimal holding return is

$$\tilde{r}_{\text{TNX},t} \approx \frac{\tilde{y}_{t-1}}{252} - D_{\text{mod},t-1} \Delta \tilde{y}_t + \frac{1}{2} C_{t-1} (\Delta \tilde{y}_t)^2,$$

which combines daily carry with the modified-duration and convexity price effects, where  $D_{\text{mod}}$  and  $C$  denote the modified duration and convexity of a 10-year par bond. Finally,  $r_{\text{TNX},t} = 100 \tilde{r}_{\text{TNX},t}$  expresses the holding return in percent, consistent with the price-series returns. Thus, dividing  $y_t$  by 100 converts a quoted yield into a decimal yield, while multiplying  $\tilde{r}_{\text{TNX},t}$  by 100 converts the resulting decimal return back into percent. For notational compactness, the joint return vector  $\mathbf{r}_t \in \mathbb{R}^4$  is referred to as “returns” throughout the remainder of the thesis.



**Figure 3: Daily returns.** Returns of  $\hat{\text{GSPC}}$ ,  $\text{GC}=\text{F}$ ,  $\text{CL}=\text{F}$ , and  $\hat{\text{TNX}}$  from August 2000 to May 2026. All series exhibit pronounced volatility clustering, with clearly visible variance spikes around the 2008 financial crisis, the 2014–2016 oil price collapse, the COVID-19 crash of March 2020, and the 2022 fixed-income repricing.

| Panel A: Distributional moments |      |        |        |         |         |          |         |  |
|---------------------------------|------|--------|--------|---------|---------|----------|---------|--|
|                                 | N    | Mean   | SD     | Skew    | Kurt.   | Min      | Max     |  |
| S&P 500                         | 6427 | 0.0252 | 1.2135 | -0.3526 | 14.0059 | -12.7652 | 10.9572 |  |
| Gold                            | 6427 | 0.0441 | 1.1231 | -0.4841 | 9.8960  | -12.0657 | 8.6432  |  |
| Oil                             | 6427 | 0.0267 | 2.6207 | -0.0918 | 18.7559 | -28.2206 | 31.9634 |  |
| 10Y Treasury                    | 6427 | 0.0152 | 0.4895 | -0.0080 | 5.4737  | -2.7307  | 4.1394  |  |

| Panel B: Diagnostic tests |            |          |                |          |                  |           |          |           |
|---------------------------|------------|----------|----------------|----------|------------------|-----------|----------|-----------|
|                           | JB         | $p_{JB}$ | $LB_{10}(r_t)$ | $p_r$    | $LB_{10}(r_t^2)$ | $p_{r^2}$ | ADF      | $p_{ADF}$ |
| S&P 500                   | 32514.6175 | < 0.0001 | 106.3594       | < 0.0001 | 5997.8174        | < 0.0001  | -19.9335 | < 0.0001  |
| Gold                      | 12962.5183 | < 0.0001 | 22.6466        | 0.0121   | 578.2513         | < 0.0001  | -32.0769 | < 0.0001  |
| Oil                       | 66376.7734 | < 0.0001 | 25.2755        | 0.0048   | 3099.2635        | < 0.0001  | -14.4659 | < 0.0001  |
| 10Y Treasury              | 1635.0027  | < 0.0001 | 18.5453        | 0.0464   | 1176.9981        | < 0.0001  | -59.4736 | < 0.0001  |

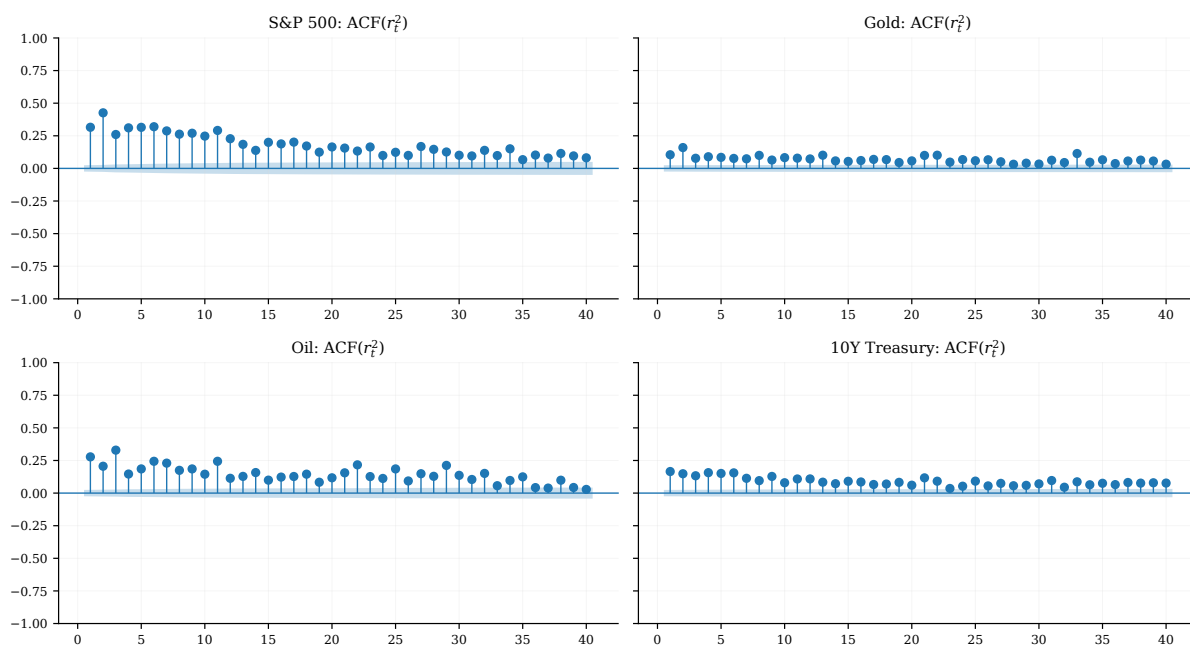
**Table 1:** Descriptive statistics of daily returns, expressed in percent. For the three price series these are log returns ( $\times 100$ ); for the 10-year Treasury it is the daily holding return implied by the change in yield through a duration–convexity approximation ( $\times 100$ ). Panel A reports distributional moments and Panel B reports full-sample diagnostic tests.

In Panel B, JB denotes the Jarque–Bera test of normality,  $LB_{10}$  the Ljung–Box test for serial correlation through lag 10, and ADF the augmented Dickey–Fuller unit-root test (Jarque and Bera, 1987; Ljung and Box, 1978; Said and Dickey, 1984). The JB results reject normality for every series. The Ljung–Box results indicate statistically significant serial dependence in the raw returns and, much more strongly, in the squared returns. The ADF results reject a unit root in every return series. Taken together, Panel B describes stationary but non-Gaussian returns with pronounced volatility dependence.

## 3.2 Empirical Properties

### 3.2.1 Volatility Clustering and Persistence

Figure 3 provides a first visual indication of volatility clustering: large return magnitudes tend to be followed by further large return magnitudes, while quiet periods cluster as well. Both absolute and squared returns are common proxies for return magnitude; the analysis below uses squared returns. To quantify this property, figure 4 reports their sample autocorrelation function for each asset.



**Figure 4: Sample autocorrelation function of squared returns.** Lags 1–40 for each asset. Slowly decaying positive autocorrelation across all series provides empirical support for the volatility-clustering property (Stylized Fact 2 above) and motivates the GARCH framework. Persistence is most pronounced for the S&P 500 and crude oil.

All four series display slowly decaying, persistently positive autocorrelation in the squared returns, well beyond conventional confidence bands even at long lags. This is the canonical empirical signature of conditional heteroskedasticity and provides the primary justification for GARCH-type modeling.

### 3.2.2 Time-Varying Cross-Asset Dependence

The motivation for moving from univariate GARCH to multivariate specifications rests on the empirical observation that cross-asset correlations vary substantially over time, particularly during periods of financial stress. Figure 5 reports rolling 120-day Pearson correlations for three economically relevant asset pairs.



**Figure 5: Rolling 120-day Pearson correlations.** Three economically relevant pairs: equity–gold ( $\hat{\text{GSPC}}-\hat{\text{GC}}=\text{F}$ ), equity–Treasury ( $\hat{\text{GSPC}}-\hat{\text{TNX}}$ ), and gold–oil ( $\text{GC}=\text{F}-\text{CL}=\text{F}$ ). The unconditional full-sample correlations  $\rho$  are reported in each panel. The rolling correlations fluctuate substantially and switch sign in several periods, providing descriptive motivation for dynamic covariance models. Complete pairwise results are reported in figure 13.

The unconditional correlations reported in each panel mask substantial time variation: the equity–gold correlation oscillates between markedly positive and markedly negative values, and the equity–Treasury correlation switches between regimes. This pattern is consistent with the flight-to-quality interpretation, under which greater equity-market uncertainty is associated with relatively strong Treasury returns and a more negative stock–bond return correlation (Connolly et al., 2005). These observations are descriptively consistent with time-varying dependence and motivate the dynamic conditional covariance models considered in this thesis; rolling sample correlations alone do not constitute a formal test against constant conditional correlation. The complete set of pairwise rolling correlations, together with Normal QQ-plots of the daily returns, is provided in section A.

### 3.3 Portfolio Construction

All risk forecasting results in the empirical analysis are reported at the portfolio level rather than for individual assets. This reflects the multivariate nature of the modeling approach: the economic relevance of conditional covariance forecasts emerges through the aggregation of risks across assets. The portfolio return at time  $t$  is defined as the weighted sum of the individual asset returns,

$$r_{p,t} = \mathbf{w}^\top \mathbf{r}_t, \quad \mathbf{w} \in \mathbb{R}^N, \quad \mathbf{1}^\top \mathbf{w} = 1,$$

and the conditional portfolio variance under a given covariance model is

$$\sigma_{p,t}^2 = \mathbf{w}^\top \boldsymbol{\Sigma}_t \mathbf{w}.$$

Throughout the empirical analysis, equal weights  $w_i = 1/N$  are imposed and held constant over

the evaluation period. This choice is deliberate: the goal of the analysis is to isolate differences in *covariance forecasting performance* across models, not to compare portfolio optimization strategies. Dynamic weighting schemes such as global minimum-variance optimization would entangle covariance estimation error with optimization-induced estimation error, obscuring the contribution attributable to the covariance model itself. Holding weights fixed across all models therefore ensures that differences in portfolio-level VaR and ES forecasts are driven exclusively by differences in the conditional covariance estimates  $\Sigma_t$ .

## 4 Model Application

### 4.1 Model Specification

Section 2 introduced a range of conditional covariance models. The empirical analysis considers a focused subset of specifications, organised in three classes: classical econometric benchmarks, purely data-driven neural architectures, and hybrid Neural-MGARCH constructions. Table 2 identifies the models used in the experiments; the paragraphs that follow specify their implemented covariance mappings and explain their role in the comparison. The accompanying implementation and replication code are available on GitHub.<sup>1</sup>

| Model                         | Output mechanism                                                                                                                                 | Estimation               |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <i>Econometric benchmarks</i> |                                                                                                                                                  |                          |
| BEKK(1,1) symmetric           | $\Sigma_t$ via quadratic-form recursion (3)                                                                                                      | QML, BEKKs (R)           |
| BEKK(1,1) asymmetric          | $\Sigma_t$ with joint-sign asymmetry term                                                                                                        | QML, BEKKs (R)           |
| DCC(1,1) scalar               | $\Sigma_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$                                                                                              | 2-step QML, rmgarch (R)  |
| ADCC(1,1) scalar              | $\Sigma_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$ , asym. corr. dynamics                                                                       | 2-step QML, rmgarch (R)  |
| <i>Neural benchmarks</i>      |                                                                                                                                                  |                          |
| LSTM                          | $\text{vech}(\mathbf{L}_t)$ , $\Sigma_t = \mathbf{L}_t \mathbf{L}_t^\top$                                                                        | AdamW on conditional NLL |
| GRU                           | $\text{vech}(\mathbf{L}_t)$ , $\Sigma_t = \mathbf{L}_t \mathbf{L}_t^\top$                                                                        | AdamW on conditional NLL |
| <i>Hybrid architectures</i>   |                                                                                                                                                  |                          |
| BEKK-LSTM                     | $\mathbf{K}_t = \mathcal{K}_{\text{asBEKK}}(\boldsymbol{\varepsilon}_{t-1}, \Sigma_{t-1})$ , $\Sigma_t = \text{Mod}(\mathbf{K}_t, \mathbf{c}_t)$ | AdamW on conditional NLL |
| Neural-BEKK                   | $\boldsymbol{\gamma}_t = \mathcal{C}(\text{Head}(\mathbf{h}_{t-1}))$ , $\Sigma_t$ via (22)                                                       | AdamW on conditional NLL |

**Table 2:** Overview of model specifications considered in the empirical analysis.

**Econometric Benchmarks.** The symmetric BEKK(1,1) specification of (3) is the primary structural baseline. A separate asymmetric BEKK(1,1) benchmark tests whether adding fixed leverage dynamics improves on that baseline. Its implementation in the BEKKs R package (Fülle et al., 2024) follows the joint-sign formulation of Grier et al. (2004): a scalar indicator activates one  $N \times N$  asymmetry matrix  $\mathbf{B}$  when the complete innovation vector  $\boldsymbol{\varepsilon}_{t-1}$  lies in the specified negative-sign orthant. This differs from the component-wise selector  $\boldsymbol{\eta}_{t-1} = \mathbf{1}\{\boldsymbol{\varepsilon}_{t-1} < \mathbf{0}\} \odot \boldsymbol{\varepsilon}_{t-1}$  of Kroner and Ng (1998), which is used by the Neural-BEKK. Thus, the empirical analysis

<sup>1</sup><https://github.com/BlankerHans/Neural-BEKK>

contains both a symmetric and an asymmetric econometric BEKK benchmark; the latter uses the asymmetry mechanism implemented by BEKKs, whereas the neural model uses component-wise negative shocks. The scalar DCC(1,1) of Engle (2002) serves as the parsimonious correlation-based benchmark and is estimated in two stages, with univariate GARCH(1,1) fits per asset in the first stage and scalar correlation parameters in the second. The scalar ADCC implementation used here restricts the more general asymmetric correlation dynamics of Cappiello et al. (2006) to one additional asymmetry parameter, so that joint negative return shocks can raise conditional correlations more strongly than positive ones. It is likewise estimated in two stages with the `rmgarch` package.<sup>2</sup> The total parameter count is  $N(N+1)/2 + 2N^2$  for the symmetric BEKK,  $N(N+1)/2 + 3N^2$  for the asymmetric BEKK,  $3N+2$  for the scalar DCC, and  $3N+3$  for the scalar ADCC.

| Model                          | Parameters ( $N = 4, h = 64$ ) | Scaling            |
|--------------------------------|--------------------------------|--------------------|
| BEKK(1,1) symmetric            | 42                             | $\mathcal{O}(N^2)$ |
| BEKK(1,1) asymmetric           | 58                             | $\mathcal{O}(N^2)$ |
| DCC(1,1) scalar                | 14                             | $\mathcal{O}(N)$   |
| ADCC(1,1) scalar               | 15                             | $\mathcal{O}(N)$   |
| LSTM (2 layers, Cholesky head) | 54,410                         | $\mathcal{O}(h^2)$ |
| GRU (2 layers, Cholesky head)  | 40,970                         | $\mathcal{O}(h^2)$ |
| BEKK-LSTM (scalar)             | 4,943                          | $\mathcal{O}(h^2)$ |
| BEKK-LSTM (vector)             | 5,138                          | $\mathcal{O}(h^2)$ |
| BEKK-LSTM (mix)                | 5,723                          | $\mathcal{O}(h^2)$ |
| Neural-BEKK (asym. diag.)      | 21,220                         | $\mathcal{O}(h^2)$ |
| Neural-BEKK (asym. full)       | 23,560                         | $\mathcal{O}(h^2)$ |

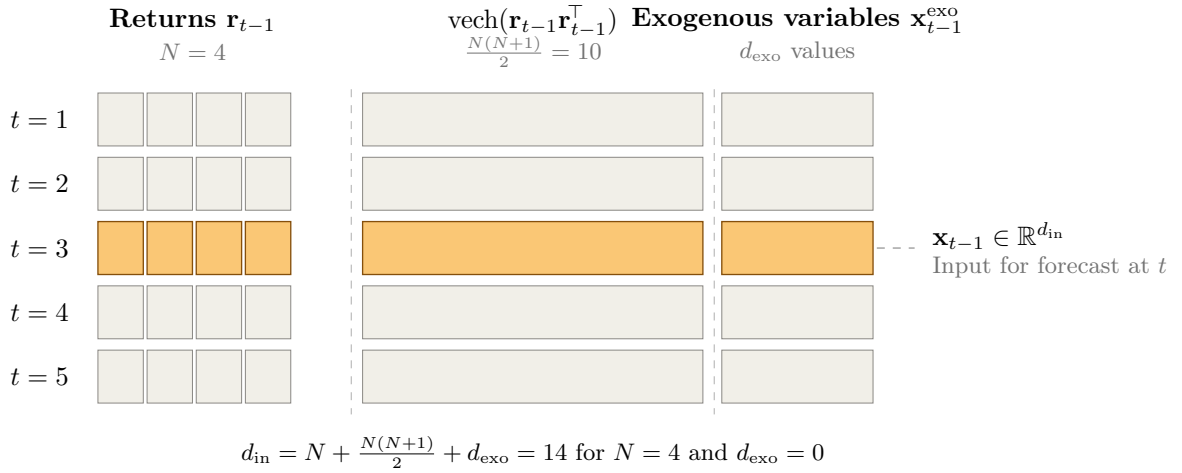
**Table 3:** Parameter counts by model, at  $N = 4$  assets and recurrent hidden dimension  $h = 64$ . The econometric counts are exact in  $N (\frac{N(N+1)}{2} + 2N^2$  for the symmetric and  $+3N^2$  for the asymmetric BEKK,  $3N+2$  for scalar DCC, and  $3N+3$  for scalar ADCC); the neural figures are the realised numbers of covariance-network parameters. The direct LSTM and GRU benchmarks use two recurrent layers with a Cholesky output head. The BEKK-LSTM variants use a custom LSTM-style state update to modulate a BEKK kernel with time-invariant coefficients, whereas Neural-BEKK uses a single-layer GRU and a parameter head that emits diagonal or full time-varying BEKK matrices. Student- $t$  variants add one separately trained scalar parameter,  $\nu$ , to the count reported here. The neural counts are dominated by recurrent weight matrices and hence scale as  $\mathcal{O}(h^2)$ .

**Neural Benchmarks.** The LSTM and GRU benchmarks parameterise the conditional covariance matrix directly by mapping the recent feature history to a half-vectorised lower-triangular Cholesky factor  $\text{vech}(\mathbf{L}_t)$ . Positive definiteness of  $\mathbf{\Sigma}_t = \mathbf{L}_t \mathbf{L}_t^\top$  is enforced by passing the diagonal entries of  $\mathbf{L}_t$  through a softplus activation. For a forecast of  $\mathbf{\Sigma}_t$ , both models receive the same

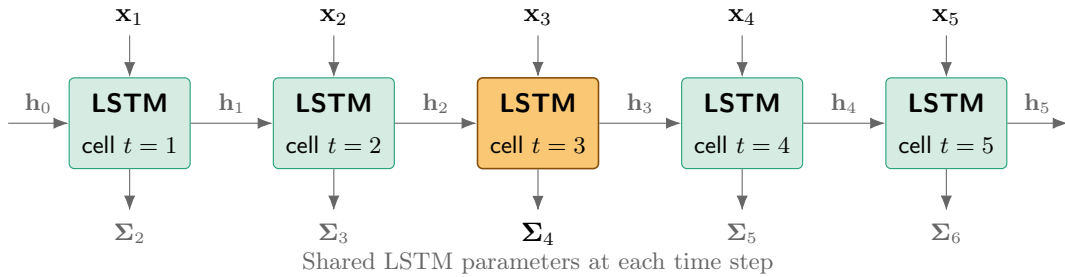
<sup>2</sup>In the present sample the estimated ADCC asymmetry parameter is effectively zero, so the ADCC correlation dynamics collapse onto those of the symmetric DCC and the two specifications produce numerically near-identical conditional covariance forecasts and out-of-sample results. ADCC is retained as a benchmark to document that, in the present fit, the correlation-level leverage extension is not activated and yields no measurable improvement in tail-risk forecasting.

lagged input vector  $\mathbf{x}_{t-1}$ , composed of the standardised return vector  $\mathbf{r}_{t-1}$ , the half-vectorisation  $\text{vech}(\mathbf{r}_{t-1}\mathbf{r}_{t-1}^\top)$  of the realised outer product. No exogenous variables enter the implemented models. With  $N = 4$ , each feature vector therefore has  $d_{\text{in}} = 4 + 4(4 + 1)/2 = 14$  elements. The outer-product features provide the recurrent network with explicit cross-asset information that would otherwise have to be inferred from the returns alone. These specifications act as flexibility-focused references against which the hybrid architectures can be compared: they impose no structural restriction on the dynamics of  $\Sigma_t$  beyond positive definiteness.

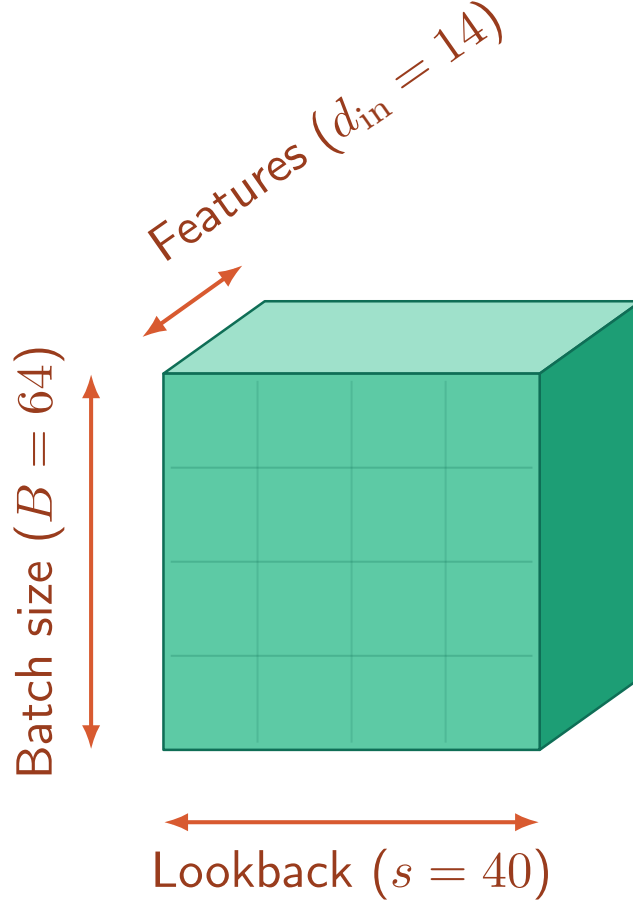
The recurrent data flow is summarised in figures 6 to 8. Figure 6 decomposes a generic input vector into returns, their half-vectorised outer product, and an optional block of predetermined exogenous variables. In the empirical specification used here,  $d_{\text{exo}} = 0$  and hence  $d_{\text{in}} = 14$ . Figure 7 illustrates schematically how recurrent weight sharing propagates information through time and shifts the input at date  $t$  into a covariance forecast for date  $t + 1$ . Figure 8 complements this time-domain view with the corresponding PyTorch `batch_first=True` tensor convention. Every recurrent benchmark and hybrid controller processes the same sequence  $\mathbf{X}_t = (\mathbf{x}_{t-s}, \dots, \mathbf{x}_{t-1})$  of feature vectors; the model classes differ in how the recurrent state and any model-specific covariance recursion map this sequence to  $\Sigma_t$ .



**Figure 6:** Input tensor structure for recurrent models.



**Figure 7:** LSTM network unrolled over five time steps. Each box represents one application of the same LSTM cell with shared parameters.



**Figure 8:** Recurrent input tensor under the PyTorch `batch_first=True` convention, with shape  $(B, s, d_{\text{in}}) = (64, 40, 14)$ .

**Hybrid Architectures.** The BEKK-LSTM specifications use a custom LSTM-style recurrent state  $\mathbf{c}_t$  to modulate an asymmetric BEKK kernel whose coefficient matrices are jointly estimated but time-invariant. In the notation used throughout the thesis, the kernel is

$$\mathcal{K}_{\text{asBEKK}}(\boldsymbol{\varepsilon}_{t-1}, \boldsymbol{\Sigma}_{t-1}; \mathbf{C}, \mathbf{A}, \mathbf{G}, \mathbf{B}) = \mathbf{C}\mathbf{C}^\top + \mathbf{A}^\top \boldsymbol{\varepsilon}_{t-1} \boldsymbol{\varepsilon}_{t-1}^\top \mathbf{A} + \mathbf{G}^\top \boldsymbol{\Sigma}_{t-1} \mathbf{G} + \mathbf{B}^\top \boldsymbol{\eta}_{t-1} \boldsymbol{\eta}_{t-1}^\top \mathbf{B},$$

with  $\mathbf{C}, \mathbf{A}, \mathbf{G}, \mathbf{B}$  time-invariant trainable matrices and the cell state used to modulate the kernel output into the final  $\boldsymbol{\Sigma}_t$ . Three modulation variants are considered: a scalar multiplicative modulation  $m_t \in \mathbb{R}$ , a vector modulation  $\mathbf{m}_t \in \mathbb{R}^N$  applied diagonally, and a convex-mixture modulation that interpolates between the BEKK kernel output and an unconstrained Cholesky-based covariance prediction. The coefficient matrices are constant across  $t$ , but the realised kernel output  $\mathbf{K}_t$  changes with  $\boldsymbol{\varepsilon}_{t-1}$  and  $\boldsymbol{\Sigma}_{t-1}$ ; additional temporal adaptation is supplied by the modulation function. Unlike the additive LSTM term of H. Wang et al. (2025) described in section 2.2.1, the recurrent state here modulates the output of a BEKK recursion with time-invariant coefficients. The variants are presented in increasing flexibility: scalar modulation, asset-specific vector modulation, and a convex mixture with a neural covariance branch. The Neural-BEKK models then move from diagonal to full time-varying BEKK matrices; a scalar

Neural-BEKK was not implemented.

#### Modulation Heads $\text{Mod}(\mathbf{K}_t, \mathbf{c}_t)$

Each variant produces a positive-definite  $\Sigma_t$  from the BEKK kernel  $\mathbf{K}_t$  and the cell state  $\mathbf{c}_t$ . Mod denotes this complete covariance mapping; the scalar  $m_t$ , vector  $\mathbf{m}_t$ , and mixture weight  $\lambda_t$  below are intermediate outputs of the respective modulation heads. The multiplicative modulation strength is set by  $\beta \in (0, 1)$ .

**Scalar.**

$$m_t = 1 + \beta \tanh(\mathbf{w}_o^\top \tanh(\mathbf{c}_t) + b_o), \quad \Sigma_t = m_t \mathbf{K}_t.$$

**Vector.**

$$\mathbf{m}_t = \mathbf{1} + \beta \tanh(\mathbf{W}_o \tanh(\mathbf{c}_t) + \mathbf{b}_o), \quad \Sigma_t = \text{diag}(\mathbf{m}_t) \mathbf{K}_t \text{diag}(\mathbf{m}_t).$$

**Convex Mixture.**

$$\begin{aligned} \mathbf{z}_t &= [\tanh(\mathbf{c}_t); \text{LN}(\text{vech}(\mathbf{K}_t))], \\ \lambda_t &= \sigma(\mathbf{w}_\lambda^\top \mathbf{z}_t + b_\lambda) \in (0, 1), \\ \mathbf{v}_t &= \text{clip}(\mathbf{W}_L \mathbf{z}_t + \mathbf{b}_L, \pm c), \\ \mathbf{L}_t^{\text{nn}} &= \text{tril}(\text{unvech}_{\text{lt}}(\mathbf{v}_t), -1) + \text{diag}(\text{softplus}(\mathbf{v}_t^{\text{diag}}) + \eta), \\ \Sigma_t^{\text{nn}} &= \mathbf{S} \mathbf{L}_t^{\text{nn}} (\mathbf{L}_t^{\text{nn}})^\top \mathbf{S}, \\ \Sigma_t &= (1 - \lambda_t) \mathbf{K}_t + \lambda_t \Sigma_t^{\text{nn}}. \end{aligned}$$

The gate  $\lambda_t$  has a smooth regime-weight interpretation: values near zero place almost all weight on the structured BEKK kernel, whereas values near one favor the neural Cholesky covariance. Intermediate values blend both regimes continuously; this is not a discrete Markov-switching model.

Unlike the kernel-replacement architectures, the Neural-BEKK makes the BEKK parameter matrices themselves time-varying functions of past information. For the forecast at  $t$ , the single-layer GRU summarises the input sequence ending at  $t-1$ . Its final hidden state is passed through the implemented parameter head—a hidden affine layer, a tanh activation, dropout, and a final affine layer—before a deterministic constraint map transforms the unconstrained head output into admissible BEKK coefficients:

$$\begin{aligned} \mathbf{h}_{t-1} &= \text{GRU}_\theta(\mathbf{x}_{t-s}, \dots, \mathbf{x}_{t-1}), \\ \mathbf{u}_t &= \text{Head}_\theta(\mathbf{h}_{t-1}), \\ \gamma_t &= \mathcal{C}(\mathbf{u}_t). \end{aligned}$$

Here  $\mathcal{C}$  denotes the constraint transformations detailed below. The resulting parameter vector differs between the two implemented variants:

$$\begin{aligned} \gamma_t^{\text{diag}} &= (\text{vech}(\mathbf{C}_t), \mathbf{a}_t, \mathbf{g}_t, \mathbf{b}_t), \\ \mathbf{A}_t &= \text{diag}(\mathbf{a}_t), \quad \mathbf{G}_t = \text{diag}(\mathbf{g}_t), \quad \mathbf{B}_t = \text{diag}(\mathbf{b}_t), \\ \gamma_t^{\text{full}} &= (\text{vech}(\mathbf{C}_t), \text{vec}(\mathbf{A}_t), \text{vec}(\mathbf{G}_t), \text{vec}(\mathbf{B}_t)). \end{aligned}$$

The conditional covariance recursion then takes the form of (22), preserving positive definiteness by construction through the BEKK quadratic form while allowing all parameter matrices to adapt to the unfolding information set. The asymmetry term in the Neural-BEKK follows the component-wise Kroner and Ng (1998) formulation, in contrast to the joint-sign asymmetry of the econometric benchmark; this difference is taken up again in the comparative discussion in

Section 5.

The stationarity restrictions stated in Section 2.1.4 are implemented directly in the Neural-BEKK forward pass. The recurrent network first produces unconstrained outputs, denoted in the code by `raw_params`. These are not used as BEKK coefficients directly. Instead, the implementation allocates a time-varying persistence budget  $\tau_t < \tau_{\max} < 1$  across the ARCH, GARCH and, if present, asymmetric components. For the diagonal Neural-BEKK this is done asset by asset:

$$\begin{aligned}\tau_{i,t} &= \tau_{\max} \sigma(\tilde{\tau}_{i,t}), \\ \omega_{i,t} &= \text{softmax}(\ell_{i,t}), \\ (a_{i,t}, g_{i,t}, b_{i,t}) &= \sqrt{\tau_{i,t} \omega_{i,t}},\end{aligned}$$

where the  $b_{i,t}$  component is omitted in symmetric specifications. Since the softmax weights are non-negative and sum to one, this construction implies

$$a_{i,t}^2 + g_{i,t}^2 + \delta_{\text{asym}} b_{i,t}^2 = \tau_{i,t} < \tau_{\max} < 1$$

for every asset and every time step. The diagonal persistence condition therefore holds by construction rather than through a post-hoc rejection step.

For the full Neural-BEKK, the code uses the same persistence-budget idea but separates the magnitude of each coefficient matrix from its direction. The network output is reshaped into raw matrices  $\tilde{\mathbf{Q}}_{k,t}$ ,  $k \in \{A, G, B\}$ , whose directions are normalised by their spectral norm,

$$\mathbf{D}_{k,t} = \frac{\tilde{\mathbf{Q}}_{k,t}}{\max\{1, \|\tilde{\mathbf{Q}}_{k,t}\|_2\}}, \quad \|\mathbf{D}_{k,t}\|_2 \leq 1.$$

A scalar budget  $\tau_t = \tau_{\max} \sigma(\tilde{\tau}_t)$  is allocated across the matrices through softmax weights  $\omega_{k,t}$ , and the BEKK matrices are formed as

$$\mathbf{Q}_{k,t} = \sqrt{\tau_t \omega_{k,t}} \mathbf{D}_{k,t}, \quad k \in \{A, G, B\}.$$

Consequently,  $\|\mathbf{A}_t\|_2^2 + \|\mathbf{G}_t\|_2^2 + \delta_{\text{asym}} \|\mathbf{B}_t\|_2^2 \leq \tau_t < \tau_{\max}$ , which is a sufficient, conservative implementation of the BEKK stationarity condition. It avoids computing Kronecker products or eigenvalues during training while still ensuring that the generated recursion remains in a stable parameter region.

**Rationale for the Selection.** The three model classes are intended to span a clear axis of inductive bias. The econometric benchmarks impose strong structural restrictions through the BEKK quadratic form and the DCC variance–correlation decomposition; the neural benchmarks impose almost no structure beyond positive definiteness; and the hybrid architectures interpolate between these extremes, with the kernel-replacement models retaining the BEKK recursion but introducing learnt modulation, and the Neural-BEKK retaining the BEKK recursion while making the recursion parameters themselves time-varying. The comparison across these classes assesses how structural priors and architectural flexibility are associated with forecasting perfor-

mance under the implemented specifications. It is not a factorial architectural ablation, because the model classes also differ in their parameterisations and estimation objectives. The comparison can also be read as a sequence of increasingly adaptive BEKK restrictions: symmetric fixed coefficients, asymmetric fixed coefficients, an asymmetric kernel with time-invariant coefficients and recurrent modulation, diagonal time-varying coefficients, and full time-varying coefficient matrices. Parameter counts are reported in table 3; the remaining training configuration is documented in section 4.3.

## 4.2 Estimation and Evaluation Framework

Using the return vector  $\mathbf{r}_t$  and information set  $\Omega_{t-1}$  defined in section 2, the empirical analysis targets one-step-ahead forecasts of the conditional covariance matrix,

$$\hat{\Sigma}_{t|t-1} = \mathcal{F}_{\hat{\theta}}(\Omega_{t-1}), \quad (25)$$

where  $\mathcal{F}_{\hat{\theta}}$  is the model-specific forecasting map after parameter estimation.

Two evaluation designs are possible: parameters can be re-estimated on rolling or expanding windows, or they can be estimated once and held fixed over a common test block. This thesis uses the second design for every model so that all models face the same refitting schedule. Performance differences under this design can reflect covariance dynamics, architecture, innovation law, and estimation procedure, but not differences in when models are re-estimated. In operational use, both econometric and machine-learning models would normally be refitted periodically. The reported results should therefore be interpreted as a controlled frozen-parameter comparison, not as a production retraining policy; rolling re-estimation is discussed in section 6.

**Sample Partition** The available sample  $\{\mathbf{r}_t\}_{t=1}^T$  is partitioned chronologically into three contiguous subsamples,

$$\underbrace{\{\mathbf{r}_t\}_{t=1}^{T_{\text{tr}}}}_{\text{training}} \cup \underbrace{\{\mathbf{r}_t\}_{t=T_{\text{tr}}+1}^{T_{\text{val}}}}_{\text{validation}} \cup \underbrace{\{\mathbf{r}_t\}_{t=T_{\text{val}}+1}^T}_{\text{test}},$$

in proportions 70%/15%/15%. The test sample is held out under identical boundaries across all models and enters neither estimation nor model selection at any point.

**Estimation Protocols by Model Class** The role of the validation sample differs between the econometric and the neural model classes, in line with their respective parameterisations.

The econometric benchmarks — the symmetric and asymmetric BEKK(1,1) specifications and the scalar DCC and ADCC models — are estimated by quasi-maximum likelihood on the combined training and validation sample  $\{\mathbf{r}_t\}_{t=1}^{T_{\text{val}}}$ . These specifications possess no hyperparameters that require a held-out monitoring sample; withholding the validation observations from the likelihood would amount to a reduction of the estimation information set without offering any methodological benefit. The BEKK specifications are estimated in R using the `BEKKs` package; DCC and ADCC are estimated using the `rmgarch` package with univariate GARCH(1,1) volatility specifications in the first stage and joint quasi-maximum likelihood estimation of the scalar

correlation parameters in the second.

The neural and hybrid architectures — the LSTM and GRU covariance benchmarks, the BEKK-LSTM kernel-replacement models, and the Neural-BEKK specification — are estimated with AdamW on the conditional negative log-likelihood, using the training sample  $\{\mathbf{r}_t\}_{t=1}^{T_{\text{tr}}}$  only. The validation sample  $\{\mathbf{r}_t\}_{t=T_{\text{tr}}+1}^{T_{\text{val}}}$  is used to monitor the loss across epochs and to trigger early stopping. Early stopping terminates optimisation after 50 consecutive epochs without an improvement of at least  $10^{-4}$  in validation NLL, and the checkpoint with the lowest validation NLL is restored. The final hyperparameters were fixed before the reported test evaluation and are listed in section 4.3. All neural and hybrid models are implemented in PyTorch.

The two protocols are not symmetric: the econometric benchmarks observe a slightly larger estimation sample in absolute terms than the neural architectures. The validation sample, however, enters the neural estimation indirectly through early stopping, so that the effective information set used to determine  $\hat{\theta}$  is bounded by  $T_{\text{val}}$  for all model classes. The out-of-sample evaluation window  $\{T_{\text{val}} + 1, \dots, T\}$  is identical across models.

**Input Preprocessing** The neural and hybrid models use feature-wise training-sample standardisation (a fixed z-score transformation, not a per-window normalisation). The mean and standard deviation are computed on the training sample only, and the same affine transformation is applied to validation and test observations to prevent look-ahead bias. For a forecast at date  $t$ , the neural input vector is  $\mathbf{x}_{t-1} = (\mathbf{r}_{t-1}, \text{vech}(\mathbf{r}_{t-1}\mathbf{r}_{t-1}^\top))$ ; the implemented feature set contains no exogenous variables. The contemporaneous return  $\mathbf{r}_t$  enters only the likelihood and evaluation loss as the prediction target. The econometric benchmarks operate directly on the raw return series, since the BEKK and DCC recursions are scale-equivariant and do not benefit from such a transformation. Conditional covariance matrices produced by the neural and hybrid models under standardised inputs are rescaled to the original return scale via

$$\hat{\Sigma}_t^{\text{raw}} = \mathbf{D} \hat{\Sigma}_t^{\text{scaled}} \mathbf{D}, \quad \mathbf{D} = \text{diag}(\hat{\sigma}),$$

with  $\hat{\sigma}$  the vector of training-sample return standard deviations, before any predicted covariance matrix enters a statistical or economic evaluation metric.

**One-Step-Ahead Forecasting under Fixed Parameters** Once  $\hat{\theta}$  has been determined, out-of-sample evaluation proceeds by generating sequential one-step-ahead forecasts over the test period. At each test date  $t \in \{T_{\text{val}} + 1, \dots, T\}$  the model receives the actual information set  $\Omega_{t-1}$  — including the realised return of the most recent period — and produces  $\hat{\Sigma}_{t|t-1}$ . Model parameters  $\hat{\theta}$  are fixed over the test block, while the conditional second moment still evolves through each model’s internal recursion: through the BEKK or DCC recursion in the realised innovations for the econometric benchmarks, through the hidden state for the recurrent neural architectures, and through both the recurrent state and the time-varying parameter matrices  $\gamma_t$  for the Neural-BEKK model.

For the recurrent and hybrid architectures, the information set  $\Omega_{t-1}$  is operationally represented

by

$$\mathbf{X}_t = (\mathbf{x}_{t-s}, \dots, \mathbf{x}_{t-1}), \quad s = 40, \quad \mathbf{X}_t \mapsto \hat{\Sigma}_{t|t-1}.$$

Thus each forecast uses the returns from  $t - 40$  through  $t - 1$  and their realised cross-products. The realised return of period  $t - 1$  is appended to this window before  $\hat{\Sigma}_{t|t-1}$  is produced, so that the forecast is strictly conditional on information available up to and including  $t - 1$ . Because each forecast is generated from a fresh pass over its own  $s$ -day window, the recurrent state is re-initialised per window rather than carried continuously across the test period; the first  $s$  test observations are absorbed into the initial window, so the recurrent and hybrid models are evaluated on the final 925 of the 965 test dates, to which the econometric benchmarks are aligned.

This protocol evaluates the ability of each implemented forecasting system to translate the unfolding information set into adaptive conditional moments through its internal dynamics rather than through repeated parameter updates. It holds the information and refitting protocol common across systems, but does not isolate architecture from the other design differences described below.

**Computational Aspects** Training wall-clock time was logged for every neural and hybrid run. Table 4 reports the median and range across the eleven seeds. The main runs used Metal Performance Shaders on a MacBook Pro with an M4 Pro; the full Neural-BEKK follow-up was trained on the CPU of the same machine. Consequently, its timing is not directly comparable with the MPS timings. Comparable fitting-time logs were not retained for the R-based econometric benchmarks, so no unsupported runtime comparison is reported for them.

| Model                     | Median (s) | Min-max (s)  |
|---------------------------|------------|--------------|
| LSTM (Normal)             | 65.4       | 49.7–75.0    |
| LSTM (Student- $t$ )      | 50.5       | 44.0–64.4    |
| GRU (Normal)              | 205.4      | 169.4–282.8  |
| GRU (Student- $t$ )       | 206.0      | 163.4–280.2  |
| BEKK-LSTM (scalar)        | 381.6      | 292.5–509.5  |
| BEKK-LSTM (vector)        | 212.3      | 168.7–283.8  |
| BEKK-LSTM (mix)           | 703.8      | 547.4–1024.5 |
| Neural-BEKK (asym. diag.) | 350.3      | 254.0–423.4  |
| Neural-BEKK (asym. full)  | 403.3      | 307.4–635.1  |

**Table 4:** Logged neural-model training time across eleven random seeds. Times include optimisation through the early-stopping epoch and restoration of the best validation checkpoint. All rows used MPS except the full Neural-BEKK, which used CPU.

### 4.3 Details of Learning

All neural and hybrid models use  $d_{\text{in}} = 14$ , hidden dimension  $h = 64$ , lookback  $s = 40$ , at most 500 epochs, AdamW with weight decay  $10^{-3}$  on the network weights, and a cosine learning-rate schedule that smoothly decreases the initial rate during training. The model-specific initial rates are reported in table 5; they are not the terminal rates after scheduling. Gradient clipping refers

to the maximum Euclidean norm of the network gradient.

| Model family                | Recurrent layers | Dropout | Batch | Initial LR         |
|-----------------------------|------------------|---------|-------|--------------------|
| LSTM (Normal/Student- $t$ ) | 2                | 0.1     | 64    | $3 \times 10^{-4}$ |
| GRU (Normal/Student- $t$ )  | 2                | 0.2     | 128   | $1 \times 10^{-4}$ |
| BEKK-LSTM (scalar/vector)   | custom cell      | 0       | 64    | $1 \times 10^{-4}$ |
| BEKK-LSTM (mix)             | custom cell      | 0       | 64    | $5 \times 10^{-5}$ |
| Neural-BEKK (diag./full)    | 1                | 0.1     | 64    | $1 \times 10^{-4}$ |

**Table 5:** Training configuration used in the reported experiments. The maximum gradient norm is 1.0 except for the full Neural-BEKK follow-up, where it is 0.5. All models use the early-stopping rule described in section 4.2.

The corresponding optimisation paths are reported in figure 15 in the appendix. Thin lines show the individual seeds and the prominent curves their pointwise medians. The panels are used to assess convergence and the train-validation gap; absolute NLL levels are not compared across Gaussian and Student- $t$  objectives. Different horizontal ranges reflect the model-specific early-stopping epochs.

Parameter estimation for the neural and hybrid models is carried out by minimizing the conditional negative log-likelihood (NLL) on the training sample, with the innovation law chosen per model. The Gaussian-innovation benchmarks—the LSTM (Normal) and GRU (Normal)—minimize the multivariate normal NLL, which assumes a conditionally multivariate normal distribution of returns:

$$\ell_t^N(\theta) = \frac{1}{2} \left[ \log \det(\Sigma_{t|t-1}) + \mathbf{r}_t^\top \Sigma_{t|t-1}^{-1} \mathbf{r}_t + N \log(2\pi) \right]. \quad (26)$$

The Student- $t$  specifications comprise the LSTM and GRU benchmarks, all three BEKK-LSTM variants, and both Neural-BEKK variants. They use the multivariate Student- $t$  NLL parameterized so that  $\Sigma_{t|t-1}$  is the conditional covariance,

$$\begin{aligned} \ell_t^t(\theta) = & -\log \Gamma\left(\frac{\nu + N}{2}\right) + \log \Gamma\left(\frac{\nu}{2}\right) + \frac{N}{2} \log((\nu - 2)\pi) + \frac{1}{2} \log \det(\Sigma_{t|t-1}) \\ & + \frac{\nu + N}{2} \log \left( 1 + \frac{\mathbf{r}_t^\top \Sigma_{t|t-1}^{-1} \mathbf{r}_t}{\nu - 2} \right). \end{aligned} \quad (27)$$

This accommodates the heavy tails of the return distribution while targeting the same conditional covariance as the Gaussian objective. For every Student- $t$  model,  $\nu$  is a scalar learnable parameter estimated jointly with the network weights but without weight decay. It is initialized at 8 and transformed so that  $2.01 < \nu < 100$ , ensuring a finite conditional covariance.

The econometric benchmarks (symmetric and asymmetric BEKK, DCC, ADCC) are estimated under Gaussian quasi-maximum likelihood. This is the implemented benchmark specification, not a claim that every econometric package is limited to Gaussian innovations: BEKKs fits the BEKK covariance recursion by Gaussian QML, while alternative distributions available for parts

of the `rmgarch` workflow were not used. Table 6 summarizes the objectives. The FZ0 score is deliberately absent because it is used only for out-of-sample forecast evaluation in section 5, not for estimation.

| Model                                     | Innovation law | Training objective |
|-------------------------------------------|----------------|--------------------|
| BEKK (sym./asym.), DCC, ADCC              | Gaussian       | Gaussian quasi-NLL |
| LSTM (Normal), GRU (Normal)               | Gaussian       | Gaussian NLL       |
| LSTM (Student- $t$ ), GRU (Student- $t$ ) | Student- $t$   | Student- $t$ NLL   |
| BEKK-LSTM (scalar/vector/mix)             | Student- $t$   | Student- $t$ NLL   |
| Neural-BEKK (asym. diag./full)            | Student- $t$   | Student- $t$ NLL   |

**Table 6:** Training objective by model. Estimation minimizes the conditional negative log-likelihood under the stated innovation law.

## 5 Model Evaluation

### 5.1 Evaluation Design

All risk forecasts are evaluated at the  $\alpha = 1\%$  tail level over the 925-observation test window. VaR and ES are relevant not only for internal risk management but also because financial regulation creates explicit incentives and obligations to measure both tail quantities (Basel Committee on Banking Supervision, 2019, January). Because ES is not elicitable in isolation (Gneiting, 2011), forecast adequacy is assessed through the VaR and ES backtests of section 2.5, while formal comparison uses the joint FZ0 score defined in equation (24).

**Risk-measure construction.** All Value-at-Risk and Expected Shortfall forecasts are produced by the FHS scheme defined in sections 2.4.1 and 2.4.2, using the same rolling window of  $W = 1000$  standardised portfolio residuals for every model.<sup>3</sup> The conditional mean is set to zero, consistent with the effectively centred daily returns. At each forecast date, only the model-specific covariance forecast and the resulting standardised-residual history differ; the portfolio weights, window length, empirical quantile rule, and available information are held fixed. At the  $\alpha = 1\%$  level the 925 out-of-sample forecasts imply an expected  $925 \times 0.01 \approx 9.25$  VaR exceedances under correct unconditional coverage, the reference against which the exceedance rates and Kupiec statistics below are read.

**Training seeds and the scope of inference.** The neural and hybrid models are trained under eleven random seeds, whereas the classical econometric models (symmetric and asymmetric BEKK, DCC, ADCC) are deterministic given the data and therefore identical across seeds. All eleven seeds share the same out-of-sample window and the same realized returns. The dispersion of a model’s performance across seeds consequently measures the *stability of the training procedure*—how sensitive the result is to the random initialization and stochastic optimization—rather

<sup>3</sup>The recurrent and hybrid models’ residual histories begin one lookback window into each split, so their initial FHS window contains marginally fewer early validation observations than the econometric models’; the subsequent rolling construction is identical.

than sampling uncertainty, and does not by itself constitute independent statistical evidence. Statistical significance is instead obtained *within* each seed, from the backtests, the Diebold–Mariano test, and the model confidence set; the seed counts reported below (“ $x/11$ ”) aggregate these within-seed decisions. The symmetric BEKK serves as the common benchmark for the loss-based comparison, being the simplest fully parameterized multivariate volatility model against which both the richer econometric specifications and the neural architectures can be measured.

## 5.2 VaR and ES Backtesting

Table 7 summarizes the VaR backtests across the eleven training seeds, Table 8 the corresponding ES backtests.

| Model                     | Hit rate | Exceptions | Expected | Kupiec | Chr. ind. | Chr. CC | DQ    |
|---------------------------|----------|------------|----------|--------|-----------|---------|-------|
| Neural-BEKK (asym. diag.) | 0.0101   | 9.3636     | 9.2500   | 0/11   | 0/11      | 0/11    | 3/11  |
| BEKK-LSTM (mix)           | 0.0104   | 9.6364     | 9.2500   | 0/11   | 0/11      | 0/11    | 3/11  |
| BEKK-LSTM (scalar)        | 0.0117   | 10.8182    | 9.2500   | 0/11   | 0/11      | 0/11    | 0/11  |
| Neural-BEKK (asym. full)  | 0.0107   | 9.9091     | 9.2500   | 0/11   | 0/11      | 0/11    | 3/11  |
| GRU (Normal)              | 0.0096   | 8.9091     | 9.2500   | 0/11   | 2/11      | 2/11    | 6/11  |
| BEKK-LSTM (vector)        | 0.0114   | 10.5455    | 9.2500   | 0/11   | 6/11      | 6/11    | 7/11  |
| GRU (Student-t)           | 0.0092   | 8.5455     | 9.2500   | 0/11   | 2/11      | 2/11    | 7/11  |
| BEKK (symmetric)          | 0.0141   | 13.0000    | 9.2500   | 0/11   | 11/11     | 11/11   | 11/11 |
| BEKK (asymmetric)         | 0.0141   | 13.0000    | 9.2500   | 0/11   | 0/11      | 0/11    | 11/11 |
| LSTM (Normal)             | 0.0118   | 10.9091    | 9.2500   | 0/11   | 0/11      | 0/11    | 3/11  |
| DCC                       | 0.0141   | 13.0000    | 9.2500   | 0/11   | 11/11     | 11/11   | 11/11 |
| ADCC                      | 0.0141   | 13.0000    | 9.2500   | 0/11   | 11/11     | 11/11   | 11/11 |
| LSTM (Student-t)          | 0.0114   | 10.5455    | 9.2500   | 0/11   | 1/11      | 1/11    | 5/11  |

**Table 7:** VaR backtesting results at the  $\alpha = 1\%$  level over 925 out-of-sample observations. Rejection columns report the number of seeds (out of 11) in which the respective test rejects at the 5% level. The Kupiec test evaluates the null of correct unconditional coverage, the Christoffersen independence test the null that VaR violations are serially independent, and the conditional-coverage test the joint null of correct coverage and independence, and the dynamic quantile (DQ) test the joint null that violations are unpredictable from lagged violations and the VaR forecast. Non-rejection is the desirable outcome. The classical models (BEKK, DCC, ADCC) are deterministic and identical across seeds.

|                           | ER   | CC   | ESR (v1) | ESR (v2) | ESR (v3) |
|---------------------------|------|------|----------|----------|----------|
| Model                     |      |      |          |          |          |
| Neural-BEKK (asym. diag.) | 0/11 | 0/11 | 6/11     | 8/11     | 0/11     |
| BEKK-LSTM (mix)           | 0/11 | 0/11 | 5/11     | 6/11     | 0/11     |
| BEKK-LSTM (scalar)        | 0/11 | 0/11 | 0/11     | 0/11     | 0/11     |
| Neural-BEKK (asym. full)  | 0/11 | 0/11 | 4/11     | 6/11     | 0/11     |
| GRU (Normal)              | 0/11 | 0/11 | 0/11     | 2/11     | 0/11     |
| BEKK-LSTM (vector)        | 0/11 | 0/11 | 1/11     | 2/11     | 0/11     |
| GRU (Student-t)           | 0/11 | 0/11 | 1/11     | 0/11     | 0/11     |
| BEKK (symmetric)          | 0/11 | 0/11 | 0/11     | 0/11     | 0/11     |
| BEKK (asymmetric)         | 0/11 | 0/11 | 0/11     | 0/11     | 0/11     |
| LSTM (Normal)             | 0/11 | 0/11 | 0/11     | 3/11     | 0/11     |
| DCC                       | 0/11 | 0/11 | 0/11     | 0/11     | 0/11     |
| ADCC                      | 0/11 | 0/11 | 0/11     | 0/11     | 0/11     |
| LSTM (Student-t)          | 0/11 | 0/11 | 0/11     | 0/11     | 0/11     |

**Table 8:** ES backtesting results across seeds (rejections at the 5% level out of 11 seeds). ER is the exceedance-residual test of McNeil and Frey (null: zero-mean residuals on violation days), CC the one-sided general conditional-calibration test of Nolde and Ziegel (Hommel-adjusted across four directed joint VaR–ES moments), and ESR the regression-based ES test of Bayer and Dimitriadis (v1/v2: bivariate intercept–slope variants, two-sided; v3: intercept-only, one-sided). Non-rejection is the desirable outcome.

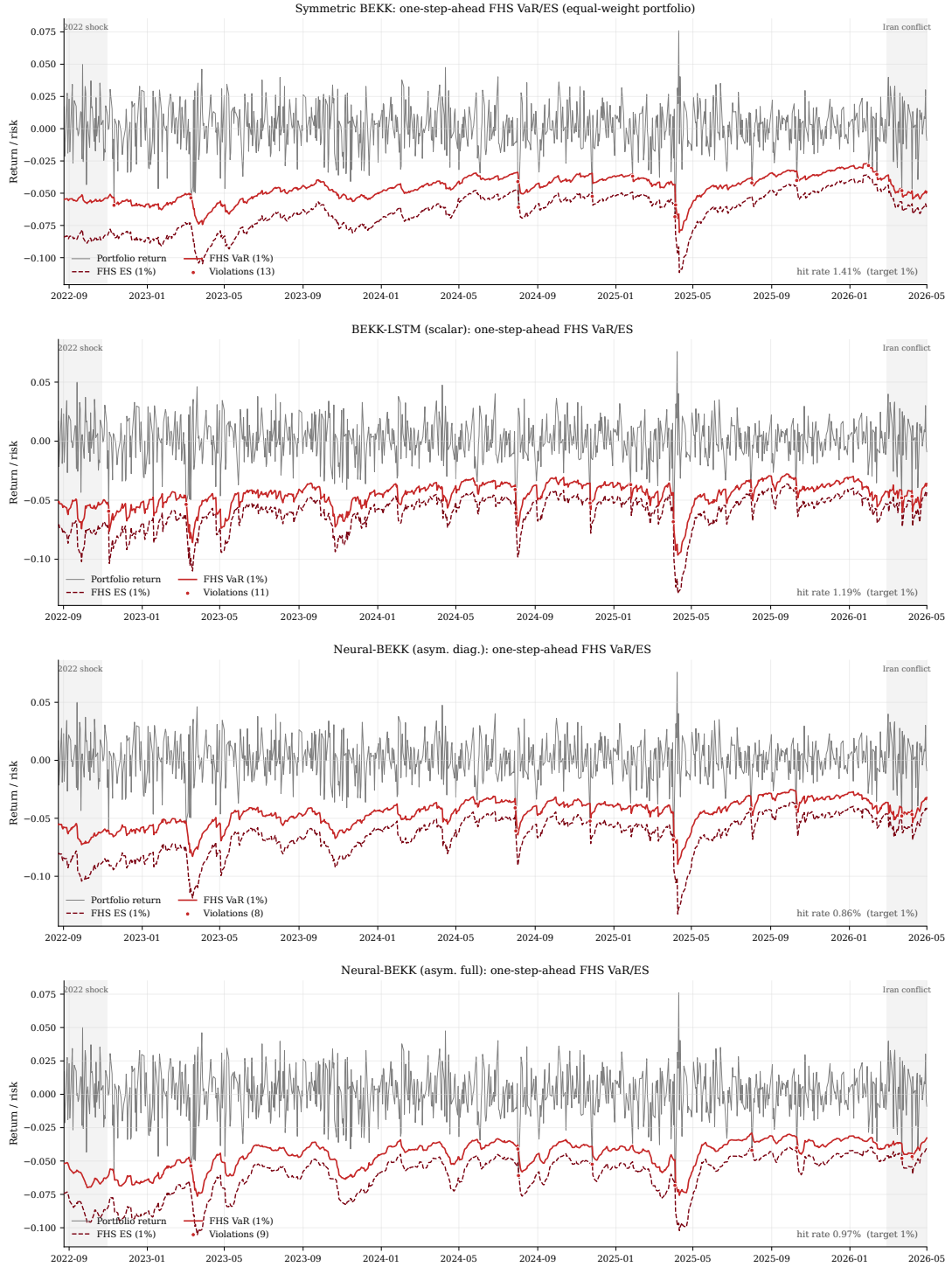
**VaR backtests.** Across all thirteen models the Kupiec test of unconditional coverage never rejects (0/11 seeds; table 7): over 925 observations the gap between the realized and nominal exceedance counts is too small to be detected by a pure frequency test, even for the classical models, which produce 13 violations against the 9.25 expected (an exceedance rate of 1.41%). The tests that condition on the *timing* of violations are far more discriminating. Among the deterministic benchmarks, the symmetric BEKK, DCC and ADCC reject the Christoffersen independence and conditional-coverage tests and the dynamic-quantile nulls in all eleven seeds, indicating first-order clustering and wider predictability of their violations. For the fitted asymmetric BEKK, the Christoffersen tests do not reject (0/11), yet the DQ test rejects (11/11). Thus its violations show no detected first-order clustering but remain predictable from the wider DQ information set; this comparison alone does not isolate the leverage term as the cause of the difference. The leading hybrid architectures improve markedly on the classical benchmarks: Neural-BEKK (asym. diag.), its full-matrix counterpart, and BEKK-LSTM (mix) reject the Christoffersen tests in no seed and the DQ test in only three, while BEKK-LSTM (scalar) passes every VaR test in every seed. The pure recurrent networks are intermediate, and the BEKK-LSTM (vector) variant is the weakest hybrid (Christoffersen 6/11, DQ 7/11); this is consistent with, but does not establish, a calibration cost from its additional per-asset modulation. Overall, the hybrids and the better-behaved neural models attain exceedance rates close to the nominal 1% with violations that are largely unpredictable. The classical models record the highest common exceedance rate (1.41%); first-order clustering is detected for the symmetric BEKK, DCC and ADCC, but not for the asymmetric BEKK.

**ES backtests.** The Expected Shortfall results in table 8 distinguish evidence of risk underestimation from departures from exact calibration. The one-sided exceedance-residual (ER) and ESR-v3 tests, both directed against systematic ES *underestimation*, reject for no model in any seed (0/11 throughout). There is therefore no evidence that any model—classical, neural or hybrid—systematically reports too small a tail expectation. The one-sided general conditional-calibration (CC) test, which combines four directed joint VaR–ES identification moments, likewise never rejects. Its non-rejection is reported separately because it is not a two-sided test of exact calibration. The only rejections arise in the two-sided ESR-v1 and ESR-v2 tests of *exact* calibration. They occur most frequently among the leading hybrids: Neural-BEKK (asym. diag.) rejects in 6 and 8 of eleven seeds, BEKK-LSTM (mix) in 5 and 6, and the full Neural-BEKK in 4 and 6. The full Neural-BEKK therefore matches the diagonal specification on the one-sided tests while producing fewer two-sided rejections. This is descriptively a cleaner exact-calibration record, although it does not offset the full model’s higher FZ0 and QLIKE losses documented below. These results do not identify the direction of the exact-calibration departures: ESR-v1 and ESR-v2 can reject because of either an intercept or a slope deviation, while non-rejection of the one-sided tests is not affirmative evidence of ES overestimation. Establishing the sign would require the ESR coefficient estimates or direct evidence on the mean exceedance residual, neither of which is reported by the present test battery. Notably, BEKK-LSTM (scalar) is rejected by none of the ES tests in any seed and attains the lowest mean FZ0 loss among the models with this clean ES backtesting record.

Figure 9 complements the rejection counts with the realised return path, FHS boundaries and violation dates for four models that anchor the comparison: the symmetric BEKK benchmark, the well-calibrated BEKK-LSTM (scalar), and the diagonal and full Neural-BEKK specifications.

### 5.3 Loss-Based Forecast Comparison

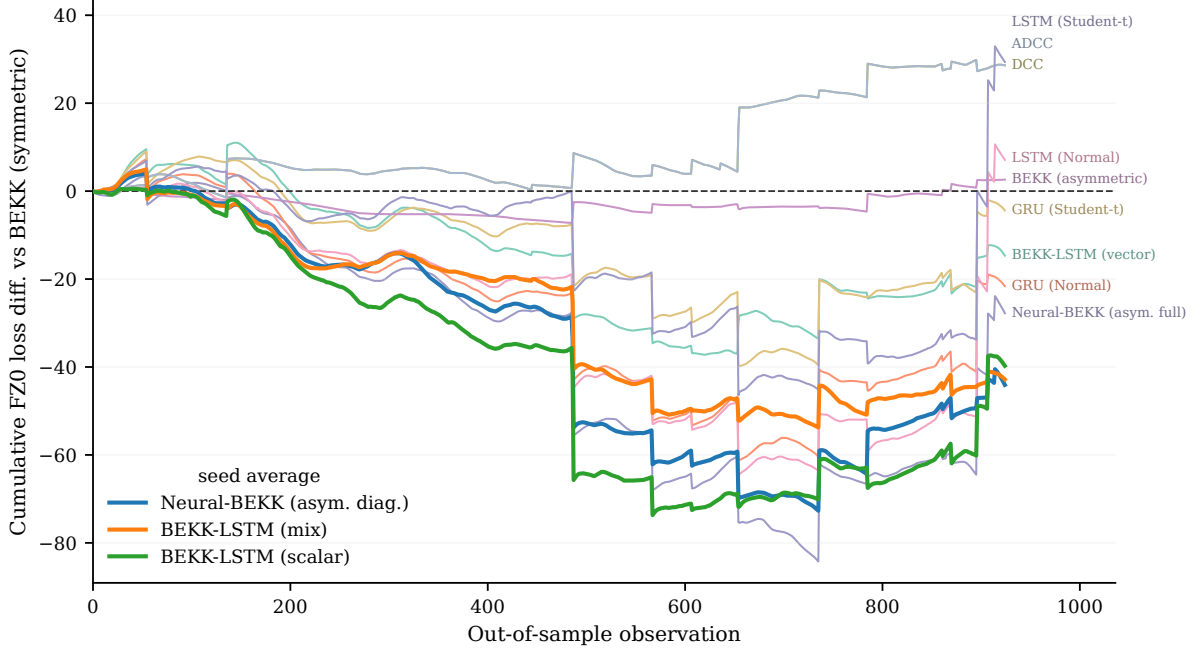
The empirical analysis uses the modified Diebold–Mariano test of Harvey et al. (1997), applied to the loss differential  $d_t = \mathcal{L}_t^{(1)} - \mathcal{L}_t^{(2)}$ . The loss  $\mathcal{L}_t$  is the joint FZ0 score defined in equation (24). Under the null hypothesis of equal predictive accuracy,  $\mathbb{E}[d_t] = 0$ . The test uses the forecast horizon  $h = 1$  because every model produces one-step-ahead forecasts. Choosing  $h > 1$  would require a different, genuinely multi-step forecasting experiment rather than an alternative tuning choice for the present test. The implementation uses the standard horizon-based truncation lag  $h - 1$ . Consequently, for  $h = 1$  no autocovariance terms enter the estimated long-run variance, and the modified statistic coincides with a one-sample  $t$ -test of  $\mathbb{E}[d_t] = 0$  referred to a  $t_{n-1}$  distribution. It is computed with an own implementation, whose statistics and  $p$ -values were cross-validated against this independent  $t$ -test reference across all models and seeds and agree to numerical precision. This horizon-based estimator does not protect the inference against residual serial dependence in the loss differential; that limitation is discussed in section 6.



**Figure 9: Selected one-step-ahead FHS VaR and ES forecast paths.** All panels cover the common 925-observation test window. The neural and hybrid paths use seed 42; the symmetric BEKK is deterministic. Red dots mark VaR violations. The remaining neural and hybrid specifications are reported in figures 16 and 17.

| Model                     | Mean loss diff. | Preferred (seeds) | $p < 0.05$ (seeds) | Mean $p$ -value |
|---------------------------|-----------------|-------------------|--------------------|-----------------|
| Neural-BEKK (asym. diag.) | -0.0476         | 11/11             | 3/11               | 0.2833          |
| BEKK-LSTM (mix)           | -0.0463         | 11/11             | 5/11               | 0.1157          |
| BEKK-LSTM (scalar)        | -0.0430         | 11/11             | 0/11               | 0.3030          |
| Neural-BEKK (asym. full)  | -0.0300         | 10/11             | 0/11               | 0.5116          |
| GRU (Normal)              | -0.0234         | 10/11             | 0/11               | 0.4881          |
| BEKK-LSTM (vector)        | -0.0159         | 9/11              | 0/11               | 0.4707          |
| GRU (Student-t)           | -0.0047         | 7/11              | 0/11               | 0.7008          |
| BEKK (asymmetric)         | 0.0028          | 0/11              | 0/11               | 0.7057          |
| LSTM (Normal)             | 0.0077          | 3/11              | 0/11               | 0.7304          |
| DCC                       | 0.0309          | 0/11              | 0/11               | 0.1148          |
| ADCC                      | 0.0309          | 0/11              | 0/11               | 0.1148          |
| LSTM (Student-t)          | 0.0317          | 0/11              | 0/11               | 0.6376          |

**Table 9:** Modified Diebold–Mariano tests of equal predictive accuracy against the symmetric BEKK benchmark, based on the FZ0 loss differential ( $h = 1$ ). A negative mean loss differential favors the row model. “Preferred (seeds)” counts the seeds in which the row model attains the lower mean loss; “ $p < 0.05$  (seeds)” the seeds with a significant two-sided test. Because all seeds share the same test observations, these counts and the mean  $p$ -value are descriptive summaries across dependent fits rather than aggregate inference.



**Figure 10:** Cumulative FZ0 loss differential relative to the symmetric BEKK benchmark, averaged across the eleven training seeds (deterministic models enter with their single fit). Negative slopes indicate periods in which a model outperforms the benchmark.

The Diebold–Mariano comparisons against the symmetric BEKK benchmark in table 9 reproduce the descriptive ordering of the mean loss differentials. Three of the five hybrid specifications

are preferred to the benchmark in every seed (11/11): Neural-BEKK (asym. diag.) by  $-0.048$ , BEKK-LSTM (mix) by  $-0.046$ , and BEKK-LSTM (scalar) by  $-0.043$ . The remaining hybrids also have negative average differentials, but less uniformly: the full Neural-BEKK is preferred in 10/11 seeds ( $-0.030$ ) and BEKK-LSTM (vector) in 9/11 ( $-0.016$ ). The pure recurrent networks are preferred less consistently—the GRU (Normal) in 10 of eleven seeds and the GRU (Student- $t$ ) in 7—while the asymmetric BEKK is statistically indistinguishable from its symmetric counterpart ( $+0.003$ , 0/11). The correlation-based DCC and ADCC record a higher mean loss in every seed ( $+0.031$ ), but equal accuracy is not rejected in any seed. The three leading hybrid specifications therefore have lower mean loss than *both* parent classes: the econometric BEKK on the one hand and the purely data-driven recurrent networks on the other. The within-seed DM evidence is sensitive to training initialisation: the test rejects equal accuracy at the 5% level in 5 of eleven fits for BEKK-LSTM (mix) and 3 for Neural-BEKK (asym. diag.), and in no fit for any other model. These frequencies do not constitute a combined significance test, and the corresponding mean  $p$ -values (0.12 and 0.28) are descriptive only. The cumulative loss differential in figure 10 drifts downward steadily but accumulates most of its advantage during volatile episodes. This timing is consistent with, but does not isolate, a benefit from the hybrids’ time-varying dynamics.

## 5.4 Synthesis and Practical Implications

Table 10 synthesizes the evaluation across the criteria developed above: the average FZ0 loss across seeds, its dispersion, the VaR exceedance rate, and the model’s inclusion in the 90% model confidence set.

| Model                     | FZ0 loss | Std (seeds) | Hit rate | MCS (90%) | MCS $p$ -value |
|---------------------------|----------|-------------|----------|-----------|----------------|
| Neural-BEKK (asym. diag.) | -2.8938  | 0.0256      | 0.0101   | 11/11     | 0.8584         |
| BEKK-LSTM (mix)           | -2.8924  | 0.0106      | 0.0104   | 11/11     | 0.8030         |
| BEKK-LSTM (scalar)        | -2.8891  | 0.0157      | 0.0117   | 11/11     | 0.8205         |
| Neural-BEKK (asym. full)  | -2.8761  | 0.0296      | 0.0107   | 11/11     | 0.7910         |
| GRU (Normal)              | -2.8695  | 0.0266      | 0.0096   | 10/11     | 0.6538         |
| BEKK-LSTM (vector)        | -2.8620  | 0.0175      | 0.0114   | 11/11     | 0.5480         |
| GRU (Student- $t$ )       | -2.8509  | 0.0205      | 0.0092   | 9/11      | 0.4010         |
| BEKK (symmetric)          | -2.8461  | —           | 0.0141   | 11/11     | 0.5044         |
| BEKK (asymmetric)         | -2.8433  | —           | 0.0141   | 11/11     | 0.5114         |
| LSTM (Normal)             | -2.8384  | 0.0263      | 0.0118   | 11/11     | 0.6468         |
| DCC                       | -2.8152  | —           | 0.0141   | 11/11     | 0.2993         |
| ADCC                      | -2.8152  | —           | 0.0141   | 11/11     | 0.2993         |
| LSTM (Student- $t$ )      | -2.8144  | 0.0212      | 0.0114   | 9/11      | 0.4133         |

**Table 10:** Main out-of-sample results across eleven training seeds, sorted by mean FZ0 loss (lower is better). “Std (seeds)” is the dispersion of the mean loss across seeds and measures training stability; deterministic models are marked “—”. “MCS (90%)” reports in how many seeds the model is contained in the 90% model confidence set of Hansen, Lunde, and Nason (2011) based on the FZ0 loss, and “MCS  $p$ -value” is the corresponding average inclusion  $p$ -value.

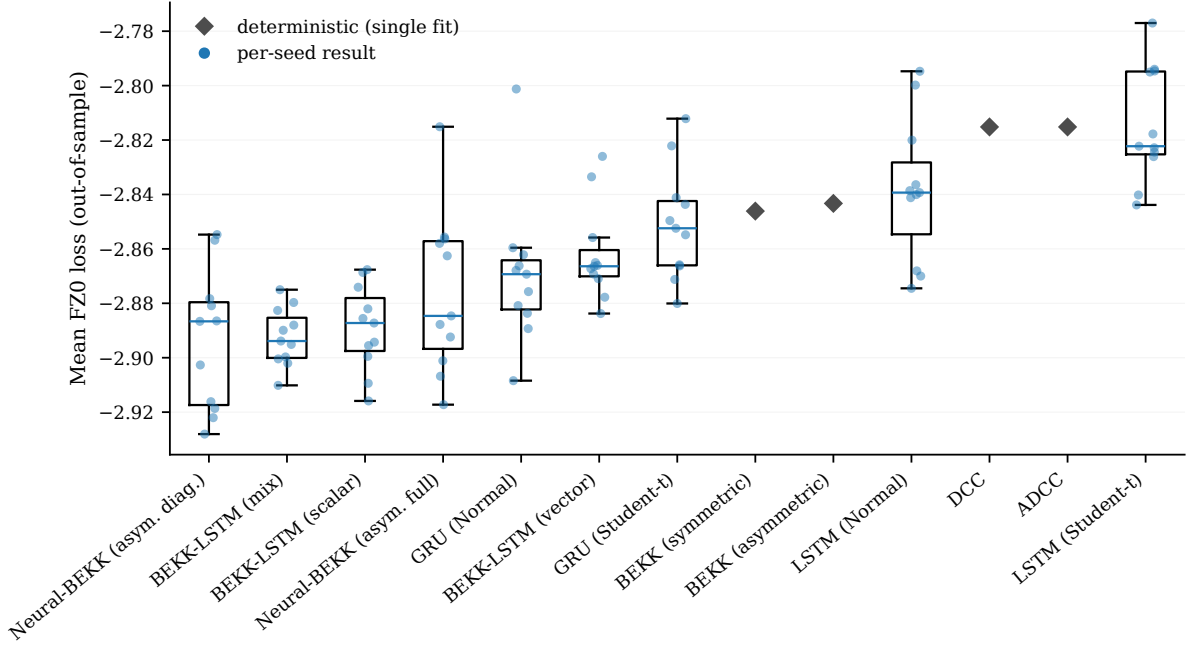
The three leading positions by mean FZ0 loss are occupied by the hybrid architectures Neural-BEKK (asym. diag.), BEKK-LSTM (mix) and BEKK-LSTM (scalar). Below them, the purely data-driven recurrent and classical model ranks overlap: both BEKK specifications record lower mean loss than the LSTM (Normal), while the LSTM (Student- $t$ ) records the highest loss overall; DCC and ADCC rank last only among the classical benchmarks. The pairwise Diebold–Mariano comparisons of table 9 reproduce this mean-loss ordering relative to the symmetric BEKK, but reject equal accuracy for only a subset of the leading hybrid fits. Two qualifications therefore temper it. First, the loss differences are small: every model is retained in the 90% model confidence set in at least nine of eleven seeds, so the FZ0 loss alone cannot isolate a single best model with statistical confidence. The descriptive average MCS inclusion  $p$ -values range from 0.86 for the leading Neural-BEKK to 0.30 for DCC and ADCC but likewise leave the broad non-separation intact. Second, the loss ranking and the calibration evidence of tables 7 and 8 are only partly aligned: the symmetric and asymmetric BEKK attain a lower loss than the LSTM (Normal) yet generate dynamically predictable violations more frequently, and the two lowest-loss hybrids show two-sided ES calibration rejections whose direction is unresolved. Predictive loss and dynamic tail calibration are therefore distinct, and partly complementary, criteria. Section 5.5 complements this comparison with a descriptive analysis of how one hybrid allocates weight between its structured and neural branches.

**Full versus diagonal Neural-BEKK.** The Neural-BEKK family is evaluated in both its diagonal and its full, non-diagonal form; the latter adds off-diagonal entries to the time-varying  $\mathbf{A}_t, \mathbf{G}_t, \mathbf{B}_t$ , letting the neural controller generate cross-asset volatility spillovers directly. The full specification remains competitive on loss and largely calibrated—it is retained in the 90% model confidence set in every seed, is preferred to the symmetric BEKK benchmark in ten of eleven seeds (table 9)—but it does not improve on the diagonal restriction: it ranks fourth by mean FZ0 loss ( $-2.876$  against  $-2.894$  for the diagonal Neural-BEKK), while the Diebold–Mariano null relative to the benchmark is not rejected in any seed. The same ordering holds under the QLIKE covariance loss. The additional off-diagonal flexibility therefore shows no observed out-of-sample benefit in this design and is associated with a small mean-loss cost. This ordering is consistent with a regularisation benefit from the diagonal restriction, but the comparison does not establish weak identification or overfitting as the mechanism.

**Implications for model choice.** What does this imply for a practitioner who must commit to a covariance-forecasting approach? No single model dominates simultaneously on all three dimensions—predictive loss, tail calibration, and training stability. If one model has to be chosen, BEKK-LSTM (scalar) is the most defensible compromise: it ranks third on loss while being the only specification whose VaR and ES backtests reject in no seed, and it is comparatively insensitive to the random training seed. Where the lowest attainable loss is the priority, Neural-BEKK (asym. diag.) and BEKK-LSTM (mix) attain the two lowest mean losses, the latter with the most stable training in the field. Their two-sided ES calibration rejections have an unresolved direction and therefore require separate validation.

A more fundamental reading follows from the model confidence set. Because it retains nearly

every specification, it documents broad statistical non-separation on loss rather than supplying an operational shortlist. A *multiple-model* strategy may still be attractive for diversifying seed-level training noise and regime sensitivity, but the confidence set cannot by itself determine its members; additional screening based on calibration, training stability, and operational considerations would be required. On balance, the evaluation identifies a descriptive mean-loss ranking, not a compact group that is statistically superior to the remaining models.



**Figure 11:** Distribution of the mean out-of-sample FZ0 loss across the eleven training seeds, by model (lower is better). Boxes show the interquartile range with the median; dots are individual seeds. Deterministic models do not depend on the seed and appear as a single value.

## 5.5 The Convex-Mixture Gate

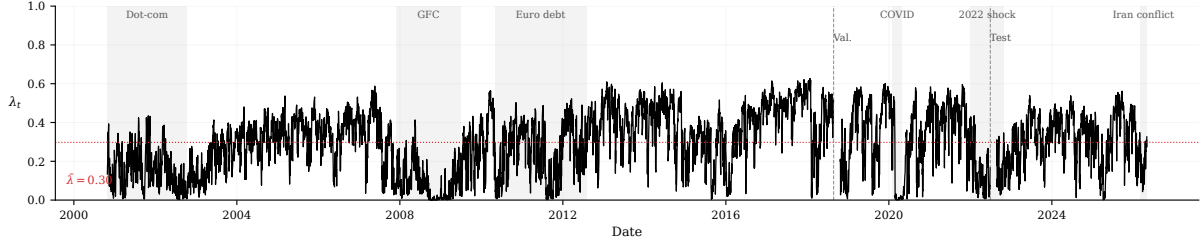
A distinctive feature of the BEKK-LSTM (mix) architecture is that its conditional covariance is an explicit convex combination of a structured BEKK kernel and a free neural covariance branch,

$$\Sigma_t = (1 - \lambda_t) \mathbf{K}_t + \lambda_t \Sigma_t^{\text{nn}}, \quad \lambda_t \in (0, 1).$$

The mixture gate  $\lambda_t$  is directly interpretable: values near zero let the BEKK kernel  $\mathbf{K}_t$  dominate, values near one hand control to the unstructured neural branch  $\Sigma_t^{\text{nn}}$ . Inspecting the realized path of  $\lambda_t$  reveals *how* the hybrid allocates between structure and flexibility, and provides a transparent, model-internal counterpart to the regularisation reading advanced above. Because the most informative stress episodes—the global financial crisis and the COVID-19 crash—precede the out-of-sample window, the gate is examined over the full sample as a descriptive diagnostic of the trained model; this is a statement about learned behaviour rather than an out-of-sample performance claim. The test-window averages reported below show that its average BEKK dominance carries over to the evaluation period, but do not by themselves establish the same regime

dependence there. The descriptive analysis uses seed 42 throughout.

For reproducibility, the shaded stress windows are defined as the dot-com decline (10 March 2000–9 October 2002), global financial crisis (1 December 2007–30 June 2009), euro-area debt crisis (1 May 2010–31 July 2012), COVID-19 crash (1 February–30 April 2020), 2022 market shock (1 January–31 October 2022), and the 2026 Iran-conflict window (28 February 2026 through the sample end).



**Figure 12:** Mixture gate  $\lambda_t$  of the BEKK-LSTM (mix) over the full sample (seed 42), with major stress episodes shaded. Low values of  $\lambda_t$  correspond to a BEKK-dominated conditional covariance, high values to a neural-dominated one. The gate is BEKK-dominant on average and falls further during crises.

*Panel A: Sample splits*

| Statistic | Overall | Train  | Validation | Test   |
|-----------|---------|--------|------------|--------|
| $N$       | 6307    | 4458   | 924        | 925    |
| Mean      | 0.2976  | 0.2920 | 0.3168     | 0.3053 |
| SD        | 0.1581  | 0.1584 | 0.1798     | 0.1296 |
| Min       | 0.0004  | 0.0005 | 0.0004     | 0.0017 |
| 5%        | 0.0270  | 0.0262 | 0.0125     | 0.0715 |
| 25%       | 0.1698  | 0.1625 | 0.1661     | 0.2110 |
| Median    | 0.3149  | 0.3079 | 0.3484     | 0.3234 |
| 75%       | 0.4243  | 0.4173 | 0.4765     | 0.4104 |
| 95%       | 0.5376  | 0.5359 | 0.5597     | 0.4858 |
| Max       | 0.6272  | 0.6272 | 0.6011     | 0.5616 |

*Panel B: Market regimes and selected stress episodes*

| Regime                  | Mean | Median |
|-------------------------|------|--------|
| Non-crisis periods      | 0.35 | —      |
| All stress episodes     | 0.17 | —      |
| Global financial crisis | 0.09 | —      |
| COVID-19 crash          | 0.11 | 0.01   |
| 2022 market shock       | 0.14 | —      |

**Table 11:** Descriptive statistics of the BEKK-LSTM convex-mixture gate  $\lambda_t$  for seed 42. Panel A reports sample-split statistics; “Overall” combines the train, validation and test forecasts. Panel B reports the averages for non-crisis observations, all stress windows shaded in figure 12, and selected stress episodes. Medians not available for the remaining rows are marked “—”.

On average the gate is markedly BEKK-dominant. Over the full sample its mean is 0.30 (median 0.31), and over the out-of-sample test window it is essentially unchanged at 0.31 (median 0.32; table 11): the model places roughly 70% of its weight on the BEKK kernel and 30% on the neural branch. The BEKK-LSTM (mix) is thus better described as a BEKK model with a neural correction than as a neural model with a BEKK admixture.

The more revealing pattern is the gate’s regime dependence. The mean gate falls from 0.35 in non-crisis periods to 0.17 during stress episodes, and the decline is sharpest in the most severe ones—a mean of 0.09 during the global financial crisis, 0.11 during the COVID-19 crash (median 0.01), and 0.14 during the 2022 market shock (table 11 and figure 12). When volatility spikes, the model assigns more weight to the structured BEKK recursion, while the neural branch receives more weight in calmer markets. This allocation is consistent with the intended division between structure and flexibility, but the regime pattern is descriptive: it shows co-movement rather than causal crisis detection and does not establish that either branch improves forecast accuracy in the regimes where it receives more weight. The gate path therefore describes learned allocation, not a causal performance mechanism.

## 5.6 Robustness Checks

The baseline conclusions are probed along four dimensions: the scoring loss, the construction of the risk measures, the tail level, and the asset universe.

**Alternative loss function (QLIKE).** The FZ0 ordering is re-examined with QLIKE, a loss that is strictly consistent for the conditional variance under a correctly specified conditional mean and is robust to conditionally unbiased but noisy volatility proxies (Patton, 2011). For the portfolio variance  $v_{p,t} = \mathbf{w}^\top \boldsymbol{\Sigma}_t \mathbf{w}$ , the implemented loss is

$$\mathcal{L}_t^{\text{pf}} = \log v_{p,t} + \frac{r_{p,t}^2}{v_{p,t}},$$

where the squared portfolio return  $r_{p,t}^2$  is the realised variance proxy. For the full covariance forecast, the matrix analogue is

$$\mathcal{L}_t^{\text{mat}} = \log \det(\boldsymbol{\Sigma}_t) + \mathbf{r}_t^\top \boldsymbol{\Sigma}_t^{-1} \mathbf{r}_t,$$

which uses the realised outer product  $\mathbf{r}_t \mathbf{r}_t^\top$  as the covariance proxy. These proxies are noisy at the daily frequency, but under the zero-mean specification their conditional expectations equal the corresponding portfolio variance and covariance matrix. The portfolio and matrix losses are not numerically comparable; only rankings within each definition are meaningful (table 15). Under portfolio QLIKE the hybrids lead and the comparison becomes discriminating: BEKK-LSTM (mix) and Neural-BEKK (asym. diag.) head the ranking and remain in the 90% model confidence set in every seed, while the two purely data-driven LSTMs are excluded in all seeds. Under matrix QLIKE, by contrast, the correlation-targeting DCC and ADCC rank first and the hybrids fall to the middle. The two views are consistent rather than contradictory: the leading hybrids

rank best for the evaluated equal-weight portfolio variance, but do not necessarily dominate in reproducing the *entire* covariance matrix. DCC and ADCC instead rank best under the criterion that evaluates their dynamic correlation structure jointly with the marginal variances.

**Parametric VaR/ES.** Replacing the filtered historical simulation by a law-matched parametric construction—a Gaussian quantile for the Gaussian-trained models and a Student- $t$  quantile with the per-seed estimated  $\nu$  for the Student- $t$  models, the conditional covariance being rescaled to the  $t$ -scale by the factor  $\sqrt{(\nu - 2)/\nu}$ —broadly preserves the ranking and sharpens its contrasts (table 16). The Student- $t$  hybrids remain close to nominal unconditional coverage (Kupiec rejects in at most one seed, exceedance rates of 1.0–1.3%). The symmetric BEKK, DCC and ADCC reproduce the clustered violations seen under FHS (Christoffersen conditional coverage and DQ reject in all eleven seeds), whereas the asymmetric BEKK now also rejects both tests in every seed. The purely data-driven LSTMs over-violate severely (exceedance rates of 4.7–5.6%, with Kupiec rejecting in all seeds, and LSTM (Normal) excluded from the model confidence set in every seed). GRU (Normal) also deteriorates, although less severely: its exceedance rate rises to 1.82%, Kupiec rejects in 9/11 seeds, and conditional coverage and DQ reject in all seeds. Thus FHS yields much closer unconditional coverage for these pure neural models than the law-matched parametric construction. The leading hybrids’ mean-loss ranking therefore does not depend on the FHS scheme, although several calibration results do depend on the risk-measure construction.

**Alternative tail level ( $\alpha = 5\%$ ).** Re-running the evaluation at  $\alpha = 5\%$  again places the hybrids at the top of the FZ0 ranking—BEKK-LSTM (mix) and the diagonal and full Neural-BEKK lead—and the 90% model confidence set again retains every model (table 17). Mean exceedance rates range from 5.24% to 5.90%, close to the nominal level descriptively. Because the reported robustness table does not contain formal VaR-test results, it does not establish that every classical specification passes those tests. The persistence of the mean-loss ranking shows that the leading hybrids’ FZ0 result is not confined to the deepest 1% tail; the broad confidence set nevertheless prevents a claim of statistically distinct performance at 5%.

**Alternative portfolio and sample.** To probe sensitivity to the asset universe and sample, the models are re-estimated and re-evaluated on a two-asset equity–bond portfolio (S&P 500 and the 10-year Treasury) whose history extends far beyond the commodity-constrained main sample. The exercise documents a *sample-design sensitivity* rather than a causal regime effect, since the training share and the test window shift jointly. The full Neural-BEKK is not included in this robustness exercise. On a 1990-onwards cohort (test 2013–2026), the diagonal Neural-BEKK attains the lowest mean FZ0 loss and all models stay inside the confidence set. On the full 1962-onwards cohort (test 2010–2026), an extreme sample shift, the parsimonious asymmetric BEKK leads instead and the BEKK-LSTM variants record higher mean FZ0 losses than the symmetric benchmark (table 18). The table reports neither pairwise DM tests nor QLIKE results for these cohorts, so it supports no significance or covariance-loss claim beyond the displayed FZ0 and MCS evidence. These cohorts use five seeds and inherit the main hyperparameters without

per-dimension retuning. The exercise therefore shows that the main-sample ordering is sensitive to the cross-section and sample design; it does not identify whether dimensionality, parameter stability, or hyperparameter transfer drives the change.

## 6 Limitations & Future Work

The empirical design of this thesis reflects a deliberate trade-off between methodological rigour, comparability across heterogeneous model classes, and computational feasibility. The present section makes the resulting limitations explicit and outlines directions for future research that the current analysis cannot accommodate.

### 6.1 Limitations

**Fixed-parameter out-of-sample evaluation.** All models are evaluated under fixed parameters on a single chronological hold-out test period, as specified in Section 4.2. The unified train-validation-test protocol places all models on the same test observations, information set, and refitting schedule, but it departs from the rolling or expanding window re-estimation conventional in the (M)GARCH literature. If the test window spans a regime in which the conditional second-moment dynamics differ from those that prevailed during training, the econometric benchmark cannot adapt and may be penalised relative to a hypothetical scheme with periodic re-estimation. Neural and hybrid parameters are likewise not updated, but their recurrent states permit sequence-dependent modulation under fixed weights, whereas the econometric coefficient matrices remain constant.

**Single test split and absence of resampling.** Out-of-sample performance is assessed on one chronological test period. A block bootstrap of the already computed loss differentials could provide dependence-aware confidence intervals without re-estimating the models. A separate expanding-origin design would assess sensitivity to the estimation window but would require repeated model fitting. Neither extension is conducted. The reported Diebold–Mariano tests (Diebold and Mariano, 1995) therefore provide pairwise inference for this single forecast path rather than resampling-based uncertainty across estimation windows. Moreover, their horizon-based long-run-variance estimator uses truncation lag  $h-1 = 0$  for the one-step forecasts (Harvey et al., 1997). It therefore reduces to the sample lag-zero autocovariance of the loss differential and does not protect the reported  $p$ -values against residual serial dependence. A HAC estimator with a data-dependent positive lag or a block bootstrap would provide a useful sensitivity check.

**Finite tail information.** At the  $\alpha = 1\%$  level, each  $W = 1000$  FHS window contributes only about ten standardized residuals to the empirical tail mean, while the 925-observation test period implies only 9.25 expected VaR violations. The resulting ES forecasts and violation-based backtests therefore rely on sparse tail information. Non-rejection should be read as absence of detected misspecification rather than proof of exact calibration; a longer evaluation sample, alternative FHS window lengths, or finite-sample simulation would provide useful sensitivity checks.

**Distributional assumptions.** Parameter estimation relies on Gaussian and Student- $t$  negative log-likelihoods. Although the main Value-at-Risk and Expected Shortfall forecasts use filtered historical simulation rather than parametric tails, the training likelihood still affects the parameter estimates and conditional covariance forecasts. Moreover, the econometric benchmarks are estimated by Gaussian quasi-maximum likelihood, whereas all hybrid specifications use a Student- $t$  likelihood. Architecture and innovation law are therefore not fully crossed, so their separate contributions to the performance differences cannot be identified. The Gaussian and Student- $t$  likelihoods also exclude conditional skewness and asymmetric tail dependence, both of which have been documented in equity returns.

**Cross-sectional scope.** The empirical analysis is restricted to a small fixed asset universe. The behaviour of the proposed Neural-BEKK architecture in higher dimensions is therefore not directly assessed; the quadratic growth of BEKK parameters in the number of assets and the well-documented curse of dimensionality of MGARCH models constrain a straightforward extrapolation of the present results to large cross-sections. The two-asset robustness exercise probes a smaller cross-section and alternative sample windows, but uses only five seeds, carries over the main hyperparameters, and omits the full Neural-BEKK. It therefore documents sensitivity rather than identifying whether dimensionality or the changed estimation sample causes the different ranking.

**Omitted exogenous information.** The implemented feature set contains the four portfolio returns and their ten contemporaneous cross-products, but no exogenous macroeconomic variables. Consequently, the empirical comparison does not test whether information on inflation, real activity, monetary conditions, or macroeconomic uncertainty improves conditional covariance forecasts. This omission matters because such variables have been linked to long-run volatility and stock–bond dependence (Asgharian et al., 2015; Engle, Ghysels, and Sohn, 2013; Yang et al., 2009).

**Stochasticity of neural training.** Neural and hybrid models are trained by stochastic gradient descent under a finite hyperparameter search budget. Reported results are therefore conditional on the chosen initialisation, optimiser settings and validation schedule. A more exhaustive hyperparameter search or an ensembling strategy across multiple random seeds could affect the reported relative performance.

**Forecast horizon.** Evaluation is restricted to one-step-ahead conditional covariance forecasts. Many risk management applications require horizon-coherent multi-step forecasts (for example ten-day Value-at-Risk for regulatory capital), which neither the BEKK recursion nor the proposed hybrid architectures produce by default in closed form.

## 6.2 Future Work

**Rolling re-estimation across model classes.** A natural extension is to implement a unified rolling-window framework with block-wise re-estimation. For the primary architecture compar-

ison, every model class should use the same estimation window and re-estimation interval  $B$ , so that refitting frequency is not confounded with model class. This design would separate within-recursion conditional dynamics, captured between updates, from the capacity to absorb structural change through parameter updates. A secondary, production-oriented scenario could re-estimate the econometric benchmarks at every forecast origin while retraining neural and hybrid models less frequently for computational reasons. Sensitivity analysis over the window length  $W$  and interval  $B$  would quantify the resulting trade-off.

**Deeper kernel-replacement controllers.** The implemented BEKK-LSTM variants use a single custom recurrent update around the econometric kernel. A direct architectural extension would stack a second recurrent layer in this controller and evaluate an analogous two-layer BEKK-GRU specification. Comparing one- and two-layer variants under a matched parameter budget would test whether hierarchical temporal features improve covariance forecasts rather than merely adding capacity; stronger regularisation may be required to limit overfitting and training instability.

**Richer distributional specifications.** Replacing the Gaussian or Student- $t$  likelihood by skewed- $t$ , multivariate generalised hyperbolic, or copula-based specifications would relax the symmetry and elliptical-dependence assumptions of the present setup. In a Neural-MGARCH context, the parameters of such distributions could themselves be made functions of past information through the recurrent backbone, yielding fully time-varying conditional densities.

**Probabilistic Neural-BEKK extensions.** The deterministic time-varying parameter matrices produced by the Neural-BEKK architecture could be extended into a probabilistic form through variational inference, Monte Carlo dropout, or deep ensembles. This would yield predictive distributions over conditional covariance matrices and, by extension, over portfolio-level tail risk measures, addressing one of the practical drawbacks of point-forecast neural approaches in risk management.

**Scalability and alternative asset universes.** Investigating the scalability of the Neural-BEKK to larger cross-sections, possibly through low-rank or factor-augmented parameter matrices, and the transferability of the architecture across asset classes (equities, fixed income, foreign exchange) would document the practical applicability of the proposed model in realistic risk management environments.

**Neural-BEKK-X and mixed-frequency inputs.** A direct Neural-BEKK-X extension would augment the recurrent input with lagged macroeconomic and financial state variables and compare it with the present closed-universe specification. This would test whether the broader information set available to a recurrent controller yields incremental forecast gains. Inflation, industrial production, policy rates, term spreads, and uncertainty measures could enter a separate low-frequency branch and be combined with the daily return features through a MIDAS-style component. The DCC-MIDAS framework of Colacito et al. (2011) provides an econometric ref-

erence for separating short-run dependence from a slowly moving correlation component. Any such experiment would need to respect publication dates and historical data vintages to prevent look-ahead bias.

**Horizon-coherent multi-step forecasts.** Extending the Neural-BEKK to produce internally consistent multi-step conditional covariance paths, either analytically through the implied recursion or via Monte Carlo simulation, is a natural and operationally relevant next step that would broaden the scope of the framework to regulatory and long-horizon risk applications.

## 7 Conclusion

This thesis set out to ask whether integrating recurrent neural networks into the BEKK framework yields multivariate volatility models that forecast portfolio tail risk *better* than, and remain statistically *adequate* relative to, both their econometric and their purely data-driven counterparts. To this end two hybridisation strategies were developed—a BEKK kernel embedded in a recurrent cell, and the fully asymmetric Neural-BEKK with time-varying parameter matrices proposed here—and evaluated bidirectionally against classical BEKK, DCC and ADCC specifications on the one side and pure LSTM and GRU covariance models on the other. All models were placed on a common information footing through a unified train–validation–test protocol, compared at the portfolio level by VaR and ES backtests, the strictly consistent joint VaR–ES score, and model confidence sets, and the neural and hybrid architectures were re-trained across eleven random seeds to quantify sensitivity to training noise.

Under this fixed-parameter design, the evidence answers the research question in the affirmative, but with calibrated rather than emphatic force. The leading hybrid specifications attain the lowest mean out-of-sample joint VaR–ES losses of the entire field and are preferred to the symmetric BEKK benchmark in every seed. They also improve VaR calibration substantially: the scalar BEKK–LSTM passes every VaR test, while the other leading hybrids reject only in a minority of seeds. By contrast, the symmetric BEKK, DCC, and ADCC fail both conditional-coverage and dynamic-quantile tests, and the asymmetric BEKK continues to fail the latter. No model shows evidence of systematic Expected Shortfall underestimation, although some leading hybrids reject the two-sided tests of exact ES calibration. Within the Neural-BEKK family, the diagonal restriction records lower mean FZ0 and QLIKE losses than the full parameterization, so the results do not support an incremental benefit from time-varying off-diagonal spillover coefficients. The overall improvement is, however, consistent rather than decisive. The within-seed Diebold–Mariano test rejects equal accuracy only for a subset of trained fits; because the fits share the same test observations, this pattern measures sensitivity to initialisation rather than aggregate significance, and the horizon-based variance estimate does not guard against residual serial dependence in the loss differential. The model confidence set retains nearly every specification, so the loss differences—although consistently ordered across seeds—are too small to single out one model with statistical confidence, and the robustness checks of section 5.6 show that the ordering itself is not invariant across loss functions and sample designs.

Beyond the headline ranking, the convex-mixture variant offered a transparent window onto how the hybrid allocates weight between its two branches. Its learnt gate is BEKK-dominant on average—placing roughly seventy percent of its weight on the structured kernel—and shifts further toward the BEKK recursion precisely during the most severe stress episodes, granting the flexible neural branch more weight only in calmer markets. This pattern is consistent with the intended division between structure and flexibility, but it is descriptive rather than causal: it does not establish that either branch generates state-dependent forecast gains. Attributing the hybrid’s performance to this allocation would require branch or gate ablations linked directly to realised forecast losses.

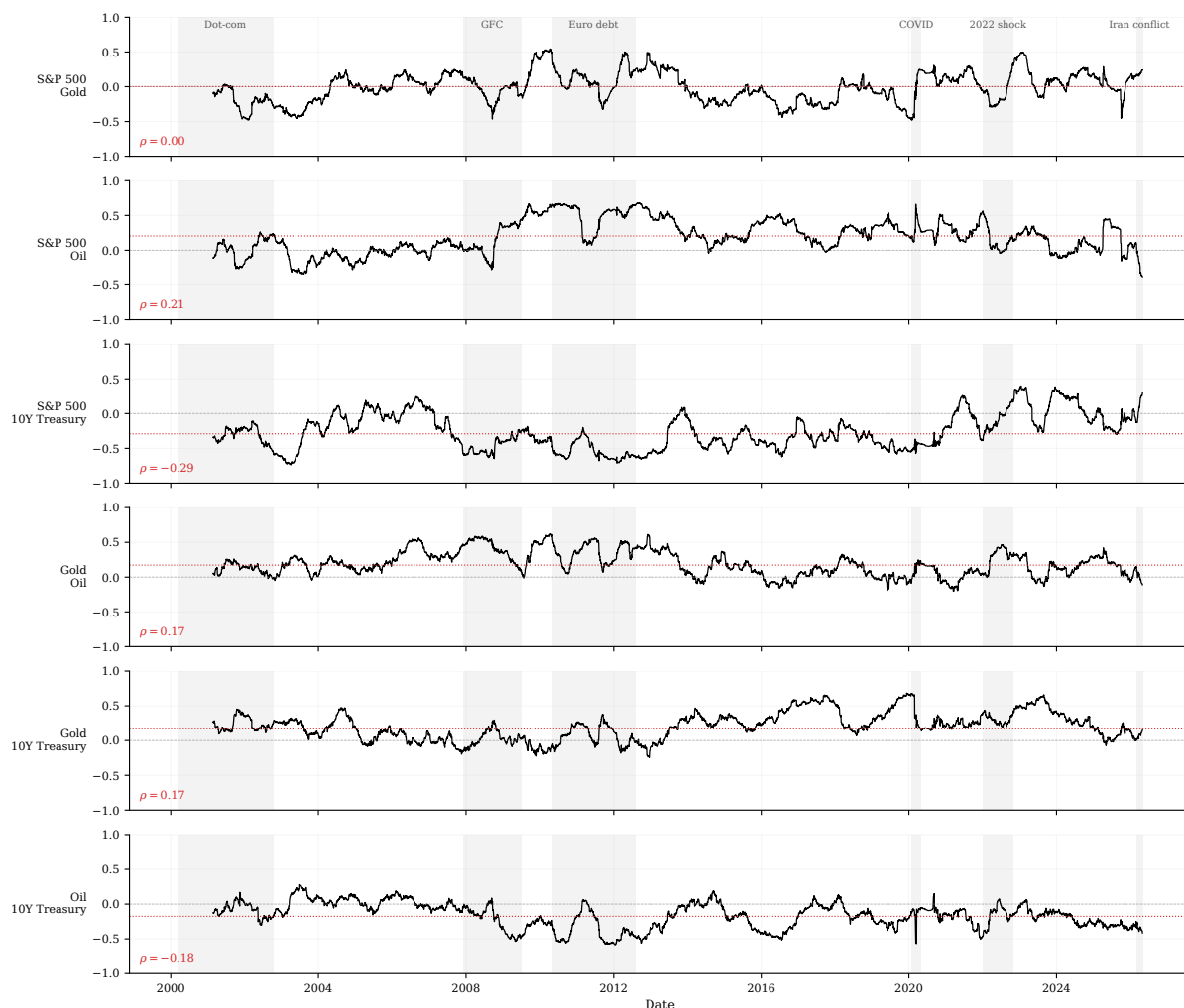
These findings do not identify a single model that dominates loss, calibration, and training stability. Where one model must nonetheless be selected, the scalar BEKK-LSTM is the most defensible compromise; the Neural-BEKK and mixture variant attain lower mean losses but require separate validation of their two-sided ES calibration rejections. The broad model confidence set therefore supplies no statistically superior shortlist.

The principal methodological contribution of this work is the fully asymmetric Neural-BEKK with time-varying parameter matrices, together with a deliberately bidirectional evaluation against both econometric and purely neural baselines that the existing literature on neural-augmented volatility models typically omits. The conclusions are necessarily bounded by the design choices discussed in section 6—a single fixed-parameter test window, a small asset universe, and elliptical innovation laws—and the most valuable extensions follow directly from them: rolling re-estimation across model classes, richer and possibly time-varying conditional densities, exogenous Neural-BEKK-X inputs, and an explicit link between the mixture gate and realised forecast losses. Taken together, the results are consistent with a benefit from embedding econometric structure in a neural volatility model: the leading hybrids attain lower mean FZ0 losses than either parent class in the main design, improve dynamic VaR calibration relative to the classical benchmarks, and are at least competitive with the pure recurrent models on calibration. Because architecture, likelihood, and parameterisation are not fully separated, this evidence supports an association rather than an isolated causal effect, and the margin over a well-specified classical benchmark remains modest.

## Appendix A: Dataset

### A.1 Additional Cross-Asset Dependence

Figure 13 reports the rolling 120-day Pearson correlations for all six pairwise combinations of the four assets, complementing the three economically relevant pairs shown in figure 5 of the main text. The unconditional full-sample correlations  $\rho$  are reported in each panel.

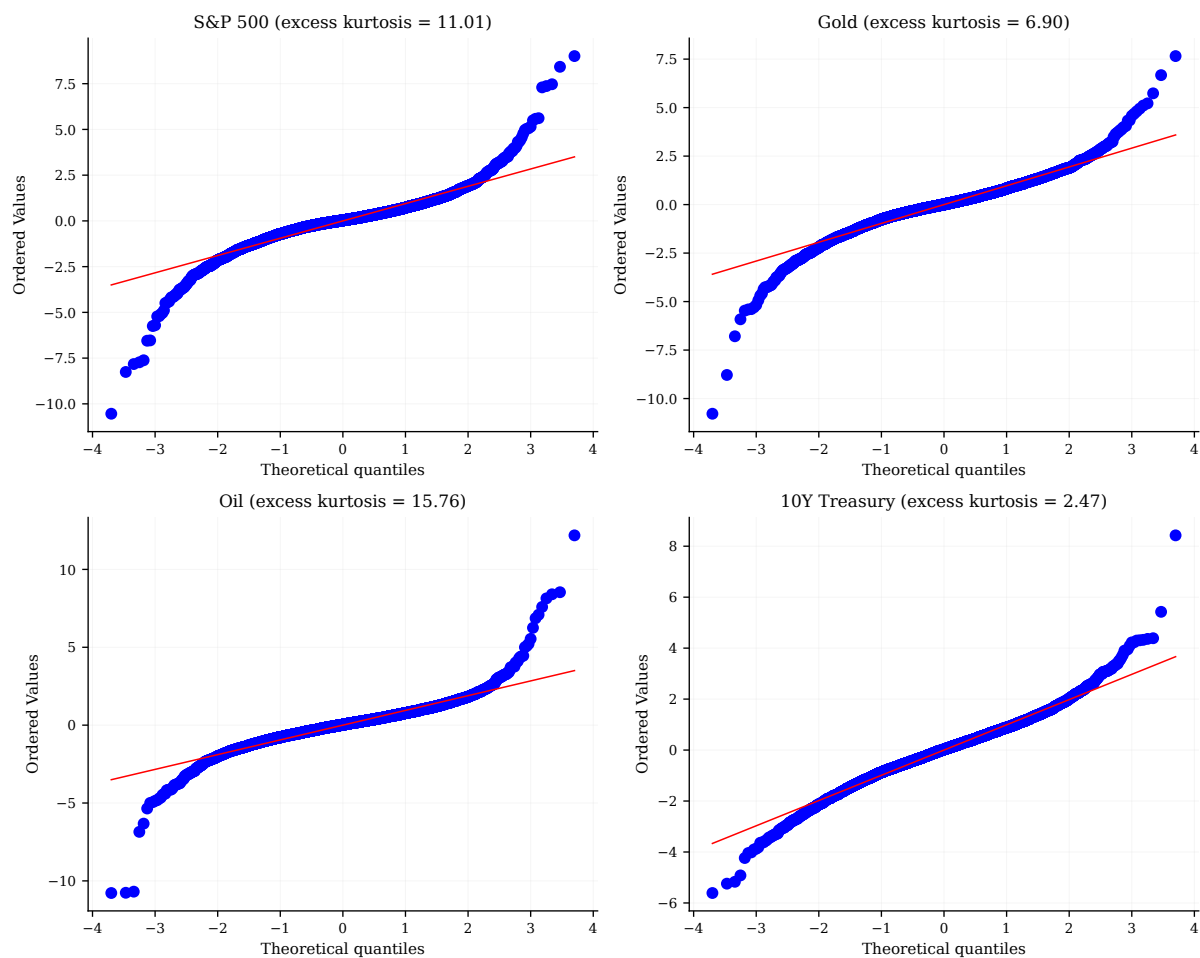


**Figure 13: Rolling 120-day Pearson correlations, all six asset pairs.** All combinations of  $\hat{GSPC}$ ,  $GC=F$ ,  $CL=F$ , and  $\hat{TNX}$ .

### A.2 Distributional Properties

Figure 14 reports normal QQ-plots of the daily returns. All four series display markedly heavier-than-Gaussian tails, with excess kurtosis ranging from 2.47 for the 10-year Treasury to 15.76 for crude oil. These departures from normality motivate Gaussian quasi-maximum-likelihood estimation for the econometric benchmarks and fat-tailed Student- $t$  likelihoods for the hybrid specifications; the pure recurrent benchmarks are estimated under both innovation laws. The law-

matched parametric risk-measure construction in section 5.6 provides an additional sensitivity check on this distributional choice.

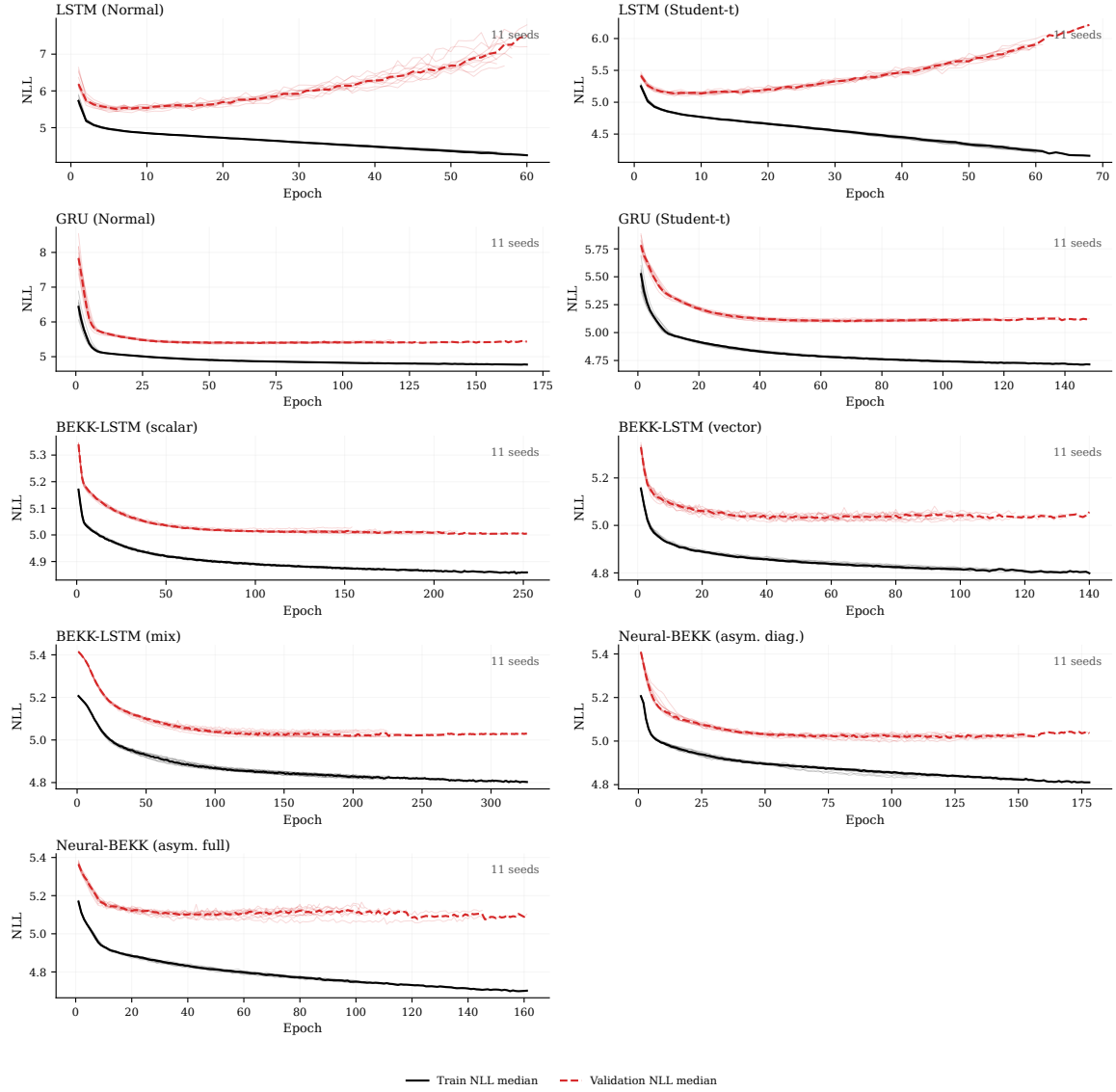


**Figure 14: Normal QQ-plots of daily returns.** Theoretical Gaussian quantiles (horizontal axis) against ordered sample values (vertical axis). The deviation from the diagonal reference line in both tails reveals pronounced fat-tailedness; the corresponding excess kurtosis values are reported in each panel title.

## Appendix B: Model Diagnostics

### B.1 Training and Validation Loss

Figure 15 reports the optimisation histories used to assess convergence and early stopping. The comparison is within panels: Gaussian and Student- $t$  NLL values are based on different conditional densities and their absolute levels are therefore not a model-ranking criterion.



**Figure 15: Training and validation NLL across eleven seeds.** Thin curves are individual runs; solid black and dashed red curves are the pointwise median training and validation losses. Panel lengths differ because optimisation stops after the model-specific early-stopping epoch.

## Appendix C: Backtesting

Section 2.5 develops the backtests and forecast-comparison procedures used in the empirical analysis. For reference, the two tables below summarise their null and alternative hypotheses (Table 12) and their exact statistics, limiting distributions, and implementations (Table 13). Throughout,  $I_t = \mathbb{1}\{r_t \leq q_{\alpha,t}\}$  is the VaR exceedance indicator,  $q_{\alpha,t} = -\text{VaR}_{\alpha,t}$  and  $e_{\alpha,t} = -\text{ES}_{\alpha,t}$  are the signed return-space quantities of section 2.5,  $x = \sum_{t=1}^N I_t$  is the number of exceedances,  $\hat{\pi} = x/N$ , and  $p_{ij} = \Pr(I_t = j \mid I_{t-1} = i)$  are the exceedance-transition probabilities. The Nolde–Ziegel identification functions are  $V_t^V = \alpha - I_t$  and  $V_t^E = e_{\alpha,t} - q_{\alpha,t} + \alpha^{-1}(q_{\alpha,t} - r_t) I_t$ , the exceedance residual is  $\xi_t = (r_t - e_{\alpha,t})/\sigma_t$ , and  $d_t = \mathcal{L}_t^{(1)} - \mathcal{L}_t^{(2)}$  is the loss differential under the FZ0 score

$$S_{\text{FZ}}(r, q, e) = \mathbb{1}\{r \leq q\} \frac{r - q}{\alpha e} + \frac{q}{e} + \log(-e) - 1.$$

**Table 12:** Backtests and forecast-comparison tests: null and alternative hypotheses. Non-rejection of  $H_0$  is the desirable outcome for the adequacy tests.

| Test                                            | Null hypothesis $H_0$                                                                                 | Alternative $H_1$                                |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <i>VaR adequacy</i>                             |                                                                                                       |                                                  |
| Kupiec (UC)                                     | $\pi = \alpha$                                                                                        | $\pi \neq \alpha$                                |
| Christoffersen (ind.)                           | $p_{01} = p_{11}$ (exceedances serially independent)                                                  | first-order dependence (clustering)              |
| Christoffersen (CC)                             | $\pi = \alpha$ and $p_{01} = p_{11}$                                                                  | coverage and/or independence fails               |
| Dynamic quantile (DQ)                           | $\boldsymbol{\delta} = \mathbf{0}$ in $\tilde{I}_t = \mathbf{x}_{t-1}^\top \boldsymbol{\delta} + u_t$ | exceedances predictable from $\mathbf{x}_{t-1}$  |
| <i>ES / joint adequacy</i>                      |                                                                                                       |                                                  |
| Exceedance residual (ER)                        | $\mathbb{E}[\xi_t \mid r_t \leq q_{\alpha,t}] = 0$                                                    | mean $< 0$ (ES underestimation; one-sided)       |
| Conditional calibration (CC; one-sided general) | all four selected instrumented moment means are non-positive                                          | at least one selected moment mean is positive    |
| ESR-v1 (strict)                                 | $(\beta_0, \beta_1) = (0, 1)$ in $r_t = \beta_0 + \beta_1 e_{\alpha,t} + u_t$                         | $(\beta_0, \beta_1) \neq (0, 1)$ (two-sided)     |
| ESR-v2 (auxiliary)                              | $(\beta_0^e, \beta_1^e) = (0, 1)$ in the joint quantile-ES system                                     | $(\beta_0^e, \beta_1^e) \neq (0, 1)$ (two-sided) |
| ESR-v3 (strict intercept)                       | $\beta_0 = 0$ for the forecast error $r_t - e_{\alpha,t}$                                             | $\beta_0 < 0$ (ES underestimation; one-sided)    |
| <i>Forecast comparison</i>                      |                                                                                                       |                                                  |
| Diebold–Mariano (mod.)                          | $\mathbb{E}[d_t] = 0$                                                                                 | $\mathbb{E}[d_t] \neq 0$                         |
| Model confidence set (MCS)                      | $\mathbb{E}[d_{ij,t}] = 0 \quad \forall i, j \in \mathcal{M}$                                         | some model has strictly lower expected loss      |

**Table 13:** Backtests and forecast-comparison tests: statistics, limiting distributions, and implementation. Here  $N$  is the number of out-of-sample forecasts and, for the modified Diebold–Mariano test at horizon  $h = 1$ ,  $\hat{\gamma}_0 = n^{-1} \sum_{t=1}^n (d_t - \bar{d})^2$  is the lag-zero sample autocovariance of the loss differential.

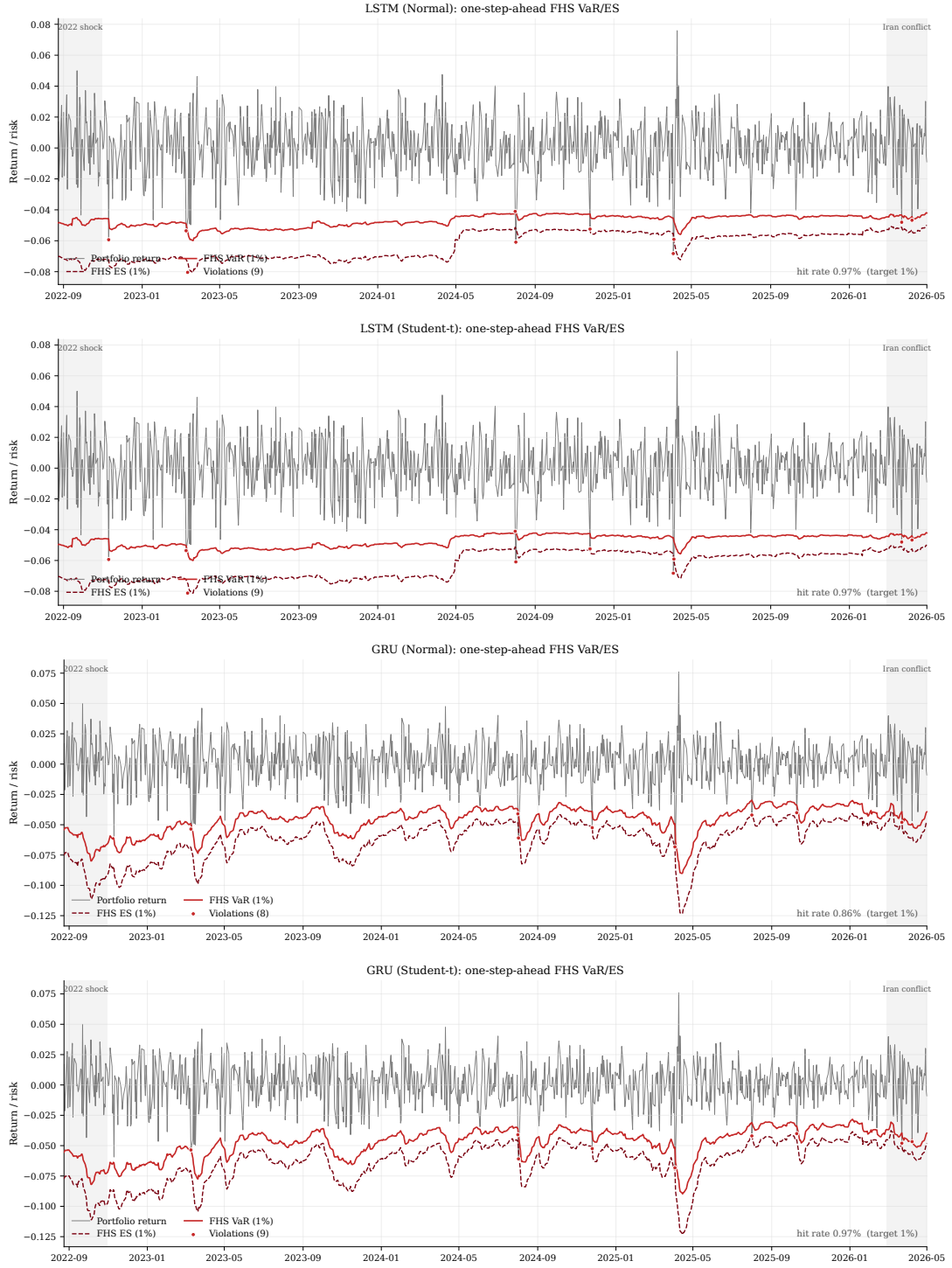
| Test                  | Statistic / regressors                                                                        | Limit      | Implementation |
|-----------------------|-----------------------------------------------------------------------------------------------|------------|----------------|
| <i>VaR adequacy</i>   |                                                                                               |            |                |
| Kupiec (UC)           | $LR_{UC} = -2 \ln [\alpha^x (1-\alpha)^{N-x} / (\hat{\pi}^x (1-\hat{\pi})^{N-x})]$            | $\chi_1^2$ | own            |
| Christoffersen (ind.) | $LR_{IND}$ , first-order Markov chain on $(I_t)$                                              | $\chi_1^2$ | own            |
| Christoffersen (CC)   | $LR_{CC} = LR_{UC} + LR_{IND}$                                                                | $\chi_2^2$ | own            |
| Dynamic quantile (DQ) | Wald on $\delta$ ; $\mathbf{x}_{t-1} = (1, I_{t-1}, \dots, I_{t-4}, q_{\alpha,t}, r_{t-1}^2)$ | $\chi_7^2$ | segMGarch      |

**Table 13:** Backtests and forecast-comparison tests: statistics, limiting distributions, and implementation (continued).

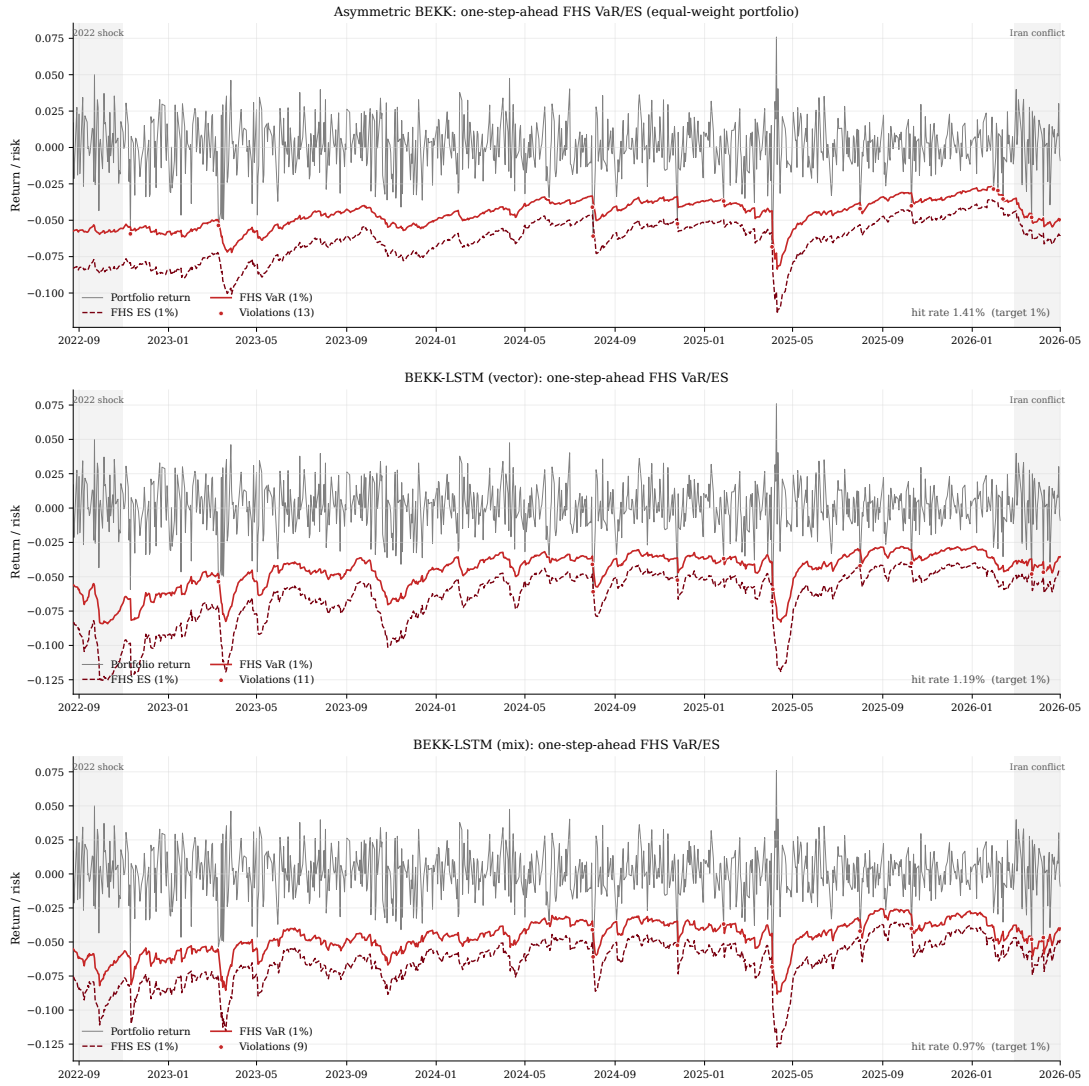
| Test                                               | Statistic / regressors                                                                   | Limit                                    | Implementation                                          |
|----------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------|
| <i>ES / joint adequacy</i>                         |                                                                                          |                                          |                                                         |
| Exceedance residual (ER)                           | sample mean of $\xi_t$ on violation days;<br>one-sided                                   | bootstrap                                | <b>esback</b> , $B = 1000$                              |
| Conditional calibration<br>(CC; one-sided general) | four standardized instrumented<br>identification moments; Hommel-<br>adjusted            | $\mathcal{N}(0, 1)$ com-<br>ponent tests | <b>esback</b>                                           |
| ESR-v1 / v2                                        | Wald on ES regression coefficients<br>(two-sided)                                        | $\chi_2^2$                               | <b>esback</b>                                           |
| ESR-v3                                             | intercept restriction, slope fixed at 1<br>(one-sided)                                   | $\mathcal{N}(0, 1)$                      | <b>esback</b>                                           |
| <i>Forecast comparison</i>                         |                                                                                          |                                          |                                                         |
| Diebold–Mariano (mod.)                             | $DM^* = \frac{\sqrt{\frac{n+1-2h+h(h-1)/n}{n}}}{h=1} \bar{d} / \sqrt{\hat{\gamma}_0/n},$ | $t_{n-1}$                                | own (vs. one-sample<br>$t$ -test)                       |
| Model confidence set<br>(MCS)                      | range statistic $T_R$ ; elimination rule;<br>stationary bootstrap                        | bootstrap                                | <b>arch</b> , 1000 reps;<br>block length 30;<br>seed 42 |

### C.1 Additional VaR and ES Forecast Paths

The selected paths in figure 9 are complemented below by the remaining direct recurrent benchmarks, the asymmetric BEKK, and the BEKK-LSTM vector and mixture variants. All stochastic specifications use the same pre-specified seed 42. DCC and ADCC are omitted because their forecasts and backtesting results are numerically near-identical to each other in this sample.



**Figure 16: FHS VaR and ES paths of the direct recurrent benchmarks.** All panels use seed 42 and the common 925-observation test window. Red dots mark VaR violations.



**Figure 17: Additional FHS VaR and ES paths.** The asymmetric BEKK is deterministic; both BEKK-LSTM panels use seed 42. All panels cover the common 925-observation test window, and red dots mark VaR violations.

## Appendix D: Risk Regulatory

**Table 14:** Regulatory risk measures in banking and insurance supervision. Sources: Basel Committee on Banking Supervision (2004, June), Basel Committee on Banking Supervision (2009, July), and Basel Committee on Banking Supervision (2019, January), European Parliament and Council of the European Union (2013), European Parliament and Council of the European Union (2019), and European Parliament and Council of the European Union (2024), and European Parliament and Council of the European Union (2009).

| Legal Framework                           | Document                                        | Risk Measure                       | Conf. Level / Horizon                    | Application                            | Additional Requirements                                                                        |
|-------------------------------------------|-------------------------------------------------|------------------------------------|------------------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------|
| Basel II (2004)                           | BCBS Framework                                  | Value at Risk (VaR)                | 99%, 10-day                              | Market Risk (Trading Book)             | Regulatory backtesting (e.g., unconditional coverage), capital multiplier linked to exceptions |
| Basel 2.5 (2009)                          | Revisions to the Basel II market risk framework | VaR + Stressed VaR                 | 99%, 10-day                              | Market Risk                            | Additional stressed calibration window required for Stressed VaR                               |
| FRTB (2016; revised 2019)                 | Fundamental Review of the Trading Book          | Expected Shortfall (ES)            | 97.5%, liquidity horizons of 10–120 days | Market Risk (Internal Models Approach) | Risk-factor modellability, P&L attribution test, liquidity horizon adjustments                 |
| Regulation (EU) No 575/2013 (CRR)         | Capital Requirements Regulation                 | VaR (pre-FRTB)                     | 99%, 10-day                              | EU banking supervision                 | Daily calculation and supervisory backtesting requirements                                     |
| CRR II / CRR III (EU FRTB implementation) | EU FRTB reporting / binding capital rules       | Expected Shortfall (ES)            | 97.5%, liquidity-adjusted horizons       | Market Risk                            | Phased implementation and transitional arrangements                                            |
| Solvency II (2009/138/EC)                 | Insurance regulation                            | VaR (Solvency Capital Requirement) | 99.5%, 1-year                            | Insurance risk                         | Standard formula or approved internal model                                                    |

## Appendix E: Robustness Checks

This appendix collects the detailed tables underlying the robustness checks of section 5.6. All figures are out-of-sample across the same eleven training seeds as the main results, except the two-asset cohorts, which use five seeds.

| Model                     | Portfolio QLIKE | MCS pf | Matrix QLIKE | MCS mat |
|---------------------------|-----------------|--------|--------------|---------|
| BEKK-LSTM (mix)           | -7.1183         | 11/11  | -27.5793     | 5/11    |
| Neural-BEKK (asym. diag.) | -7.1158         | 11/11  | -27.6432     | 9/11    |
| BEKK-LSTM (scalar)        | -7.1068         | 11/11  | -27.5790     | 0/11    |
| BEKK (symmetric)          | -7.1027         | 11/11  | -27.8651     | 11/11   |
| BEKK (asymmetric)         | -7.1016         | 10/11  | -27.8230     | 11/11   |
| ADCC                      | -7.0977         | 10/11  | -27.9111     | 11/11   |
| DCC                       | -7.0977         | 10/11  | -27.9111     | 11/11   |
| BEKK-LSTM (vector)        | -7.0968         | 9/11   | -27.5718     | 6/11    |
| Neural-BEKK (asym. full)  | -7.0926         | 9/11   | -27.3958     | 11/11   |
| GRU (Student-t)           | -7.0926         | 6/11   | -27.6136     | 10/11   |
| GRU (Normal)              | -7.0908         | 6/11   | -27.4920     | 10/11   |
| LSTM (Normal)             | -6.7933         | 0/11   | -26.0849     | 0/11    |
| LSTM (Student-t)          | -6.7768         | 0/11   | -26.0692     | 0/11    |

**Table 15: QLIKE covariance-loss comparison (4-asset main sample, 11 seeds).** Mean QLIKE (lower is better) and 90% model-confidence-set inclusion under the portfolio loss  $\mathbf{w}^\top \boldsymbol{\Sigma}_t \mathbf{w}$  and the full-matrix loss. The two columns are not numerically comparable; only the ranking within each is meaningful. Sorted by portfolio QLIKE.

| Model                     | FZ0 loss | Hit rate | Kupiec | Chr. CC | DQ    | MCS (90%) |
|---------------------------|----------|----------|--------|---------|-------|-----------|
| BEKK-LSTM (mix)           | -2.9385  | 0.0107   | 0/11   | 0/11    | 2/11  | 11/11     |
| Neural-BEKK (asym. diag.) | -2.9344  | 0.0113   | 0/11   | 0/11    | 1/11  | 11/11     |
| Neural-BEKK (asym. full)  | -2.9227  | 0.0125   | 1/11   | 0/11    | 1/11  | 11/11     |
| BEKK-LSTM (scalar)        | -2.9167  | 0.0127   | 0/11   | 0/11    | 0/11  | 11/11     |
| BEKK-LSTM (vector)        | -2.8978  | 0.0097   | 0/11   | 2/11    | 3/11  | 9/11      |
| GRU (Student-t)           | -2.8894  | 0.0132   | 0/11   | 3/11    | 3/11  | 10/11     |
| BEKK (symmetric)          | -2.8730  | 0.0141   | 0/11   | 11/11   | 11/11 | 11/11     |
| BEKK (asymmetric)         | -2.8680  | 0.0162   | 0/11   | 11/11   | 11/11 | 11/11     |
| DCC                       | -2.8383  | 0.0151   | 0/11   | 11/11   | 11/11 | 11/11     |
| ADCC                      | -2.8383  | 0.0151   | 0/11   | 11/11   | 11/11 | 11/11     |
| GRU (Normal)              | -2.8292  | 0.0182   | 9/11   | 11/11   | 11/11 | 11/11     |
| LSTM (Student-t)          | -2.2889  | 0.0474   | 11/11  | 11/11   | 11/11 | 4/11      |
| LSTM (Normal)             | -1.8721  | 0.0555   | 11/11  | 11/11   | 11/11 | 0/11      |

**Table 16: Law-matched parametric VaR/ES ( $\alpha = 1\%$ , 11 seeds).** Gaussian quantiles for the Gaussian-trained models and Student- $t$  quantiles (per-seed  $\nu$ ) for the Student- $t$  models, replacing the filtered historical simulation. Columns report mean FZ0 loss, exceedance rate, and the number of seeds (out of 11) in which the Kupiec, Christoffersen conditional-coverage and dynamic-quantile tests reject.

| Model                     | FZ0 loss | Hit rate | MCS (90%) |
|---------------------------|----------|----------|-----------|
| BEKK-LSTM (mix)           | -3.2491  | 0.0524   | 11/11     |
| Neural-BEKK (asym. diag.) | -3.2426  | 0.0542   | 11/11     |
| Neural-BEKK (asym. full)  | -3.2332  | 0.0554   | 11/11     |
| BEKK-LSTM (scalar)        | -3.2315  | 0.0572   | 11/11     |
| BEKK-LSTM (vector)        | -3.2277  | 0.0545   | 11/11     |
| GRU (Student-t)           | -3.2259  | 0.0544   | 11/11     |
| GRU (Normal)              | -3.2236  | 0.0566   | 11/11     |
| BEKK (symmetric)          | -3.2194  | 0.0541   | 11/11     |
| ADCC                      | -3.2179  | 0.0541   | 11/11     |
| DCC                       | -3.2179  | 0.0541   | 11/11     |
| BEKK (asymmetric)         | -3.2173  | 0.0541   | 11/11     |
| LSTM (Student-t)          | -3.2006  | 0.0590   | 11/11     |
| LSTM (Normal)             | -3.1996  | 0.0589   | 11/11     |

**Table 17: FZ0 ranking at  $\alpha = 5\%$  (11 seeds).** Mean FZ0 loss, exceedance rate, and 90% MCS inclusion. The top of the ranking is unchanged relative to  $\alpha = 1\%$  and the MCS again retains every model. Formal VaR-test results are not reported for this check.

| Model                     | FZ0 (1962) | MCS (1962) | FZ0 (1990) | MCS (1990) |
|---------------------------|------------|------------|------------|------------|
| Neural-BEKK (asym. diag.) | -2.4845    | 5/5        | -2.5149    | 5/5        |
| BEKK-LSTM (vector)        | -2.4481    | 2/5        | -2.5102    | 5/5        |
| BEKK (asymmetric)         | -2.5153    | 5/5        | -2.5052    | 5/5        |
| BEKK-LSTM (scalar)        | -2.4559    | 4/5        | -2.5049    | 5/5        |
| BEKK (symmetric)          | -2.5137    | 5/5        | -2.5048    | 5/5        |
| ADCC                      | -2.4915    | 5/5        | -2.4970    | 5/5        |
| DCC                       | -2.4906    | 5/5        | -2.4970    | 5/5        |
| GRU (Student-t)           | -2.4689    | 3/5        | -2.4938    | 5/5        |
| BEKK-LSTM (mix)           | -2.4627    | 4/5        | -2.4870    | 5/5        |
| GRU (Normal)              | -2.4914    | 5/5        | -2.4844    | 5/5        |
| LSTM (Student-t)          | -2.4932    | 5/5        | -2.4669    | 5/5        |
| LSTM (Normal)             | -2.4967    | 4/5        | -2.4662    | 5/5        |

**Table 18: S&P 500/10Y-Treasury cohorts (5 seeds).** Mean FZ0 loss and 90% MCS inclusion for the full-history 1962 cohort (test 2010–2026) and the modern 1990 cohort (test 2013–2026). Sorted by the 1990 FZ0 loss. The full Neural-BEKK is not included in this exercise.

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## Declaration of Authorship

### *Hybrid Neural–MGARCH Models for Volatility and Financial Risk Forecasting*

I hereby declare that the present paper is entirely my own work and without the use of any unauthorised assistance. Any content which has been taken verbatim or paraphrased from other sources has been identified as such. This paper has not been submitted in any form whatsoever to an examining body. Previously published work has been cited as such.

Göttingen, June 30, 2026

Henning Siept Dinkela

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## **Erklärung zur Nutzung von ChatGPT und vergleichbaren Werkzeugen im Rahmen von Prüfungen**

In der hier vorliegenden Arbeit habe ich ChatGPT oder eine andere KI wie folgt genutzt:

- ☐ gar nicht
- ☒ bei der Ideenfindung
- ☒ bei der Erstellung der Gliederung
- ☐ zum Erstellen einzelner Passagen, insgesamt im Umfang von ...% am gesamten Text
- ☒ zur Entwicklung von Software-Quelltexten
- ☒ zur Optimierung oder Umstrukturierung von Software-Quelltexten
- ☒ zum Korrekturlesen oder Optimieren
- ☐ Weiteres, nämlich: ...

Ich versichere, alle Nutzungen vollständig angegeben zu haben. Fehlende oder fehlerhafte Angaben werden als Täuschungsversuch gewertet.